How well do deep mutational scans resolve top-ranked variant batches?

Ranking uncertainty in protein engineering and machine-learning benchmarks

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[Working draft: Numerical results have been checked against the saved analysis tables. Outstanding references, fitting details, and publication metadata are marked TODO.]

Abstract

Protein engineers use measurements of protein fitness, performance in a specified experimental test, to choose batches of variants for further study. Machine-learning researchers use the same measurements to train and evaluate predictors of fitness from sequence. Deep mutational scans provide these measurements for many variants, and sorting the reported scores gives a convenient ranking. But the selected batch depends on both the calculation of fitness from sequencing counts and uncertainty in those counts. In the present work, we quantify these two sources of variation in the leading ranks of three protein systems. We fit models to the counts, generate sampled fitness scores, and measure rank intervals, how often each candidate is selected, and overlap between batches. Changing the fitness calculation changes batch membership in the examples examined, and noise within a fixed analysis can replace a substantial fraction of a batch. The extent of this turnover differs across landscapes and selection budgets, so an exact list of leading variants can conceal substantial uncertainty about which candidates to select.

1 Introduction

Protein engineers want to identify variants with high protein fitness: good performance in a specified experimental test (Yang et al., 2019). For example, an engineer may select a hundred variants for validation, while a machine-learning researcher asks whether a predictor identifies the same candidates. Deep mutational scans measure many variants, sequences that differ from a reference protein, in parallel (Fowler and Fields, 2014). Their fitness scores guide batch selection and serve as labels for learning to predict fitness from sequence. Benchmarks such as FLIP, ProteinGym, and FLIP2 evaluate predictions against these measurements (Dallago et al., 2021; Notin et al., 2023; Didi et al., 2026).

A released *fitness landscape* is usually a table of sequences and scores. Sorting those scores gives the *reported ordering*, and taking the highest-scoring variants gives a candidate batch. This is a natural way to use the experiment, but it leaves two questions unanswered. First, would another calculation of fitness select the same variants? Fitness depends on the test, or *assay*, and on choices such as normalization and how repeated measurements are combined. Second, how certain is the selection under a fixed calculation? In these experiments, sequencing counts how often each variant is observed before and after selection or sorting. Those finite counts leave uncertainty in the fitness estimates, which an exact sorted list does not display.

Methods for estimating fitness already account for measurement uncertainty. Enrich2 and DiMSum estimate errors from counts and repeated measurements (Rubin et al., 2017; Faure et al., 2020); model-based enrichment estimation shares information between related sequences (Busia and Listgarten, 2023). Experimental-design analyses also study how read depth and sampling choices affect precision (Matuszewski et al., 2016). But an error estimate for each score does not tell an analyst how many members of a selected batch could change. That depends on the competing scores and how their uncertainties vary together. We use *ranking resolution* to describe uncertainty in the ranks and batch membership of leading variants. Our question is how much the score uncertainty changes these ranking and selection conclusions.

In the present work, we study scans of three protein systems: protein binding through PDZ interactions, tryptophan production by TrpB, and DNA cleavage by NucB (Zarin et al., 2024; Johnston et al., 2024; Thomas et al., 2025). We fit statistical models directly to the sequencing counts and sample fitness scores or alternative sequencing outcomes under those models. We rank each sampled set of scores and select its leading batch. We compare reported and fitted orderings (Section 4), measure rank uncertainty and batch turnover within each analysis (Section 5), and examine their relationships with count support and competing scores (Section 6). The reported top 100 and top 1,000 and batch sizes from 10 to 1,000 define the scope of these comparisons.

Changing the fitness calculation changes the selected batch in the examples we examine. Within a fixed analysis, noise can also replace a substantial fraction of its members. The amount of turnover differs across landscapes and does not always decrease as batch size grows. Lower input counts are associated with wider score intervals among the leading candidates in every analysis. These results quantify uncertainty in the experimental lists used for engineering and prediction benchmarks, before introducing a fitness predictor. The landscapes studied here contribute to FLIP2; evaluating the effect on its particular predictor comparisons remains future work. [TODO: Add versioned code and data links once their public availability is confirmed.]

2 From sequencing counts to ranking decisions

Fitness depends on the experiment. A protein is a chain of amino acids, and its sequence helps determine what it binds or which chemical reactions it helps carry out. Changing the sequence can affect these properties differently. A test of binding to another protein and a test of enzyme activity therefore ask different questions about the same variant. There is no single assay-independent fitness that both must recover. In this paper, fitness is the score defined by the experimental test and the calculation applied to its measurements (Fowler and Fields, 2014; Johnston et al., 2024). A high score means good performance according to that test and calculation, not universal usefulness.

From variants to sequencing counts. The experiments begin with a *library*, a collection of DNA sequences encoding different protein variants. Researchers test these variants together and connect each protein's performance to the DNA that identifies it. First, sequencing samples the library before selection, giving an input count for each variant. The experiment then favors variants according to the property being tested. Finally, sequencing samples the selected population, giving output counts. Comparing input and output accounts for the fact that variants need not have been equally common at the start.

The physical selection differs between assays. In a growth selection, cells carrying different variants reproduce at rates affected by the tested protein property. In a fluorescence screen, a signal reports activity and a sorting device collects cells or droplets above a chosen signal threshold, called a *gate*. A study can sequence several timepoints or several sorted populations, rather than just one

output. The next section describes the particular assays used here.

A sequencing *read* is an observation of a DNA fragment assigned to a variant. The variant's count is the number of reads assigned to it, and the total assigned reads in a sequenced population are its sequencing depth. A read identifies a variant; it does not directly measure that protein's binding strength or reaction rate. The data used to calculate fitness are therefore counts, not observations of an exact score or rank.

From counts to a reported score. A common starting point is *enrichment*: the fraction of output reads assigned to a variant divided by its fraction of input reads. Fractions account for differences in the total number of reads collected before and after selection. The calculation may then compare the variant with a reference sequence, combine repeated measurements, or summarize changes across timepoints. The unmutated reference sequence is often called wild type; other known sequences can serve as experimental controls. These choices define the score and can affect the resulting ranking.

Why counts leave uncertainty. The ratio alone hides how much data supports it. Consider a hypothetical comparison with equal input and output sequencing depths: one variant has 10 output reads for 1 input read, and another has 10,000 output reads for 1,000 input reads. Both ratios are ten. But one additional input read halves the first ratio and barely changes the second. The same estimated enrichment can therefore come with very different uncertainty. A large experiment can contain both well-observed and sparsely observed variants.

Zero counts require additional choices because division by zero is undefined and an observed zero need not mean a variant was absent from the population. For example, adding a small count before taking a ratio or excluding a poorly observed variant changes the calculation. When only the final score is used downstream, neither these choices nor the amount of supporting data is apparent. Enrich2 and DiMSum retain uncertainty when estimating fitness from counts (Rubin et al., 2017; Faure et al., 2020); we examine how that uncertainty affects ranks and selected batches.

Repeated measurements. Repeating the biological experiment with another population gives a biological replicate. Sequencing the same population again instead measures variation in the sequencing step. Later timepoints and different sorting gates provide other observations, but do not repeat the same experiment. These distinctions matter when combining counts: disagreement between observations can reflect both limited sampling and differences in the populations or conditions. The models retain the observation structure rather than treating every count vector as a biological replicate.

From scores to ranks and batches. A score describes a variant on the experiment's scale; a rank locates it relative to the other candidates. A top-100 batch contains the hundred highest-scoring candidates, regardless of their order within that batch. Two candidates can exchange positions many times while both remain selected. A smaller movement near the selection cutoff can instead replace a member. We therefore examine batch membership separately from internal ordering.

It also matters how scores vary together. If every sampled score increases by the same amount, none of the ranks changes. If scores are close and vary relative to each other, even small changes can alter their order. Separate error bars do not describe this difference. We keep the sampled scores for all variants together and rank each complete set, so the analysis retains their shared variation.

The decisions we study. We compare the reported top 100 and top 1,000 with the fitted analyses, and examine batch sizes from 10 to 1,000. The top 100 is the main worked example, not a universal experimental budget. Detailed displays include members of the proposed batch and outside candidates that could plausibly replace them; Section 3 defines these groups precisely. Restricting the display does not remove competitors from the ranking: each displayed variant is still ranked against all available competitors in that sample.

3 Assay data and uncertainty analysis

Data and scope. We analyze five PDZ landscapes, nine TrpB libraries with three varied sites, one TrpB library with four varied sites, and four NucB engineering generations. For each landscape, we have variant sequences, reported fitness scores, and sequencing counts indexed by replicate, timepoint, or sorting gate. We fit the counts and obtain sampled scores for all candidates together. Ranking these samples gives a distribution of ranks and selected batches for each analysis. The assays differ in what fitness measures and how selection produces the output counts.

PDZ: binding measured through growth. The PDZ data measure binding between two proteins (Zarin et al., 2024). PDZ3 is a domain, a structural unit within the human protein PSD95, that binds a short amino-acid segment of the protein CRIPT. The source study changes amino acids in the binding partners and measures how those changes affect their interaction. A candidate in a joint library therefore specifies a pair of interacting sequences, not necessarily a mutation in PDZ3 alone.

The experiment couples binding to yeast growth. In a protein complementation assay, the binding partners bring complementary reporter fragments together; productive interaction supports growth under selection with the drug methotrexate. Cells carrying different variants compete in the same culture. Sequencing before and after this competition measures which variants increased in relative abundance. The reported DiMSum score summarizes that change across three biological replicates. It is a binding-dependent growth score, not a direct measurement of a binding dissociation constant. Protein abundance, folding, and binding can all affect this readout. We retain the source labels cis, trans-1, trans-2, CRIPT-C, and CRIPT-N for the five prepared landscapes; Appendix A.1 records the available library details.

TrpB: catalysis measured through growth. TrpB is the enzyme subunit that makes the amino acid tryptophan from indole and serine. Johnston et al. (2024) express TrpB variants in bacteria whose native tryptophan-synthesis genes have been deleted. With indole supplied, the introduced enzyme's production of tryptophan supports bacterial growth. Cells carrying active variants can increase in the pooled culture, and sequencing tracks their abundance at the start and at later timepoints.

The three-site libraries A–I vary different triples of amino-acid positions in an engineered TrpB background; they are different candidate collections, not replicate measurements of one collection. We also analyze a four-site library. The reported fitness estimates summarize growth inferred from abundance changes, with normalization using nonfunctional stop-codon variants and the highest estimated growth. Our models instead fit effective growth parameters to the time-indexed counts. Neither score is a purified enzyme's catalytic rate constant: growth also depends on the amount of functional enzyme in the cell and the assay environment. The prepared three-site libraries have one shared input count vector and library-specific output replicate structures; the four-site landscape has two input observations.

NucB: catalysis measured through fluorescence sorting. NucB is a nuclease, an enzyme that cleaves DNA. Thomas et al. (2025) engineer it through four successive generations of designed and screened variants. Here fluorescence reports nuclease activity, and sorting collects populations whose signals exceed chosen thresholds. Unlike PDZ and TrpB, the selection readout is passage through an activity gate, not growth during a competition. Sequencing the unsorted input and each selected population identifies how strongly each variant is enriched by that gate. A variant can have many output reads because it was common initially, because it was more likely to pass the gate, or both; the input counts help distinguish these cases.

The source analysis uses enrichment relative to control sequences to assess activity. We retain enrichment at the highest selected gate as the reported score and fit the counts from all available gates. The fitted score describes the location of a variant's fluorescence distribution (Appendix A.4). Generations G1–G4 contain different candidate collections and screening protocols, so we analyze them separately. A *fiducial* is a known control sequence included in the experiment. The appended premature-stop fiducials encode truncated proteins; we retain their counts where required for normalization but exclude them from candidate rankings. The fitted count model can therefore include controls that are excluded from ranking.

Count models. Let $\mathcal{D}$ denote the observed counts and their replicate, timepoint, or gate identities. Write θ for the model parameters and $p(\mathcal{D} \mid \theta)$ for the count likelihood. Each sequenced population has a vector of variant probabilities. Conditional on those probabilities and the recorded total number of reads, its count vector follows a multinomial distribution. This retains the dependence between variant counts that share a sequencing total. Assay-specific models determine how input probabilities and protein performance change the output probabilities.

Growth models. For PDZ and TrpB, exponential or sigmoid growth functions map input abundances and variant-specific rates to expected output abundances. The Exponential and Lomax specifications assign different priors to a positive central growth rate. The Gamma specification assigns a Gamma prior to an inverse-rate parameter, which is the rate of the exponential distribution for replicate growth rates. We name each specification rate-family–growth-function, such as Exponential–Exponential. Appendices A.2 and A.3 give the equations and hyperparameters.

PDZ has three replicate-level rate vectors linked by a shared variant-level parameter. TrpB libraries A–C use one rate per variant across their observed times; libraries D–I and the four-site model allow replicate-specific rates. TrpB observation times are divided by the last modeled time, so these growth parameters describe change over the modeled assay duration rather than per hour.

Fluorescence models. For NucB, the model assigns each variant a fluorescence location and a distribution of observed signals around it. A gate likelihood combines the input composition with the probability that the signal exceeds the gate's threshold, then normalizes across variants before evaluating a multinomial count likelihood. The Exponential and Lomax specifications differ in their prior on the positive fluorescence-scale location. Generation-specific input populations and gate thresholds determine which observations share parameters. Appendix A.4 defines those probabilities, the normal and lognormal signal options, and all threshold values.

Fitness scores. Let N be the number of eligible ranking candidates and $i \in \{1, \ldots, N\}$ index them. We retain the prepared candidate set for each analysis, including the source-aligned PDZ and TrpB candidates and excluding the appended NucB controls. Candidate counts and exclusions are recorded in Appendix A.8.

Table 1: Analyses used for the ranking summaries. A run is one landscape, model specification, and resampling scheme. Retained samples are draws per run, not effective sample sizes. Models fitted to one landscape share data.

Scheme	Landscapes	Runs	Retained samples
Parameter	18	49	16,000–32,000
Optimized-fit counts	19	60	967–1,000
Posterior counts	18	49	15,839–32,000

The score $h_i(\theta)$ ranks variant i from a fitted parameter vector. For TrpB models without replicate effects, $h_i = \log b_i$, where $b_i > 0$ is the variant's growth rate. For Exponential or Lomax growth hierarchies, $h_i = \log m_i$, where m_i is the replicate-rate center. It is a conditional mean for normal replicate variation and a conditional median for lognormal variation. For the Gamma inverse-rate hierarchy, $h_i = -\log \kappa_i$, where $1/\kappa_i$ is the conditional mean replicate rate. For NucB, $h_i = L_i$, where L_i is the fitted log-fluorescence location. Higher scores are preferred in all three assays, but the units differ between protein systems.

We also calculate scores directly from counts, denoted $g_i(\mathcal{D})$. The PDZ calculation combines replicate log enrichments relative to wild type using inverse-variance weights. It approximates, rather than reruns, the published DiMSum analysis. The TrpB calculation uses the deposited stop-variant and maximum-score normalization before averaging timepoints and replicates. The NucB calculation uses the highest selected gate's enrichment. The TrpB and NucB calculations reproduce the source scores from observed counts; Appendix B gives formulas, tolerances, and filtering rules.

Fitting. We combine the count likelihood with a prior $p(\theta)$ and draw from $p(\theta \mid \mathcal{D}) \propto p(\mathcal{D} \mid \theta)p(\theta)$ using Hamiltonian Monte Carlo (HMC). We also obtain a single parameter vector $\hat{\theta}$ by gradient-based optimization of the joint density, including likelihood and prior terms. We call this an *optimized fit*. Appendix A.5 gives the objective and per-fit settings.

Sampling fitness. One *run* combines a landscape, model specification, and resampling scheme. Let S be its number of retained samples, indexed by $s \in \{1, \ldots, S\}$, and $\boldsymbol{\phi}^{(s)} = (\phi_1^{(s)}, \ldots, \phi_N^{(s)})$ its joint sampled score vector. We use three recipes. Parameter resampling draws $\theta^{(s)} \sim p(\theta \mid \mathcal{D})$ and sets $\phi_i^{(s)} := h_i(\theta^{(s)})$. Optimized-fit count resampling draws $\mathcal{D}^{(s)} \sim p(\cdot \mid \hat{\theta})$ and sets $\phi_i^{(s)} := g_i(\mathcal{D}^{(s)})$. Posterior count resampling draws $\theta^{(s)} \sim p(\theta \mid \mathcal{D})$, then $\mathcal{D}^{(s)} \sim p(\cdot \mid \theta^{(s)})$, and again sets $\phi_i^{(s)} := g_i(\mathcal{D}^{(s)})$. The count-based schemes recompute fitness without refitting the model. Each score vector retains the dependence between candidates within one draw.

Parameter draws describe uncertainty in the fitted score. Count draws describe alternative sequencing outcomes at the recorded depths under the fitted model. They use the count-derived score and do not reproduce every source of variation in a new physical experiment. We apply the same ranking summaries to all three recipes, keeping them separate because they differ in both score calculation and uncertainty. Table 1 lists the available analyses; the four-site TrpB landscape has optimized-fit count samples only.

Ranking summaries. Let $R_i^{(s)}$ be the descending rank of $\phi_i^{(s)}$ among ranking-eligible candidates with finite scores in sample s. Rank one is best, and exact score ties receive their average rank. When the focal candidate's score is undefined, its rank is also undefined. We write $Q_a(\phi_i)$ and $Q_a(R_i)$ for empirical quantiles, at probability a, of the finite samples $\{\phi_i^{(s)}\}$ and $\{R_i^{(s)}\}$, respectively. Thus

unsuperscripted scores and ranks inside Q_a refer to sample collections, with quantiles calculated separately for each candidate.

We define the *median-score reference* by taking $\widetilde{\phi}_i := Q_{0.50}(\phi_i)$ for each candidate and ranking these medians to obtain $\widetilde{r}_i$. Ranking median scores need not give the median sampled ranks. Below, the *reference top 100* means the hundred candidates with the highest median scores. We summarize rank uncertainty using $[Q_{0.05}(R_i), Q_{0.95}(R_i)]$, its width in rank positions, and the two endpoints minus $\widetilde{r}_i$. These signed deviations describe movement relative to the median-score reference. Its rank need not lie inside the sampled-rank interval. The intervals describe the retained sample distributions; they are not a claim of frequentist coverage.

Candidate selection. For a selection budget k, let $T_k^{(s)}$ be the set of the k highest-scoring eligible variants with finite scores in sample s. We estimate inclusion probabilities by $\hat{q}_i(k) := S^{-1}\sum_{s=1}^{S}\mathbf{1}\{i \in T_k^{(s)}\}$, where the indicator equals one if the candidate is selected and zero otherwise. An undefined score counts as no selection; the denominator includes every retained sample. Let $T_k^{\star}$ and T_k^{rep} denote the sets induced by the median-score reference and reported orderings, respectively. We report three shared fractions:

$$\widehat{C}_k := \frac{1}{S(S-1)}\sum_{s\neq s'}\frac{|T_k^{(s)} \cap T_k^{(s')}|}{k}, \tag{1}$$

$$\widehat{C}_k^{\star} := \frac{1}{S}\sum_{s}\frac{|T_k^{(s)} \cap T_k^{\star}|}{k}, \qquad \widehat{O}_k^{\mathrm{rep}} := \frac{1}{S}\sum_{s}\frac{|T_k^{(s)} \cap T_k^{\mathrm{rep}}|}{k}. \tag{2}$$

The first measures agreement between two resampled selections; the second compares samples with their median-score reference; the third compares samples with the reported batch. We examine batch budgets $k \in \{10, 20, 50, 100, 200, 500, 1000\}$. At these budgets, a shared fraction is not the frequency of exact set agreement: an overlap of 0.8 for a top-100 batch means 80 shared members on average, not an 80% chance that the complete batches match. We retain fixed-size sets using the deterministic tie convention of the stored selections, rather than thresholding the average sampled ranks. Appendix C describes finite-score rules and the exact overlap calculation. Table 2 reports all three quantities for the main examples, alongside the intersection of the two fixed reference sets.

Analysis cohorts. Reported-order comparisons use the reported top 100 and top 1,000. Detailed rank-interval summaries use the median-score reference top 100. Our *decision cohort* contains that reference top 100 and every other candidate with $\hat{q}_i(100) \geq 0.01$. We call the latter candidates *challengers*; they can lie outside the reference top 1,000. A displayed candidate is always ranked against all finite eligible competitors in its sample, not just against the other displayed candidates.

Comparisons and aggregation. As a reference for set overlap, two independent uniformly selected size-k subsets of N candidates share fraction k/N in expectation. We report individual runs and summarize model specifications within landscapes by their median, so an experiment with more fitted models does not count as more experimental evidence. Comparisons across resampling schemes use the same named landscape and model specification where available. Candidate sets can differ slightly because the score calculations have different exclusion rules (Appendix C).

Table 2: Distinct top-100 comparisons for the three examples. PDZ trans-1 and TrpB libB use Exponential–Exponential; NucB G3 uses Exponential. Fixed is the intersection of the median-score reference and reported sets. Sample pair, Reference, and Reported are mean numbers shared by a sampled set with another sampled set, the median-score set, and the reported set, respectively. All lists contain 100 variants, so counts also equal percentages.

Example	Scheme	Fixed	Sample pair	Reference	Reported
PDZ trans-1	Parameter	52	59.80	70.59	47.51
PDZ trans-1	Optimized counts	38	69.00	77.74	35.70
PDZ trans-1	Posterior counts	53	49.48	62.00	44.10
TrpB libB	Parameter	33	50.96	63.44	29.81
TrpB libB	Optimized counts	34	32.12	45.89	25.75
TrpB libB	Posterior counts	34	25.87	38.33	23.84
NucB G3	Parameter	68	93.49	95.36	68.14
NucB G3	Optimized counts	67	87.02	91.12	66.79
NucB G3	Posterior counts	68	87.79	91.18	67.13

4 Does the fitness construction change the ordering?

The reported and fitted scores select different top-100 batches. Under parameter resampling, the median-score reference shares 52 candidates (52%) with the reported top 100 for PDZ trans-1, 33 (33%) for TrpB library B, and 68 (68%) for NucB generation 3 under the example specifications in Table 2. These intersections compare two fixed lists. They measure disagreement between score calculations before we consider how the selected batch varies across samples.

The score calculations need not estimate the same assay summary. For NucB, the reported comparator uses the highest selected gate, while the fitted parameter score uses counts from all available gates. For PDZ, our count-based score approximates the original DiMSum calculation. Disagreement therefore does not establish that the reported scores are wrong.

We next compare reported ranks with the sampled rank distributions. These diagnostics depend on both the fitted scores and their uncertainty. They are distinct from the fixed-list intersections above.

Reported ranks within sampled distributions. For each candidate, we calculate the fraction of finite sampled ranks that are better than its reported rank, giving ties half weight. A fraction near zero means that the reported rank is better than most sampled ranks; a fraction near one means it is worse. Figure 1a–c shows the cumulative distribution of these fractions over the reported top 100. For PDZ trans-1, the concentration near zero means that some reported leaders usually move down under the fitted analysis. The diagonal is a descriptive guide, not a calibration target: exact rank recovery would place every candidate at 0.5. Appendix C gives the formula and finite-sample conventions.

Direct rank concordance. Agreement varies across landscapes (Fig. 1d–f). Under Exponential–Exponential parameter resampling, the Spearman correlation between reported and median sampled ranks in the reported top 100 is 0.943 for PDZ cis but 0.310 for PDZ trans-1. In the TrpB library-B Exponential–Exponential example, the correlations are 0.050–0.056 across the three schemes, compared with 0.503–0.584 for NucB generation 3 under Exponential. Expanding the reported cohort

to 1,000 increases these ranges to 0.252–0.262 and 0.921–0.926, respectively (Fig. S1). Agreement over a larger leading cohort can therefore hide disagreement within its top 100.

5 What is the impact of including noise?

5.1 Rank uncertainty within leading batches

Rank intervals for the reference top 100 often extend beyond the selected batch. For TrpB library B under Exponential–Exponential parameter resampling, the median central 90% rank width is 281 positions, and 84 of the 100 candidates (84%) have an upper endpoint beyond rank 100. For NucB generation 3 under Exponential, the median width is 38 positions and 18 candidates (18%) have an upper endpoint beyond 100.

Figure 2a–c displays intervals for the first 20 candidates in each example's median-score reference. The blue points mark median sampled ranks and the black ticks mark reference ranks; these can differ because taking a median and ranking do not commute. Intervals can also extend much farther toward worse ranks than toward better ones.

The broad rank intervals in TrpB library B persist across the noise constructions examined (Fig. 2d–f): the median width among the reference top 100 ranges from 281 to 860 positions, and 84–92 candidates (84–92%) have an upper endpoint above 100. These widths summarize individual rank distributions, not an interval for an average rank. Whether the rank movement changes batch membership depends on where it occurs relative to the selection cutoff.

5.2 Turnover in selected batches

For the TrpB library-B specification above, two resampled top-100 batches share only 26–51 candidates (26–51%) on average across the three schemes (Fig. 3e). Thus about 49–74 members of either batch are replaced in the other, despite a fixed model and selection budget within each analysis.

NucB generation 3 has much higher batch agreement: two sampled top-100 batches share 87–93 candidates (87–93%) on average (Fig. 3f). Its uncertain internal ranks therefore coexist with a comparatively stable candidate group. Table 2 gives the overlaps for each scheme. These quantities describe the fitted resampling distributions, not observed agreement between new physical experiments.

Candidates near the cutoff. Figure 3a–c shows reference ranks 91–100 and the ten outside candidates with the largest inclusion probabilities under parameter resampling. In TrpB library B, several outside candidates are selected about as often as candidates just inside the reference top 100. The cutoff therefore separates candidates with similar selection frequencies. NucB generation 3 shows a clearer difference between the two groups.

Internal stability and agreement with a reference. Agreement between sampled batches need not imply agreement with the reported batch. Under parameter resampling for NucB generation 3, a sampled top-100 set shares 95.36 candidates on average with its median-score reference but only 68.14 with the reported set (Table 2). For TrpB library B, the corresponding values are 63.44 and 29.81, while two sampled sets share 50.96 candidates. NucB mainly differs from the reported list; TrpB also has substantial variation within the fitted analysis. Overlap with the reported answer alone would not distinguish these cases.

Batch size changes the conclusion, but increasing the budget does not always increase the shared fraction. After taking the median across model specifications within each landscape, $\widehat{C}_{1000} > \widehat{C}_{100}$

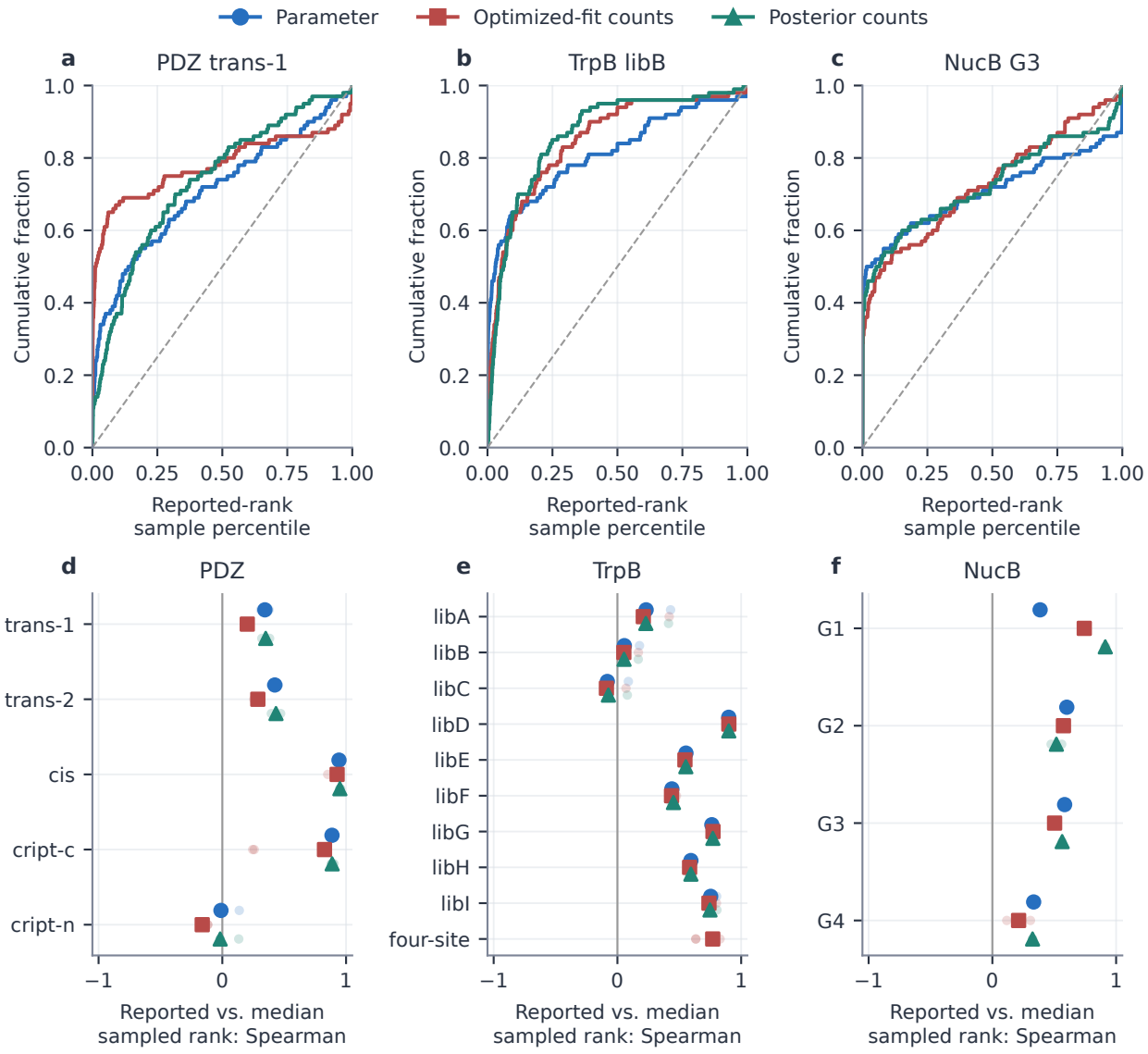


Figure 1: Agreement with the reported top-100 ordering (Section 4). Panels (a)–(c) show empirical cumulative distributions of reported-rank sample percentiles for PDZ trans-1 and TrpB libB under Exponential–Exponential specifications and NucB G3 under the Exponential specification. Blue, red, and green denote parameter, optimized-fit count, and posterior count resampling. The dashed diagonal is descriptive, not a calibration target. Panels (d)–(f) show within-cohort Spearman correlations for all analyzed landscapes. Small faint points denote model runs; larger circles, squares, and triangles denote the median across models within each landscape and scheme. The collections contain 49, 60, and 49 runs, respectively.

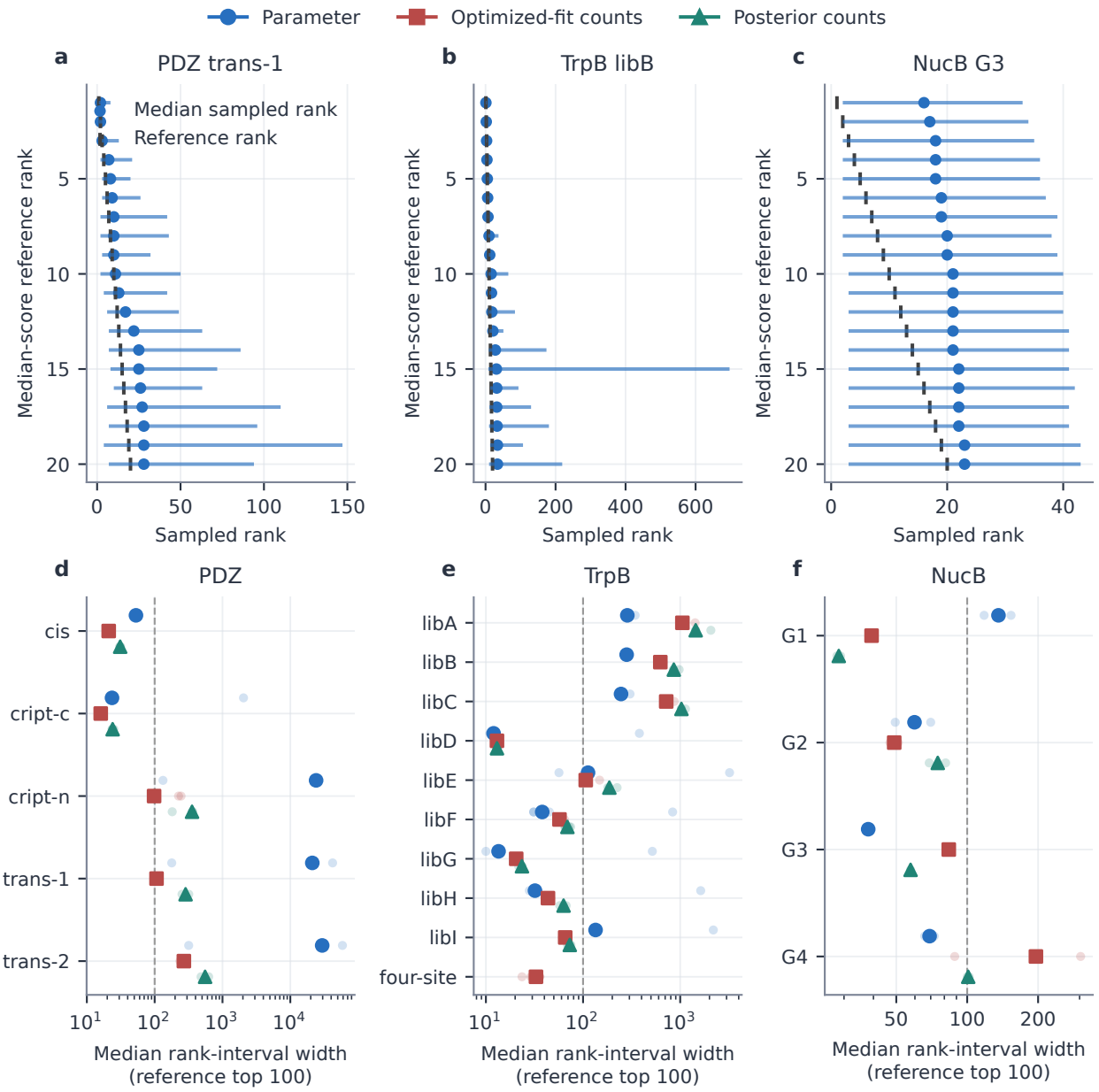


Figure 2: Rank uncertainty within leading batches (Section 5.1). Panels (a)–(c) show the first 20 candidates in the median-score ordering for the PDZ trans-1 and TrpB libB Exponential–Exponential specifications and the NucB G3 Exponential specification. Horizontal intervals span the 5th–95th sampled-rank percentiles under parameter resampling; blue points mark median sampled ranks and black ticks mark median-score reference ranks. Horizontal scales differ across examples. Panels (d)–(f) summarize the median interval width among each run's reference top 100. Faint points denote runs; larger symbols denote medians across models within a landscape. Blue circles, red squares, and green triangles denote the three resampling schemes. The logarithmic horizontal axes report rank positions; the dashed line marks a width of 100. These widths describe sample distributions, not validated coverage against a known biological ordering.

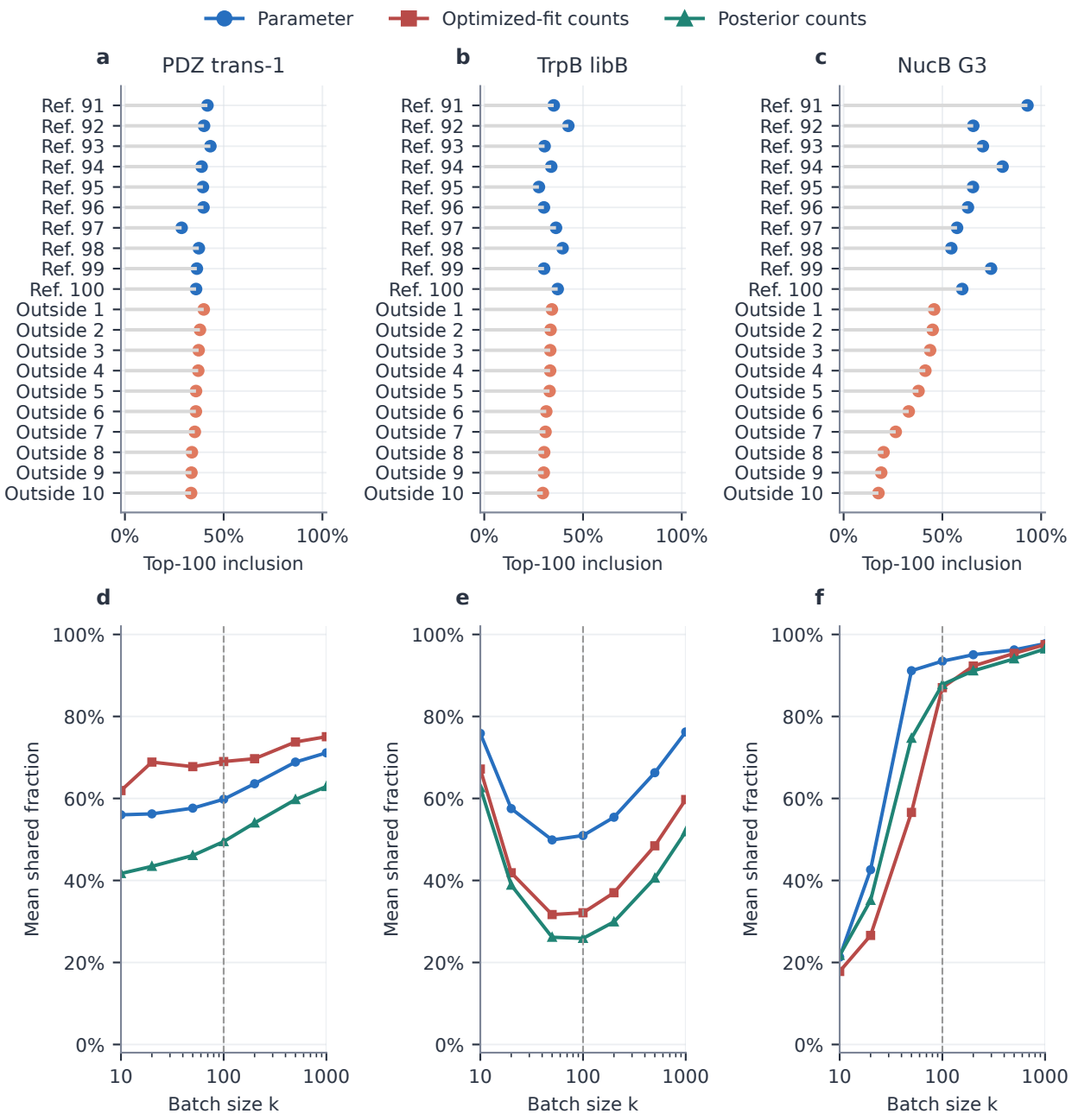


Figure 3: Turnover in selected batches (Section 5.2). Panels (a)–(c) show top-100 inclusion probabilities under parameter resampling for median-score reference ranks 91–100 (blue) and the ten highest-probability outside candidates (orange). Outside candidates are ordered by decreasing inclusion probability; their row numbers are not their sampled ranks. Panels (d)–(f) show mean shared fractions between distinct sampled top-k sets for the three resampling schemes, with batch sizes k from 10 to 1,000. Each curve is one run, not an average over landscapes. PDZ trans-1 and TrpB libB use Exponential–Exponential models; NucB G3 uses Exponential. The example parameter and posterior-count runs have 16,000 samples and the optimized-fit count runs have 1,000. The vertical guide marks $k = 100$. A shared fraction of 0.5 means that two size-k sets share $k/2$ variants on average, not that the whole sets match half the time.

in 13 of 18 landscapes (72%) for parameter resampling, 14 of 19 (74%) for optimized-fit count resampling, and 13 of 18 (72%) for posterior count resampling. The same five TrpB libraries, D through H, decrease under all three schemes (Table 5). Adjusting overlap for the uniform-subset reference using $(\widehat{C}_k - k/N)/(1 - k/N)$ leaves the counts and the five exceptions unchanged. This compares the endpoints 100 and 1,000; it does not establish monotonicity between them.

6 Why do we see these patterns?

Lower input counts are associated with wider score intervals in all 158 decision-cohort analyses. But that association alone does not say which candidates can replace members of a selected batch. Figure 4 follows the counts, score uncertainty, and selection frequencies of every reference member and challenger in three examples. Their different patterns show why the candidate-level data matter alongside the overall correlations.

Count support and score precision. For each variant, we measure score width by $W_i^{\phi} := Q_{0.95}(\phi_i) - Q_{0.05}(\phi_i)$. To compare within-run score geometry across different numerical scales, let S_{run} be the interquartile range (IQR) of the median scores among the reference top 1,000. We use $U_i := W_i^{\phi}/S_{\text{run}}$ as normalized score uncertainty. The denominator puts score widths on the scale of differences between leading candidates. It is not an uncertainty estimate for one candidate. Write n_i^{in} and n_i^{out} for total observed input and output reads over the retained observations, with shared input counts included once. Appendix D defines the sums for each assay, including timepoints, gates, and resequencing.

Within each decision cohort, the Spearman correlation between input counts and U_i is negative in all 158 runs; the output-count correlation is negative in 157. Figure S2 shows these correlations separately by landscape and scheme. Multiple runs use the same experimental counts, so the 158 results are not independent replications. Output counts also depend on selection, whereas input counts are measured before it. Appendix F shows the point-level relationships for every run, without binning or fitting trend lines.

Locating plausible replacements. Let $\widetilde{\phi}_{(100)}$ be the score at the reference rank-100 cutoff. The normalized margin $X_i := (\widetilde{\phi}_i - \widetilde{\phi}_{(100)})/S_{\text{run}}$ locates each candidate relative to that cutoff. Positive margins identify reference top-100 members, subject to exact ties; negative margins identify candidates outside the reference set. Each row of Fig. 4 shows score width against input counts, output counts, and this margin, followed by inclusion frequency against the same margin. Filled blue points are reference members and open orange points are challengers. The same candidates appear in all four panels of a row.

PDZ trans-1. Score width declines as either count increases in this example (Fig. 4a,b), although the displayed widths span a relatively narrow range on the common scale. Reference members and challengers have overlapping widths (panel c). Selection frequency changes gradually across the cutoff (panel d), and 319 outside candidates have inclusion frequency at least 0.01. The proposed top 100 therefore has many plausible replacements, even with a consistent count–width relationship.

TrpB library B. The count–width relationship spans a much wider range here (Fig. 4e,f). Many challengers with median scores below the cutoff have wider score intervals than reference members (panel g). Some lie several run-IQR units below the cutoff but still enter the top 100 in at least 1% of samples (panel h). There are 416 challengers in total. Selecting only from the median-score

ordering would leave these candidates out of the proposed batch, despite their nonzero inclusion frequencies under resampling.

NucB generation 3. The pattern differs in NucB (Fig. 4i–l). Many reference members have wider score intervals than the challengers, reversing the margin–width pattern seen in TrpB (panel k). Yet 77 of the 100 reference members (77%) have inclusion frequency at least 0.99, and only 28 outside candidates meet the challenger threshold (panel l). A broad individual score interval does not by itself imply uncertain batch membership. Inclusion depends on the relative scores in each joint sample, not on the width of one candidate's interval alone.

These examples connect the batch overlaps in Fig. 3 with the candidates that produce them. Count support, score margins, and inclusion frequencies describe different parts of that result; none can replace the others. The assays and fitted models differ between rows, so the comparison does not isolate an effect of counts or score spacing.

We also fit descriptive regressions of rank width on score width, local crowding, their interaction, and reference rank. Local crowding is the negative logarithm of normalized score spacing, so larger values mean closer competitors. The coefficients vary across analyses, and we do not find a consistently positive interaction. Appendix D reports the specifications and coefficient counts. They summarize these fits; they do not estimate how extra sequencing would change the ranks.

What normalization does and does not remove. Dividing by S_{run} does not change within-run Spearman correlations. Nonlinear monotone score transformations can, however, change widths and spacing while preserving every sampled ranking. We therefore interpret these score summaries on their stated scales and use sampled ranks and selections to measure the ranking consequences.

Dependence on modeling assumptions. For the example specifications, TrpB has lower batch agreement than NucB under all three sampling recipes. Other comparisons are much more sensitive to model choice. In PDZ trans-1, parameter-resampling top-100 overlap is 59.80% under Exponential–Exponential and 0.27% under Gamma–Exponential. We cannot interpret the larger uncertainty as a property of the experiment without checking the corresponding model. Appendix F shows all runs separately, and Appendix A.6 reports the available diagnostics and missing predictive checks.

7 Discussion

We found that fitness calculation and measurement uncertainty can change the composition of leading batches in deep mutational scans.

Limitations. The estimated uncertainty is conditional on the assay models and retained fits. A model that does not reproduce the data can give misleading rank intervals and inclusion frequencies. Predictive checks remain incomplete; six posterior fits have tree-depth saturation, and nine lack stored native-score convergence summaries (Appendix A.6). The different sampling recipes also share modeling assumptions, so agreement between them is not independent validation of an experiment's uncertainty. The studied landscapes are not a representative sample of all scans.

Protein engineering. An engineer choosing a batch can use inclusion frequencies to distinguish candidates that are repeatedly selected from those with plausible replacements. The TrpB example shows that these replacements can involve much of the batch. But turnover alone is not a loss in

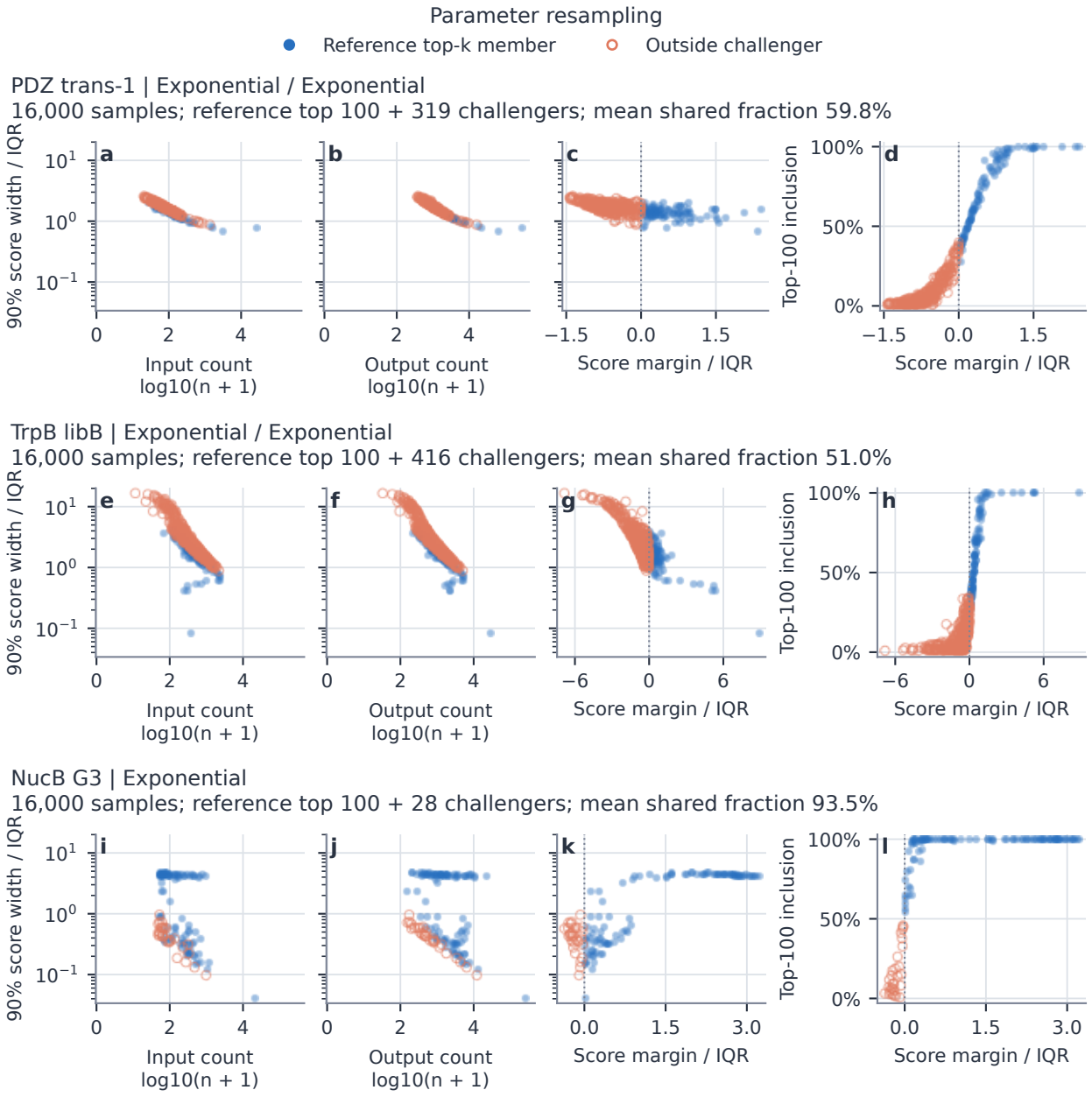


Figure 4: Count support and plausible batch replacements (Section 6). Rows show PDZ trans-1 and TrpB libB under Exponential–Exponential, and NucB G3 under Exponential, each with 16,000 parameter samples. Every reference top-100 member (blue, filled) and outside challenger with inclusion frequency at least 0.01 (orange, open) is retained: 419, 516, and 128 candidates, respectively. From left to right, columns show central 90% score width against input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion frequency against that margin. Widths and margins are divided by the run's top-1,000 median-score interquartile range (IQR). Count axes show $\log_{10}(n+1)$ and width axes are logarithmic. Count and width limits are shared across rows; margin limits are shared within a row but differ between rows. Dotted lines mark zero margin. No points are binned or subsampled, and no regression lines are fitted. Appendix D gives the example and display conventions.

engineering utility: the alternative variants may have similar activity. Comparing their activity or the expected outcome of a specified follow-up experiment is a separate question. The median-score list is a reference for this analysis, not an optimal selection rule for every experimental objective.

Machine-learning benchmarks. Fitness scores serve both as training labels and as references for evaluating predictions. Here we quantify uncertainty in the leading experimental ranks and batches, not its effect on any predictor. For landscapes used in FLIP2, a direct next step is to hold predictions fixed and evaluate them across sampled score sets within each task's candidate subset. This would measure how label uncertainty affects rank correlation, retrieval, and comparisons between predictors.

Further checks. Two kinds of evidence would strengthen the interpretation. First, predictive checks and suitable held-out measurements could test whether the fitted models reproduce the relevant experimental variation. Second, controlled read thinning or leave-one-replicate-out fits could test whether changing count support changes batch stability. Both require analyses of altered or additional observations; subsampling existing model draws only assesses Monte Carlo variability.

The current results concern candidate identity at a specified batch size. For the TrpB example, a top-100 list leaves many plausible replacements; for NucB, much more of that list remains selected. Reporting those selection frequencies alongside fitness scores tells a reader what an exact ordering leaves uncertain.

Data and code availability

PDZ processed counts and scores are available through GEO accession GSE265816. TrpB data and analysis code are deposited at `https://doi.org/10.22002/h5rah-5z170`. NucB data and source calculations are available from `https://github.com/google-deepmind/nuclease_design`. The rank-analysis code consumes prepared sampled-score matrices and aligned count features; fitting the original models is not required to recreate its tables and figures. The accompanying repository includes a small pipeline example, which is distinct from the real-data analyses reported here. [TODO: Add versioned fitting and analysis repositories and a public archive for the prepared samples, features, and fit diagnostics.]

Author contributions

[TODO: Confirm contributions with all authors.]

Competing interests

[TODO: Obtain declarations from all authors.]

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A Supplementary assay and fitting details

The models and fitting procedures use the `flipv3` module of SciStanPy at revision `09c439f675ea`, the version pinned by the companion analysis repository. We first describe the observations and model equations, then report the fitting records and available diagnostics. Source defaults, requested command settings, and values confirmed by saved artifacts are distinguished throughout. TODO entries mark missing records or checks.

A.1 Experimental records and candidate identities

The source studies are the PDZ interaction assays of Zarin et al. (2024), the TrpB active-site libraries of Johnston et al. (2024), and the NucB engineering generations of Thomas et al. (2025). We retain their landscape labels rather than counting fitted models as new experiments. The PDZ inputs are cis, trans-1, trans-2, CRIPT-C, and CRIPT-N. TrpB contains three-site libraries A–I and a four-site library; the labels refer to distinct variant libraries, not different resampling seeds. NucB generations G1–G4 are successive engineering collections with different screening protocols.

Physical assays and reported labels. The PDZ experiment measures binding between PSD95-PDZ3 and CRIPT through protein complementation in yeast. Its source protocol (GSE265816) documents three biological growth competitions for each library. Input populations are grown without the selective drug; the output competition uses methotrexate at 200 μg/mL for approximately five generations. DNA from the input and output cultures is sequenced, and DiMSum produces replicate-combined binding-dependent fitness estimates and standard errors. The deposited protocol also describes normalization between library blocks using stop-codon and synonymous variants, followed by linear scaling between libraries using shared variants. These source score-processing steps are distinct from fitting our count models.

The TrpB libraries use an engineered background derived from the *Thermotoga maritima* enzyme. Bacteria lacking the native *trpA* and *trpB* genes express a library of introduced TrpB sequences. With indole supplied, synthesis of tryptophan supports growth; sequencing observes the resulting changes in library composition (Johnston et al., 2024). The nine three-site libraries together cover twenty positions, while the four-site library varies positions 183, 184, 227, and 228. A stop codon terminates translation prematurely, providing a nonfunctional control for the source growth-score normalization. The count model does not estimate separate catalytic, expression, and stability effects.

The NucB generations are successive rounds of engineering, not four biological replicates of a fixed library. Their input and selected populations have different fluorescence thresholds and sequencing structures (Thomas et al., 2025). The source release contains pre-sort and post-sort counts, enrichment estimates relative to known controls, and activity labels combined across rounds. Our model uses the generation-specific count observations; it does not treat the combined categorical activity labels as count data. Source low-throughput measurements of purified enzymes are separate validation experiments and do not enter the rank calculations here.

[TODO: Complete the residue-level TrpB three-site designs. Specify the constructs and mutated sites in PDZ cis, trans-1, trans-2, CRIPT-C, and CRIPT-N. Verify the NucB fluorescent substrate, compartment protocol, and incubation conditions against the full experimental methods. Add the per-task correspondence to FLIP2 without treating its subsets as additional experiments.]

Count axes. The model input arrays preserve the following observation structures. A repeated sequencing library is distinguished from a new biological population. The last array dimension is

always the fitted variant axis; source controls can belong to this axis without being eligible ranking candidates.

Assay	Input observations	Output observations
PDZ	Three replicate vectors	Three paired vectors
TrpB A–C	One shared vector	One vector per timepoint
TrpB D–I	One shared vector	Two replicates per timepoint
TrpB four-site	Two replicate vectors	Two replicates per timepoint
NucB G1	Five vectors, three populations	Seven selected count vectors
NucB G2	Three vectors, two populations	Three selected count vectors
NucB G3–G4	One vector per generation	Four selected count vectors

Generation 1 sequences the first two input populations twice and the third once; generation 2 sequences the first input twice and the second once.

A.2 PDZ count model and latent fitness

Observation model. Let M denote the number of variants on the fitted count axis, including any rows that are excluded from ranking. This is distinct from the number N of ranking-eligible variants in the main text. For each of the three PDZ replicates $r \in \{1, 2, 3\}$, let $\mathbf{n}_r^{\text{in}} = (n_{1r}^{\text{in}}, \ldots, n_{Mr}^{\text{in}})$ and $\mathbf{n}_r^{\text{out}}$ be its input and output count vectors. Write $D_r^{\text{in}} := \sum_{i=1}^{M} n_{ir}^{\text{in}}$ and $D_r^{\text{out}} := \sum_{i=1}^{M} n_{ir}^{\text{out}}$ for their observed sequencing totals. The model conditions on these totals.

Let $\boldsymbol{\pi}_r^0$ be the vector of input probabilities for replicate r, with nonnegative entries summing to one. A growth function gives positive unnormalized output abundances a_{ir} from the input probabilities and replicate-level growth parameters. Define $\pi_{ir}^1 := a_{ir} / \sum_{j=1}^{M} a_{jr}$. The count model is

$$\mathbf{n}_r^{\text{in}} \mid \theta \sim \text{Multinomial}(D_r^{\text{in}}, \boldsymbol{\pi}_r^0), \tag{3}$$

$$\mathbf{n}_r^{\text{out}} \mid \theta \sim \text{Multinomial}(D_r^{\text{out}}, \boldsymbol{\pi}_r^1). \tag{4}$$

Here θ includes the input probabilities, the growth parameters, and any shared parameters. Conditional on those parameters, the input and output observations contribute separate likelihood terms. Within a library, the multinomial distribution retains dependence between variant counts through their fixed total. The implementation works with log probabilities and normalizes the output abundances on that scale.

The source defines separate input probability vectors for the three PDZ replicates and a hierarchy linking their variant-specific growth rates. Each input vector has a symmetric Dirichlet prior, $\boldsymbol{\pi}_r^0 \sim \text{Dirichlet}(\alpha_0 \mathbf{1}_M)$, where $\mathbf{1}_M$ is a vector of M ones and $\alpha_0 > 0$ is the per-variant concentration. The source class `ExpDirichlet` stores $\log \boldsymbol{\pi}_r^0$; its name denotes a log parameterization, not a different input-count likelihood.

Growth functions. Let b_{ir} be the effective growth parameter for variant i in replicate r, and let $\tau \geq 0$ be elapsed time on the model's scale. PDZ has one post-selection observation, for which $\tau = 1$. For exponential growth the unnormalized abundance is

$$a_{ir}(\tau) := \pi_{ir}^0 \exp(b_{ir}\tau).$$

For sigmoid growth, the implementation instead uses

$$a_{ir}(\tau) := \pi_{ir}^0 \frac{1 + \exp(b_{ir}c)}{1 + \exp[b_{ir}(c - \tau)]}, \qquad c \sim \text{Gamma}(a_c, b_c).$$

Here c is a shared positive offset. We use the shape–rate convention for every Gamma distribution in this supplement, so its mean is shape divided by rate. Both growth functions start at $a_{ir}(0) = \pi^0_{ir}$. Output probabilities normalize $a_{ir}(\tau)$ over variants separately for each observed library. We use these functions as assay models of abundance change, not as mechanistic descriptions of every stage of the biological assay.

Exponential and Lomax rate families. For each variant, let $m_i > 0$ be its central growth parameter and $\eta_i := \log m_i$. The Exponential specification assigns $m_i \sim \text{Exponential}(\beta)$, with rate β and mean $1/\beta$. The Lomax specification assigns a distribution with scale $\lambda > 0$, shape $a_L > 0$, and density

$$p(m_i \mid \lambda, a_L) = \frac{a_L}{\lambda}(1 + m_i/\lambda)^{-(a_L+1)}, \qquad m_i > 0.$$

SciStanPy's `ExpExponential` and `ExpLomax` variables store η_i, including the appropriate transformed density. They do not mean that η_i itself has an ordinary Exponential or Lomax distribution.

There are two available replicate models. The normal option uses $b_{ir} \mid m_i \sim \text{Normal}(m_i, \sigma_b^2)$. Its central parameter is the conditional mean rate, and the replicate rates may be negative. The lognormal option uses $\log b_{ir} \mid \eta_i \sim \text{Normal}(\eta_i, \sigma_b^2)$. Here m_i is the conditional median rate; the conditional mean is $m_i \exp(\sigma_b^2/2)$. Since σ_b is shared, both induce the same ordering of these central rates. The hierarchy can be represented using standard-normal latent variables or using the replicate variables directly. The command flag `-centered` selects the latter representation where the model exposes that option. The command records, rather than the default alone, determine whether a run uses normal or lognormal replicate variation.

Gamma inverse-rate family. The Gamma specification assigns an inverse-rate parameter $\kappa_i \sim \text{Gamma}(a_\kappa, b_\kappa)$ and $b_{ir} \mid \kappa_i \sim \text{Exponential}(\kappa_i)$. The conditional mean growth rate is $m_i := 1/\kappa_i$. The stored parameter is `inv_r_mean`, and the score extractor applies $h_i(\theta) := -\log \kappa_i$. For the Exponential and Lomax hierarchies, the stored variable is `log_r_mean` and $h_i(\theta) := \eta_i$. Thus the score labeled `log_mean_growth` orders the central growth parameter; in the lognormal hierarchy it is not literally the logarithm of the arithmetic mean replicate rate.

Hyperparameters. The source defaults are $\alpha_0 = 0.1$, $\beta = 10$, $\lambda = 1$, $a_L = 2$, $\sigma_b = 0.1$, $(a_\kappa, b_\kappa) = (2, 0.5)$, and $(a_c, b_c) = (4, 8)$. The job commands can override them. In particular, the recorded PDZ commands set $\alpha_0 = 5$ and select normal replicate variation for the Exponential and Lomax hierarchies. We retain per-run command settings in the fit inventory rather than replacing overrides with the class defaults.

A.3 TrpB count model and latent fitness

Time and observation structure. TrpB uses the same growth functions and multinomial observation model as PDZ, but has multiple post-selection timepoints. Let t_ℓ be an observed time, indexed by ℓ, and define $\tau_\ell := t_\ell / \max_{\ell'} t_{\ell'}$. Effective growth rates are therefore expressed per normalized assay duration, not per hour. The model factory removes optical-density columns before model construction; the likelihood uses sequencing counts and normalized times.

Libraries A–C have one input count vector and one output vector per timepoint. They use a single input probability vector $\boldsymbol{\pi}^0$ and one rate b_i per variant across the observed times. The prior is Exponential or Lomax on b_i, with the definitions given above; the flat Gamma alternative is not implemented. Their latent score is $h_i(\theta) := \log b_i$, obtained from the stored positive variable `r`.

Libraries D–I have one shared input count vector and replicate-specific output vectors at each retained time. They use one $\boldsymbol{\pi}^0$ across output replicates and hierarchical b_{ir} rates, with the Exponential, Lomax, or Gamma inverse-rate construction above. For each time and replicate, the output vector has a multinomial likelihood with its own observed total and normalized grown probabilities. The single input observation contributes one likelihood term; it is not repeated once per output replicate. Their score is η_i or $-\log \kappa_i$, as for PDZ.

The four-site model has two input count vectors and replicate-specific outputs. Both input vectors share one latent input probability vector, while the output rates use the same hierarchical families. The prepared paper collection contains optimized-fit count resampling, but no parameter-posterior or posterior-count analysis, for this landscape.

Model and scoring timepoints. The model uses the times supplied by the data loader; the count-derived fitness calculation can use a different retained subset. In particular, the four-site loader supplies 12, 16, 20, 24, 28, and 36 hours, while the deposited count-score reconstruction excludes the 12-hour observation. Appendix B distinguishes those score rules from the count likelihood. The input Dirichlet concentration and other hyperparameters have the defaults above, with overrides and centered/noncentered choices listed for each job.

A.4 NucB gate model and latent fitness

Latent fluorescence. For variant i, let L_i be a log-fluorescence location and $v_i := \exp(L_i) > 0$ its fluorescence-scale center. The Exponential family assigns $v_i \sim \text{Exponential}(\beta)$; the Lomax family assigns the scale–shape distribution defined above. Both use a log parameterization internally. The paper's parameter score is $h_i(\theta) := L_i$, not a count enrichment and not raw measured fluorescence. For generation 1 the stored variable is `absolute_mean_log_fluorescence`; for generations 2–4 it is `log_fluorescence`. The extractor retains those values without exponentiating them. The data-product label `model_fluorescence` therefore denotes a log-scale parameter in these analyses.

Gate probabilities. Let $z_g > 0$ be the fluorescence threshold for gate g, and $\sigma_F > 0$ a shared spread parameter. For a lognormal observation model, the probability of exceeding the threshold for a variant with location L is

$$s_g(L, \sigma_F) := 1 - \Phi\left(\frac{\log z_g - L}{\sigma_F}\right),$$

where Φ is the standard-normal cumulative distribution function. Generations 3 and 4 also expose a normal option, $s_g(L, \sigma_F) := 1 - \Phi[(z_g - \exp L)/\sigma_F]$. The recorded generation-3 command uses that normal option; generations 1 and 2 use the lognormal survival model, and the recorded generation-4 command leaves the lognormal default in place.

For gate g arising from input population r, the variant probabilities are

$$\pi_{ig}^{\text{gate}} := \frac{\pi_{ir}^0 s_g(L_{ir}, \sigma_F)}{\sum_{j=1}^{M} \pi_{jr}^0 s_g(L_{jr}, \sigma_F)}.$$

Here L_{ir} is the population-specific location when the model includes one, and otherwise equals L_i. Each observed gate count vector is multinomial with these probabilities and its observed sequencing total. The gate likelihoods use survival probabilities above thresholds, not a partition of all cells into mutually exclusive fluorescence bins. Repeated sequencing of an input or selected population shares its underlying probability vector but contributes its own count likelihood and total.

Generations. Generation 1 has three input-population vectors, $\boldsymbol{\pi}_r^0 \sim \text{Dirichlet}(\alpha_0 \mathbf{1}_M)$, and population-specific log locations $L_{ir} \mid L_i, \sigma_E \sim \text{Normal}(L_i, \sigma_E^2)$. The experimental spread satisfies $\sigma_E \sim \text{HalfNormal}(s_E)$, meaning the absolute value of a mean-zero normal draw with standard deviation s_E. Low- and high-threshold selected counts use the corresponding location and input population. Generation 2 has two input vectors and a shared L_i across gates; the 86 and 97.5 gates use the first input population, and the 93 gate uses the second. Generations 3 and 4 each have one input vector and four thresholded output count vectors, all sharing L_i.

All generations assign $\sigma_F \sim \text{Gamma}(a_F, b_F)$. The source names this parameter `codon_noise`; that name is a modeling interpretation, not evidence that the observed spread can be attributed only to codon usage. Defaults are $(a_F, b_F) = (2, 2)$ and $s_E = 0.5$. The command records use $\alpha_0 = 5$ for NucB and request Exponential rates $10, 12.5, 15, 17.5$ in generations 1–4, respectively, through the flag `exp_beta`. However, the inspected factory forwards that keyword while the Exponential mixin accepts `beta`; the unmatched keyword is absorbed without overriding the default. Under this source revision the effective rate is therefore 10, unless `beta` is passed through another entry point. We record the requested setting and this source-level behavior separately. The Lomax shape keyword matches its constructor argument; the recorded shapes are $2, 4, 6, 8$, respectively, at scale one. Only specifications present in the curated fit inventory contribute to the paper's results.

Threshold values. The immutable data loader supplies the following thresholds on its fluorescence scale. These are assay calibration constants, not fitted rank cutoffs. For generation 1, the three pairs correspond to the three input populations. For generations 2–4, values follow the gate order used in the count arrays.

Generation	Gate labels	Threshold values
G1	Low, by population	0.100, 0.115, 0.100
G1	High, by population	0.501, 0.600, 0.370
G2	86, 93, 97.5	0.201, 0.318, 0.385
G3	59, 80, 95, 99	0.118, 0.217, 0.285, 0.346
G4	70, 90, 98, 99.5	0.124, 0.199, 0.306, 0.384

Controls and interpretation. Appended negative-control rows enter the multinomial normalization and fitted axis but not the candidate ranking. Gate thresholds determine the connection between the latent location and the assay; none of these parameters is a universal biochemical fitness measure. The highest-gate enrichment score used for the count-resampling analyses is defined separately in Appendix B. It is not identical to L_i, even though both are oriented so that higher scores correspond to higher inferred assay activity.

A.5 Optimization and posterior simulation

Posterior simulation. The `mcmc_flip.py` entry point constructs the selected model, generates and compiles its Stan program, and runs Hamiltonian Monte Carlo through CmdStan. It then converts samples and generated quantities to NetCDF and runs the diagnostic routines. The command-line defaults are eight chains, 2,000 warmup iterations and 2,000 retained iterations per chain, with seed 1025. The job command list overrides warmup and retained iterations for some models. The saved artifact attributes, rather than the defaults, are the source for actual chain counts, iteration counts, and Stan versions in the fit inventory. The representative artifacts inspected here record Stan 2.38.0 and maximum tree depth 10. Absence of an explicit adaptation option in the launcher is not a record of its effective per-fit value.

Optimized fit. The source calls the second entry point `mle_flip.py`. Its PyTorch objective sums log densities of both observed nodes and latent parameter nodes. In the parameterization evaluated by that backend, it maximizes a joint-density objective of the form

$$\hat{\theta} := \arg\max_{\theta} \left\{ \sum_{o \in \mathcal{O}} \log p(d_o \mid \mathrm{pa}(o)) + \sum_{v \in \mathcal{V}} \log p(\theta_v \mid \mathrm{pa}(v)) \right\}.$$

Here $\mathcal{O}$ indexes observed count nodes with values d_o, $\mathcal{V}$ indexes latent stochastic nodes, and $\mathrm{pa}(\cdot)$ denotes a node's parents in the model. Thus the fitted mode includes priors and hierarchical densities; it is not the maximizer of the count likelihood alone. Density modes also depend on the chosen parameterization. We use *optimized-fit count resampling* in the prose and preserve `mle_predictive` as the existing data identifier. This correction changes the description, not the stored fits or results.

Optimization uses Adam, with source defaults of learning rate 0.001, at most 100,000 full-data steps, and stopping after 10 steps without an improvement in the recorded objective. The routine retains a saved best state when early stopping is triggered and writes the loss trajectory. The launcher passes seed 1025 unless overridden. These are reproducible source settings; the inventory flags where a historical artifact or loss record is unavailable to confirm an individual execution. Optimization convergence requires the loss and restart records listed below.

Resampled counts. Posterior-count samples are generated using sampled model parameters and the corresponding fixed library totals. The optimized-fit workflow exports the fitted distributions and produces count realizations using its sampling routine without reoptimizing on each realization. Appendix B then maps those counts to the same family-specific reconstruction functions used throughout the rank analysis. Preserving one joint draw across all variant and observation axes is necessary for coherent rank uncertainty.

A.6 Convergence and predictive checks

Available sampler evidence. All 49 posterior artifacts were accessible for this inventory. Their recorded chain count, warmup, retained iterations, and seed agree with the corresponding job commands. All have eight chains, seed 1025, Stan 2.38.0, and a recorded maximum tree depth of 10. None records a divergent transition. Six fits record tree-depth saturation: TrpB jobs 20, 25, 26, 38, 43, and 44 have 4,000, 16,000, 16,000, 1,692, 12,454, and 14,000 saturated transitions, respectively. These are warnings about trajectory length and computational efficiency; zero divergences does not remove the need to examine them.

Forty artifacts include convergence and effective sample size summaries for the stored score parameter. Across their fitted variant axes, the largest finite recorded convergence statistic $\hat{R}$ is approximately 1.0067, the smallest bulk effective sample size is 1,876.9, and the smallest tail effective sample size is 889.2. These diagnostics apply to the native stored score parameter, including fitted controls when present. They are not effective sample sizes for ranks or top-k membership. The nine raw-artifact fallbacks lack these stored summaries and are shown as unavailable. The code's variable-level diagnostic defaults flag $\hat{R} \geq 1.01$ and effective sample sizes at most 100 per chain. We report the recorded values rather than assigning one pass/fail label to the whole analysis.

Energy diagnostic. For each saved chain with energy sequence $E_1, \ldots, E_T$, we compute the ratio

$$B := \frac{(T-1)^{-1} \sum_{t=2}^{T} (E_t - E_{t-1})^2}{T^{-1} \sum_{t=1}^{T} (E_t - \overline{E})^2}, \qquad \overline{E} := T^{-1} \sum_{t=1}^{T} E_t.$$

This is the energy-based fraction of missing information (BFMI) estimate using the displayed variance convention. The minimum over all inspected chains is 0.637. We calculate this from energy traces because the inspected source's stored flag named `low_ebfmi` compares individual energy values with a threshold rather than calculating this chain-level ratio. The inventory does not interpret that flag as a valid BFMI test.

What these checks do not establish. The inventory records diagnostic availability and values for the retained fits without applying new exclusions. Sampler diagnostics assess simulation from the fitted distribution, not whether that distribution describes the experiment. Predictive checks remain necessary, particularly for specifications that produce very different rank uncertainty from the same counts.

A.6.1 Reserved predictive-check exhibits

[TODO: Add per-fit observed-versus-predictive checks for count support, zero counts, replicate or timepoint variation, and gate enrichment. Define each discrepancy statistic and show the relevant biological examples. This subsection is intentionally open; the sampler inventory is not a substitute for these checks.]

A.6.2 Reserved exception and optimization checks

[TODO: Recompute missing native-score diagnostics for raw fallbacks; quantify rank- and inclusion-specific Monte Carlo precision; review tree-depth saturation and the strongly specification-sensitive PDZ fits; recover the optimization loss curves and any restart evidence for all optimized fits. Document the original retention decisions separately from any new sensitivity analysis. No new fits or filters have been applied in this draft.]

A.7 Score reconstruction and missing-data audit

The TrpB and NucB reconstructions reproduce the source finite scores and coverage at the numerical tolerances stated in Appendix B. The fit mapping records retained samples and exclusions. The PDZ reconstruction is an approximation because the exact original DiMSum configuration has not been recovered. [TODO: Add the per-assay reconstruction comparisons, original DiMSum revision and configuration if recovered, finite-score support distributions, and sensitivity of reported-rank summaries to missingness and ties.]

A.8 Reproducibility and complete supplementary exhibits

The two tables below enumerate all 49 posterior fits and all 60 optimized fits used by the 158 analysis runs. Each posterior fit supplies both a parameter and a posterior-count view. The separate machine-readable inventory records each analysis identifier, its fitted artifact, its source job command, all parsed options, retained samples and exclusions, and available native-score diagnostics. Source hashes identify the model and launcher files used for this description. Historical job commands are records of requested settings; saved artifact attributes confirm the reported sampling counts, but do not embed every prior option or the complete software environment.

Model codes combine the rate-family initial (E: Exponential, L: Lomax, G: Gamma inverse-rate) and growth-function initial (E: exponential, S: sigmoid). NucB has only the fluorescence-family code E or L. A source artifact may have been retired after its prepared score matrix was validated; this is distinct from a missing paper analysis. Three original optimized artifacts remain locally available, while all 60 prepared optimized-count score matrices are available for the rank analysis.

Table 3: All posterior fits used by the parameter and posterior-count analyses. Iterations are per chain from artifact metadata. D denotes a diagnosed artifact; R a raw artifact. The maximum stored native-score R-hat uses all fitted variants, not only the decision cohort; – means unavailable. Depth counts samples reaching the recorded maximum tree depth. BFMI is recomputed from energy traces.

Landscape	Model	Job	Chains	Warmup	Draws	File	Diverg.	Depth	max $\hat{R}$	BFMI
nuc-G1	E	55	8	4000	4000	D	0	0	1.0026	0.708
nuc-G1	L	56	8	2000	4000	R	0	0	–	0.739
nuc-G2	E	57	8	2000	2000	D	0	0	1.0054	0.883
nuc-G2	L	58	8	4000	3000	D	0	0	1.0037	0.901
nuc-G3	E	59	8	2000	2000	D	0	0	1.0067	0.895
nuc-G4	E	61	8	2000	2000	D	0	0	1.0028	0.885
nuc-G4	L	62	8	2000	2000	D	0	0	1.0046	0.89
pdz-cis	EE	1	8	6000	2000	D	0	0	1.0024	0.939
pdz-cis	ES	4	8	6000	2000	D	0	0	1.0027	0.937
pdz-cis	LE	3	8	6000	2000	D	0	0	1.0028	0.893
pdz-cript-c	EE	70	8	2000	2000	D	0	0	1.0040	0.949
pdz-cript-c	ES	73	8	2000	3000	D	0	0	1.0026	0.924
pdz-cript-c	GE	72	8	2000	3000	R	0	0	–	0.847
pdz-cript-c	LE	71	8	2000	3000	R	0	0	–	0.908
pdz-cript-n	GE	78	8	4000	3000	R	0	0	–	0.824
pdz-cript-n	GS	81	8	2000	2000	D	0	0	1.0027	0.86
pdz-cript-n	LS	80	8	2000	2000	D	0	0	1.0039	0.899
pdz-trans-1	EE	82	8	2000	2000	R	0	0	–	0.926
pdz-trans-1	GE	84	8	2000	2000	D	0	0	1.0035	0.852
pdz-trans-2	EE	88	8	2000	2000	D	0	0	1.0035	0.929
pdz-trans-2	GS	93	8	2000	2000	D	0	0	1.0029	0.867
trpb-libA	EE	7	8	2000	2000	D	0	0	1.0027	0.787
trpb-libA	ES	9	8	2000	2000	D	0	0	1.0027	0.821
trpb-libA	LE	8	8	2000	2000	D	0	0	1.0029	0.644
trpb-libB	EE	11	8	2000	2000	D	0	0	1.0023	0.811
trpb-libB	ES	13	8	2000	2000	D	0	0	1.0028	0.758
trpb-libB	LE	12	8	2000	2000	D	0	0	1.0026	0.673
trpb-libC	EE	15	8	2000	2000	R	0	0	–	0.775
trpb-libC	ES	17	8	2000	2000	D	0	0	1.0024	0.805
trpb-libC	LE	16	8	2000	2000	D	0	0	1.0027	0.637
trpb-libD	EE	19	8	2000	2000	D	0	0	1.0024	0.868
trpb-libD	GE	21	8	2000	2000	D	0	0	1.0022	0.708
trpb-libD	LE	20	8	2000	2000	D	0	4000	1.0041	0.815
trpb-libE	EE	25	8	6000	2000	R	0	16000	–	0.812
trpb-libE	GE	27	8	6000	2000	D	0	0	1.0025	0.72
trpb-libE	LE	26	8	8000	2000	D	0	16000	1.0025	0.802
trpb-libF	EE	31	8	2000	2000	D	0	0	1.0020	0.908
trpb-libF	ES	34	8	2000	2000	D	0	0	1.0022	0.846
trpb-libF	GE	33	8	2000	2000	D	0	0	1.0018	0.701
trpb-libF	LE	32	8	2000	2000	D	0	0	1.0026	0.808
trpb-libG	EE	37	8	6000	2000	R	0	0	–	0.852
trpb-libG	GE	39	8	4000	2000	D	0	0	1.0025	0.717
trpb-libG	LE	38	8	4000	2000	D	0	1692	1.0028	0.775
trpb-libH	EE	43	8	2000	2000	R	0	12454	–	0.88
trpb-libH	GE	45	8	4000	2000	D	0	0	1.0023	0.71
trpb-libH	LE	44	8	2000	2000	D	0	14000	1.0028	0.8
trpb-libI	EE	49	8	2000	2000	D	0	0	1.0025	0.885
trpb-libI	GE	51	8	2000	2000	D	0	0	1.0025	0.767
trpb-libI	LE	50	8	2000	2000	D	0	0	1.0027	0.84

Table 4: All optimized fits used by the count-resampling analyses. Concentration, observation-family option, and centered (C) or noncentered (NC) flag come from the recorded job commands; flags may not affect every model class. Available indicates whether the original fitted artifact remains locally accessible, not whether its prepared scores are available. Model codes are defined in the text.

Landscape	Model	Job	α_0	Variation option	Flag	Available
nuc-G1	E	55	5	lognormal	C	no
nuc-G1	L	56	5	lognormal	C	no
nuc-G2	E	57	5	lognormal	C	no
nuc-G2	L	58	5	lognormal	C	no
nuc-G3	E	59	5	normal	C	yes
nuc-G4	E	61	5	lognormal	C	no
nuc-G4	L	62	5	lognormal	C	no
pdz-cis	EE	1	5	normal	NC	no
pdz-cis	ES	4	5	normal	NC	no
pdz-cis	LE	3	5	normal	NC	yes
pdz-cript-c	EE	70	5	normal	NC	no
pdz-cript-c	ES	73	5	normal	NC	no
pdz-cript-c	GE	72	5	lognormal	NC	no
pdz-cript-c	LE	71	5	normal	NC	no
pdz-cript-c	LS	74	5	normal	NC	no
pdz-cript-n	EE	76	5	normal	NC	no
pdz-cript-n	GE	78	5	lognormal	NC	no
pdz-cript-n	GS	81	5	lognormal	NC	no
pdz-cript-n	LE	77	5	normal	NC	no
pdz-cript-n	LS	80	5	normal	NC	no
pdz-trans-1	EE	82	5	normal	NC	no
pdz-trans-1	GE	84	5	lognormal	NC	no
pdz-trans-2	EE	88	5	normal	NC	no
pdz-trans-2	GE	90	5	lognormal	NC	no
pdz-trans-2	GS	93	5	lognormal	NC	no
pdz-trans-2	LS	92	5	normal	NC	no
trpb-four-site	EE	64	5	normal	NC	yes
trpb-four-site	ES	67	5	normal	NC	no
trpb-four-site	GE	66	5	lognormal	NC	no
trpb-four-site	GS	69	5	lognormal	NC	no
trpb-four-site	LE	65	5	normal	NC	no
trpb-four-site	LS	68	5	normal	NC	no
trpb-libA	EE	7	0.1	lognormal	NC	no
trpb-libA	ES	9	0.1	lognormal	NC	no
trpb-libA	LE	8	0.1	lognormal	NC	no
trpb-libB	EE	11	0.1	lognormal	NC	no
trpb-libB	ES	13	0.1	lognormal	NC	no
trpb-libB	LE	12	0.1	lognormal	NC	no
trpb-libC	EE	15	0.1	lognormal	NC	no
trpb-libC	ES	17	0.1	lognormal	NC	no
trpb-libC	LE	16	0.1	lognormal	NC	no
trpb-libD	EE	19	0.1	normal	NC	no
trpb-libD	GE	21	0.1	lognormal	NC	no
trpb-libD	LE	20	0.1	normal	NC	no
trpb-libE	EE	25	0.1	lognormal	C	no
trpb-libE	GE	27	0.1	lognormal	NC	no
trpb-libE	LE	26	0.1	normal	NC	no
trpb-libF	EE	31	0.1	normal	NC	no
trpb-libF	ES	34	0.1	normal	NC	no
trpb-libF	GE	33	0.1	lognormal	NC	no
trpb-libF	LE	32	0.1	normal	NC	no
trpb-libG	EE	37	0.1	normal	NC	no
trpb-libG	GE	39	0.1	lognormal	NC	no
trpb-libG	LE	38	0.1	normal	NC	no
trpb-libH	EE	43	0.1	normal	NC	no
trpb-libH	GE	45	0.1	lognormal	NC	no
trpb-libH	LE	44	0.1	normal	NC	no
trpb-libI	EE	49	0.1	normal	NC	no
trpb-libI	GE	51	0.1	lognormal	NC	no

Landscape	Model	Job	α_0	Variation option	Flag	Available
trpb-libI	LE	50	0.1	normal	NC	no

[TODO: Supply versioned public archives and complete environment locks, and assemble the remaining fit-specific predictive and sensitivity exhibits. Full-landscape uncertainty estimation, read-depth interventions, and FLIP2 predictor evaluation remain outside this draft.]

B Count-based fitness calculations

PDZ. Let n_{ir}^{in} and n_{ir}^{out} denote counts for variant i in replicate r, and let w index the wild-type reference. Let $\mathcal{R}_i$ be the set of replicates for which all four counts for i and w are positive. For $r \in \mathcal{R}_i$, define the wild-type-relative log enrichment f_{ir} and its working variance v_{ir} by

$$f_{ir} := \log \frac{n_{ir}^{\text{out}}}{n_{ir}^{\text{in}}} - \log \frac{n_{wr}^{\text{out}}}{n_{wr}^{\text{in}}}, \tag{5}$$

$$v_{ir} := \frac{1}{n_{ir}^{\text{in}}} + \frac{1}{n_{ir}^{\text{out}}} + \frac{1}{n_{wr}^{\text{in}}} + \frac{1}{n_{wr}^{\text{out}}}, \qquad g_i(\mathcal{D}) := \frac{\sum_{r\in\mathcal{R}_i} f_{ir}/v_{ir}}{\sum_{r\in\mathcal{R}_i} 1/v_{ir}}. \tag{6}$$

Only valid replicates enter the sums. We define the score as undefined when $\mathcal{R}_i$ is empty, so undefined logarithms never enter the weighted average. This inverse-Poisson-variance calculation is a DiMSum-style approximation. The exact source DiMSum revision and configuration have not been recovered, so we do not describe it as an exact reproduction of the published score (Faure et al., 2020; Zarin et al., 2024).

TrpB. Following the deposited calculations of Johnston et al. (2024), we form log output/input ratios separately for each retained timepoint and replicate. We subtract the same-observation mean for stop-codon variants and divide by the difference between the same-observation maximum and that stop-variant mean. The final score averages valid normalized observations. The three-site libraries use one input count per variant; the four-site landscape uses the mean of two input-replicate counts. Inputs must be at least 10 and required outputs positive; if an observation has two output replicates, both must be positive. Library-specific timepoints follow the deposited notebooks; the four-site calculation excludes 12 hours and retains 16, 20, 24, 28, and 36 hours. Culture-density and time factors cancel from the within-observation stop/maximum normalization. The source count calculations reproduce the published finite scores and coverage at numerical tolerance 10^{-12}.

NucB. For the highest selected fluorescence gate, we calculate

$$g_i(\mathcal{D}) := \frac{n_i^{\text{gate}}/D^{\text{gate}}}{n_i^{\text{in}}/D^{\text{in}}},$$

where D^{gate} and D^{in} are the source normalization totals, including controls when specified by the source. The selected gates are generation-specific (Thomas et al., 2025). A zero gate count gives zero enrichment; a zero input count makes the score undefined. Source controls contribute to the appropriate totals but not to ranking candidates. Generation 1 retains the source total correction for sequences omitted from the fitted axis; this common multiplicative correction does not alter ranks. The observed count calculations reproduce the source enrichment values at tolerance 10^{-12}.

C Ranking and computational conventions

Finite samples and reference support. We store sampled scores as single-precision arrays and retain samples with at least 1,000 finite candidate scores. Table 1 gives the resulting sample-count ranges. Rank intervals and reported-rank percentiles use the finite observations for each candidate; undefined scores do not become zero scores. Inclusion counts use membership in the retained size-k sets, with the retained sample count as denominator. Count-based median-score references require at least two finite scores per candidate; the parameter collections require at least one. The stored sample identities preserve alignment across parameters and count draws. Source controls and other excluded candidates remain outside the ranking universe.

Ranks and selections. Exact tied scores receive average ranks for interval summaries. Fixed-size top-k selection uses the deterministic selection order, with source block order retained where needed to reproduce the prepared results. Because the two conventions answer different questions at a tie, we calculate set membership from stored selections rather than from average rank thresholds. The median-score reference orders median scores and resolves its exact ties deterministically. Rankings always use the eligible full competitor pool; the reported cohorts and displayed summaries remain top-focused.

Exact shared fractions. Let $c_i(k) := \sum_s \mathbf{1}\{i \in T_k^{(s)}\}$ be the number of retained sets containing variant i. Then the sample-to-sample overlap above is exactly

$$\widehat{C}_k = \frac{\sum_i c_i(k)[c_i(k)-1]}{S(S-1)k}.$$

We use this identity instead of drawing random pairs of samples. It is an empirical summary of distinct retained samples; dependence among Monte Carlo draws is not removed by the identity. We likewise obtain reference overlap by averaging inclusion frequencies over each fixed reference set.

Reported-rank percentiles. Let r_i^{rep} be candidate i's reported rank and $\mathcal{S}_i$ the indices of its finite sampled ranks. The fraction plotted in Fig. 1 is

$$u_i := \frac{1}{|\mathcal{S}_i|} \sum_{s \in \mathcal{S}_i} \left[\mathbf{1}\{R_i^{(s)} < r_i^{\mathrm{rep}}\} + \tfrac{1}{2}\mathbf{1}\{R_i^{(s)} = r_i^{\mathrm{rep}}\} \right].$$

We plot its empirical cumulative distribution over the reported top 100 or top 1,000, omitting candidates with no finite sampled rank. Exact rank recovery gives $u_i = 0.5$ for every candidate, so a uniform diagonal is not a calibration target. These percentiles and the reported-versus-median-rank correlations describe agreement with a fixed reference, not errors against a known biological ranking.

Implementation checks. The repository tests compare streamed rankings with dense calculations, exercise ties and candidate exclusions, and check table and figure outputs. These are checks of the analysis implementation, not a demonstration of frequentist coverage or biological model adequacy. A controlled synthetic coverage study is not part of the present draft.

D Descriptive association analyses

The unaggregated relationships behind this section are shown for every landscape, model, and resampling scheme in Appendix F. Its scatterplots contain no fitted regression lines. The regression coefficients summarized here are separate descriptive calculations; the point displays let the reader assess patterns that a single coefficient or an across-landscape median can obscure.

Recall that U_i is normalized score width and $\widetilde{r}_i$ is the median-score reference rank. Local score spacing is the average change in median score per rank across up to five neighbors on each side of a reference rank, divided by S_{run}. Endpoints use the available neighbors. We define crowding as minus the base-10 logarithm of this spacing, using a small numerical floor for zero spacings. Larger values indicate closer competitors on this scale.

Figure 4 examples and plotted data. Figure 4 uses the parameter-resampling versions of the same three examples as Figs. 1–3: PDZ trans-1 and TrpB library B under Exponential–Exponential, and NucB generation 3 under Exponential. No Gamma specification enters this figure. The examples illustrate different candidate-level patterns; they are not a representative sample of model specifications. The complete per-analysis displays, including Gamma fits, remain in Appendix F.

Each row includes every member of the median-score reference top 100 and every outside candidate with $\hat{q}_i(100) \geq 0.01$. The three rows contain 1,063 candidates in total. Every candidate appears in all four panels of its row, without binning, point subsampling, or pooling across analyses. Each point summarizes one candidate's score samples; it is not an individual sample. The stored input counts, output counts, score medians, interval endpoints, and inclusion frequencies are unchanged.

Counts are transformed as $\log_{10}(n+1)$, retaining zero counts at zero. Normalized widths use a logarithmic axis. All widths in these three examples are positive, so no plot-only floor affects the displayed values. Count and width limits cover all points and are common to the three rows. The two margin panels use the same limits within a row, but the limits differ between rows; apparent transition steepness should therefore not be compared without reading the axes. Inclusion frequencies use a common zero-to-one scale.

The exported variant table retains every plotted row and its coordinates, and the run index records example identities, cohort sizes, sample counts, and mean shared fractions. Inclusion frequencies and shared fractions come from the joint sampled rankings against all finite eligible competitors, not from a candidate's marginal interval or a ranking restricted to the displayed cohort. Because challengers are selected using their inclusion frequencies, relationships in this cohort do not describe all variants in the landscape.

Observed count features. The features are fixed from the observed data and shared across resampling schemes for a landscape. PDZ input and output support sum the three paired replicates separately. TrpB input support uses the single input count once for three-site libraries and the sum of the two input counts for four-site; output support sums every retained output replicate and model timepoint. NucB input support sums the arrays `ic1`, `ic2`, and `ic3` where present, over all their observation axes. Output support sums the low- and high-gate arrays for G1, the three gate arrays for G2, and all four output rows for G3 and G4. Resequenced libraries contribute their recorded reads to these descriptive totals. None of these sums turns sequencing repeats, different timepoints, or different gates into independent biological replicates. Log input support means $\log(1+n_i^{\mathrm{in}})$; the count scatterplots use $\log_{10}(1+n)$ for display.

Observation discordance. For PDZ paired replicates, let e_{ir} be the difference of smoothed log output and input frequencies for variant i and replicate r. Specifically,

$$e_{ir} := \log \frac{n_{ir}^{\text{out}} + \delta}{D_r^{\text{out}} + M\delta} - \log \frac{n_{ir}^{\text{in}} + \delta}{D_r^{\text{in}} + M\delta}, \qquad v_{ir}^{\text{obs}} := \frac{1}{n_{ir}^{\text{in}} + \delta} + \frac{1}{n_{ir}^{\text{out}} + \delta},$$

where $\delta = 0.5$ and D_r are totals on the feature-construction count axis of length M. Set $w_{ir} := 1/v_{ir}^{\text{obs}}$ and $\bar{e}_i := \sum_r w_{ir} e_{ir} / \sum_r w_{ir}$. The reduced heterogeneity is

$$H_i := \frac{1}{R-1} \sum_{r=1}^{R} w_{ir}(e_{ir} - \bar{e}_i)^2,$$

where $R = 3$ for PDZ. It measures replicate scatter relative to a working Poisson log-count variance. The smoothing is part of this diagnostic feature; it does not change the no-pseudocount fitness reconstructions in Appendix B.

For TrpB observations with two output replicates, we instead use smoothed log output frequencies and variance $1/(n_{i\ell r}^{\text{out}} + \delta)$, remove a separate weighted mean within each timepoint ℓ, and sum the weighted squared residuals over timepoints. With J timepoints and $R = 2$ replicates, the denominator is $J(R-1)$. This removes between-timepoint changes before measuring within-timepoint disagreement. TrpB A–C have no paired output contrast; NucB has no verified comparable feature in the current analysis. Their discordance value is missing, not zero. The regression includes this predictor only when more than three decision-cohort values are available, then uses complete cases.

Within the reference top-100-plus-challengers cohort, the score-precision regression uses $\log(1 + U_i)$ as response and log input support and $\log(1 + \widetilde{r}_i)$ as predictors, adding the count-aware observation discordance feature where available. The rank regression uses $\log[1 + Q_{0.95}(R_i) - Q_{0.05}(R_i)]$ as response and U_i, crowding, their product, and $\log(1 + \widetilde{r}_i)$ as predictors. We fit linear regressions after standardizing the response and each predictor on complete cases. Coefficients are summarized within landscapes by their median across models. Their signs describe these fitted associations; the interaction means that a main-effect coefficient is not a universal effect at every value of the other predictor.

The score-uncertainty coefficient is positive in 17 of 18, 18 of 19, and 17 of 18 landscape summaries for parameter, optimized-fit count, and posterior count resampling, respectively. The crowding coefficient is positive in 15 of 18, 17 of 19, and 14 of 18. The interaction is positive in 3 of 18, 7 of 19, and 6 of 18. These are counts of fitted coefficient signs, not uniformly positive conditional slopes. The predictors and rank widths summarize the same samples, so the regressions are descriptive rather than independent tests of an effect of additional sequencing.

E Supporting figure views

Table 5: Mean shared top-k fractions (percent), summarized by the median over model specifications within each landscape. P, O, and PC denote parameter, optimized-fit count, and posterior-count resampling; the suffix denotes k. Missing entries indicate no analyzed posterior fit for that landscape. This is an endpoint comparison, not a claim of monotonicity over every intermediate k.

Landscape	P100	P1000	O100	O1000	PC100	PC1000
nuc-G1	68.0	85.3	88.7	97.3	91.9	96.4
nuc-G2	91.7	97.4	89.3	97.2	85.7	96.1
nuc-G3	93.5	97.8	87.0	97.5	87.8	96.5
nuc-G4	94.7	98.4	60.3	97.5	81.6	97.0
pdz-cis	82.0	91.9	92.2	96.2	88.5	94.5
pdz-cript-c	91.8	96.0	93.8	95.4	91.4	94.4
pdz-cript-n	0.6	4.9	73.4	83.9	36.9	64.0
pdz-trans-1	30.0	36.8	67.9	74.1	46.4	60.9
pdz-trans-2	23.0	32.8	49.6	67.5	31.6	51.0
trpb-four-site	–	–	87.6	95.1	–	–
trpb-libA	51.8	63.1	28.4	41.7	24.0	36.3
trpb-libB	50.9	76.0	32.2	59.7	25.9	51.9
trpb-libC	52.2	73.3	32.5	55.2	27.7	48.8
trpb-libD	98.0	78.8	97.6	76.6	97.4	71.8
trpb-libE	66.3	49.5	60.3	46.2	54.6	39.0
trpb-libF	82.3	65.4	72.6	57.6	69.4	52.5
trpb-libG	82.3	59.9	81.1	60.3	77.7	53.4
trpb-libH	80.3	41.1	74.0	39.9	69.0	35.6
trpb-libI	64.4	84.7	74.5	83.6	70.5	80.8

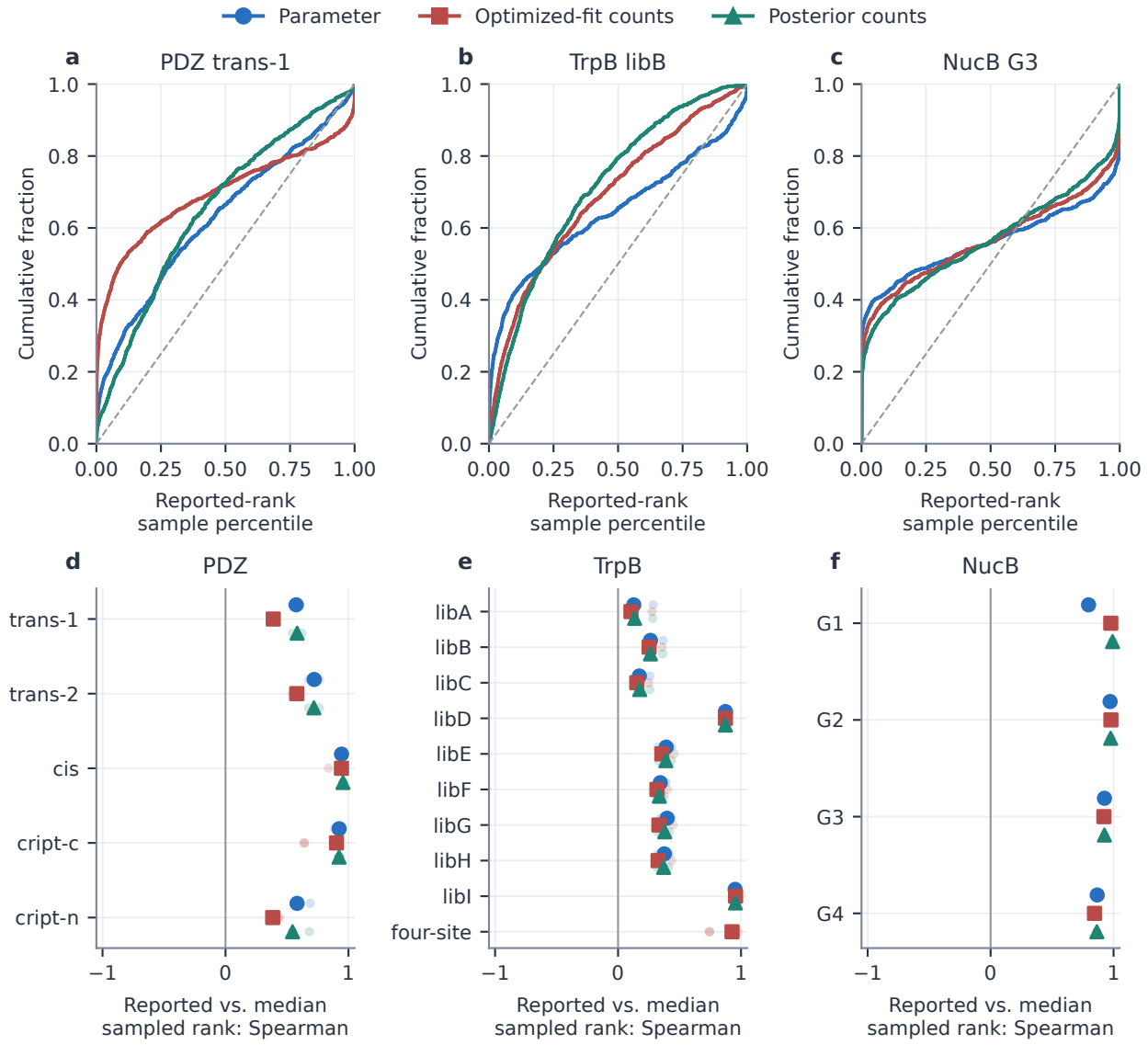


Figure S1: Agreement within the reported top-1,000 cohort. Panels (a)–(c) show reported-rank percentile distributions for PDZ trans-1 and TrpB libB under Exponential–Exponential and NucB G3 under Exponential. Blue, red, and green denote parameter, optimized-fit count, and posterior count resampling. Panels (d)–(f) show reported-versus-median-sampled-rank Spearman correlations across all analyzed landscapes; faint points are model runs and larger symbols are medians across models within each landscape. The displayed ranks are still ranks against all finite eligible competitors, not ranks recalculated only within the cohort.

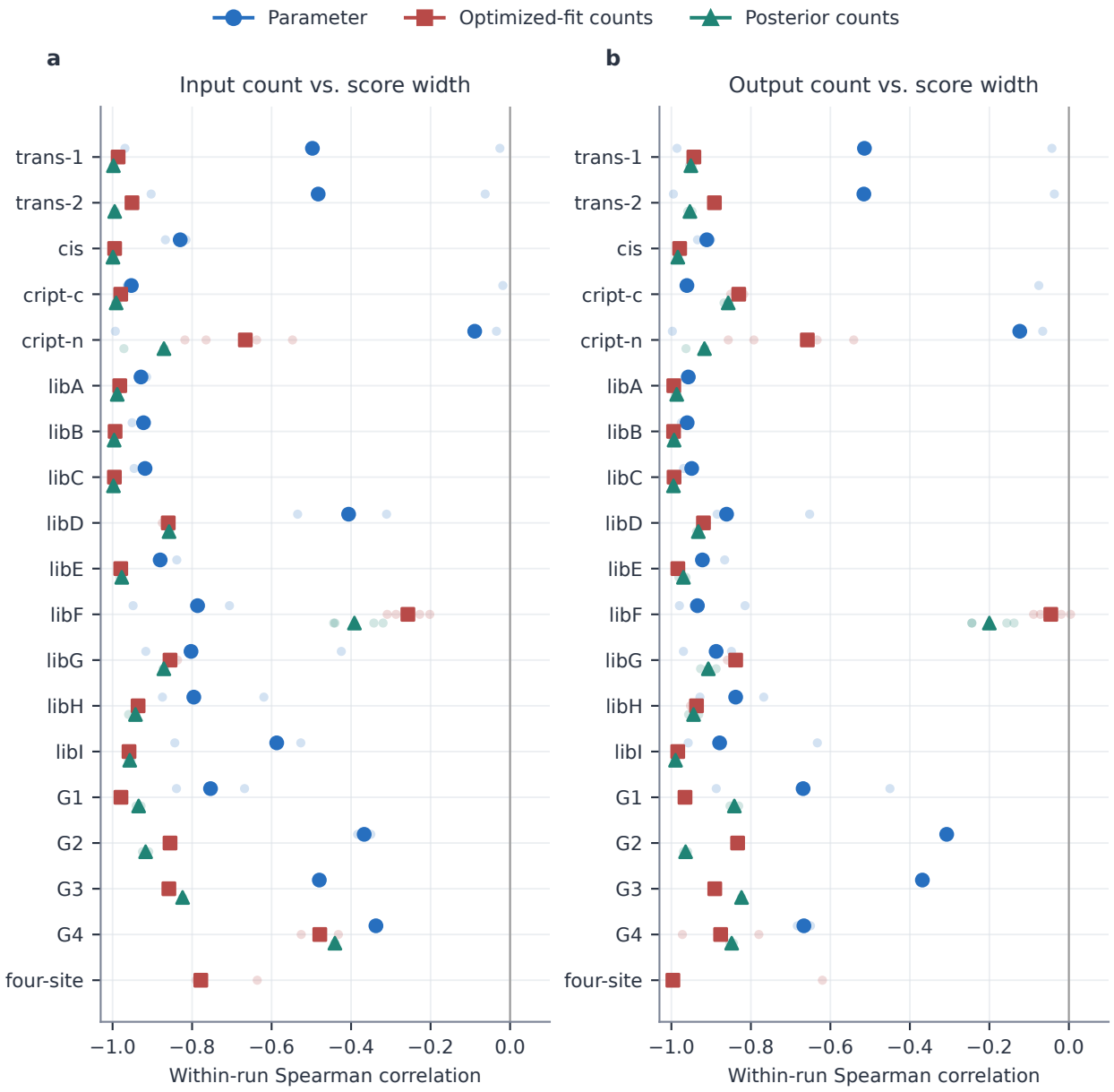


Figure S2: Count support and score uncertainty across all analyzed landscapes. Panels show within-run Spearman correlations of input counts (left) and output counts (right) with normalized central 90% score width. Each run uses its reference top 100 plus challengers with inclusion probability at least 0.01. Faint points are model runs; larger symbols are medians across models within landscape and scheme. Blue circles, red squares, and green triangles represent parameter, optimized-fit count, and posterior count resampling. These are associations in selected cohorts, not independent experimental effects.

F Per-analysis count support and selection uncertainty

We show each of the 158 analysis runs separately, grouped by protein system and resampling scheme. Table 6 locates the figures; the row headings give the landscape, model specification, retained sample count, decision-cohort size, and mean shared top-100 fraction. Each row contains the median-score reference top 100 and all outside candidates with top-100 inclusion frequency at least 0.01. These are the same top-focused cohorts used in Section 6, not samples of all eligible variants.

From left to right, the panels show normalized score width against input support, output support, and the score margin from the top-100 cutoff; the last panel shows top-100 inclusion against that same margin. Counts and score widths are defined in Appendix D. The score-width and margin denominator is each run's interquartile range (IQR) of the top-1,000 median scores. Count axes display $\log_{10}(1+n)$, and score widths use logarithmic axes. Count limits and score-width limits are common within each protein-system and scheme collection, including across pages. The two margin panels share a scale within a row; their scales can differ between runs because their score ranges differ.

All points are retained. Exact-zero score widths are displayed at half the smallest positive normalized width in the corresponding protein-system and scheme collection; the numerical tables retain zero. The accompanying figure index records those plot-only floors and the number of affected points. Blue filled points denote reference top-100 members and orange open points denote challengers. The dotted vertical lines mark a score margin of zero. No regression lines or smoothing curves are fitted. These figures display within-analysis associations, not causal effects of changing read depth and not model-adequacy checks against new experimental observations.

Table 6: Index of per-analysis count-support and selection summaries. Every one of the 158 analysis runs appears on exactly one page.

Scheme	Family	Runs	Figures
Parameter resampling	PDZ	14	S3–S7
Parameter resampling	TrpB	28	S8–S17
Parameter resampling	NucB	7	S18–S20
Optimized-fit count resampling	PDZ	19	S21–S27
Optimized-fit count resampling	TrpB	34	S28–S39
Optimized-fit count resampling	NucB	7	S40–S42
Posterior count resampling	PDZ	14	S43–S47
Posterior count resampling	TrpB	28	S48–S57
Posterior count resampling	NucB	7	S58–S60

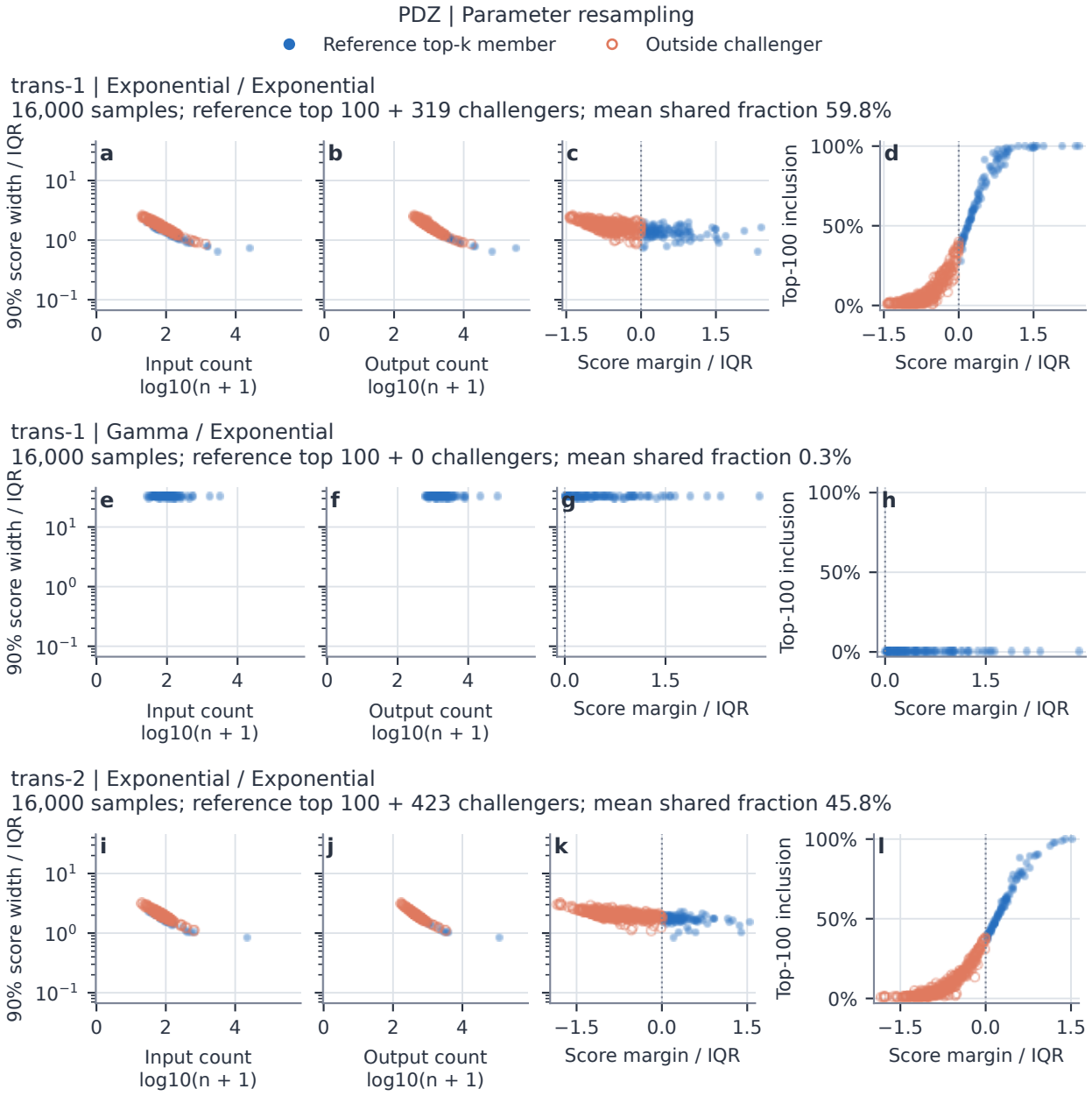


Figure S3: PDZ decision-cohort summaries: parameter resampling. Rows, top to bottom: pdz-trans-1, Exponential–Exponential; pdz-trans-1, Gamma–Exponential; pdz-trans-2, Exponential–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

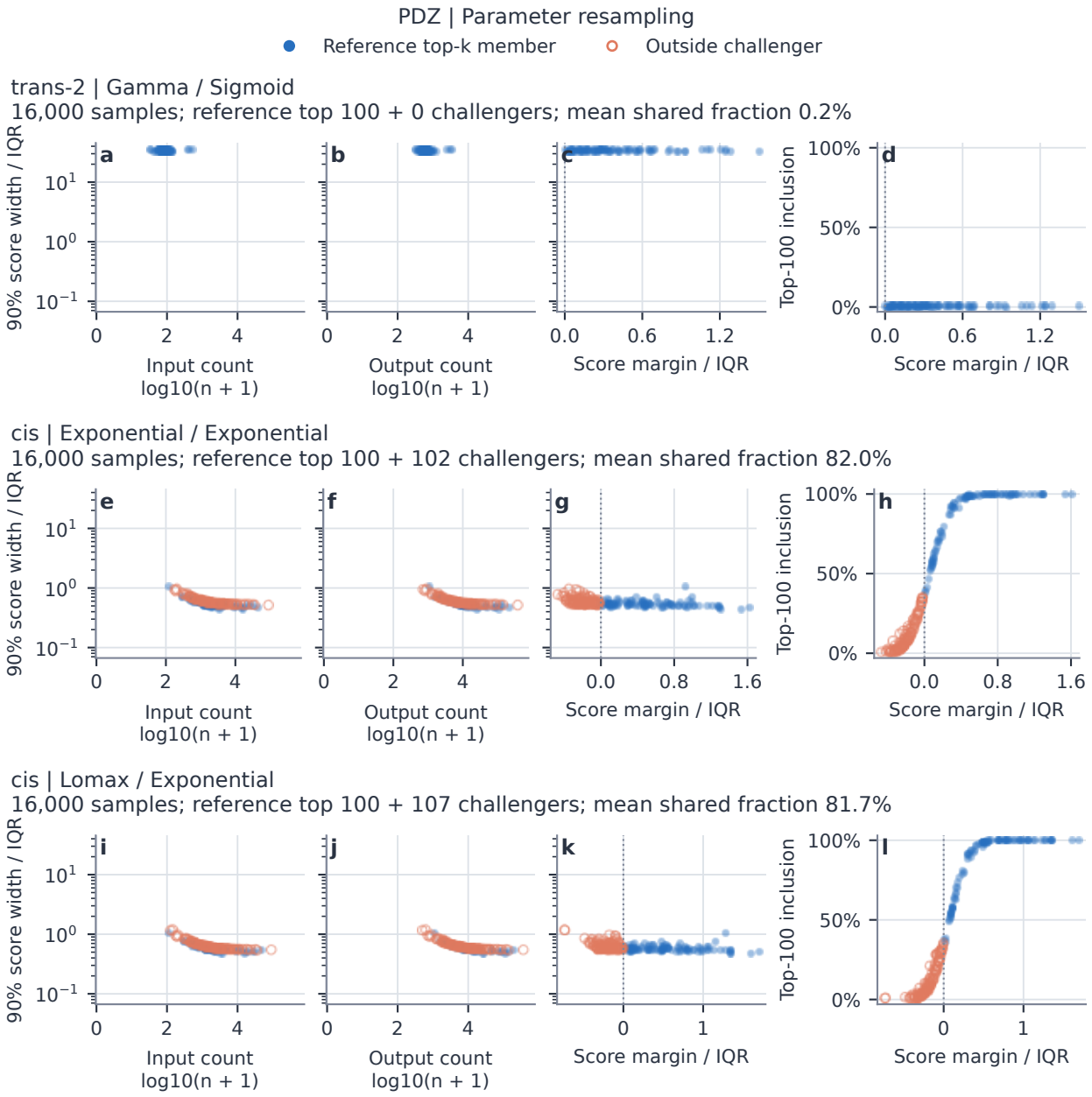


Figure S4: PDZ decision-cohort summaries: parameter resampling. Rows, top to bottom: pdz-trans-2, Gamma–Sigmoid; pdz-cis, Exponential–Exponential; pdz-cis, Lomax–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

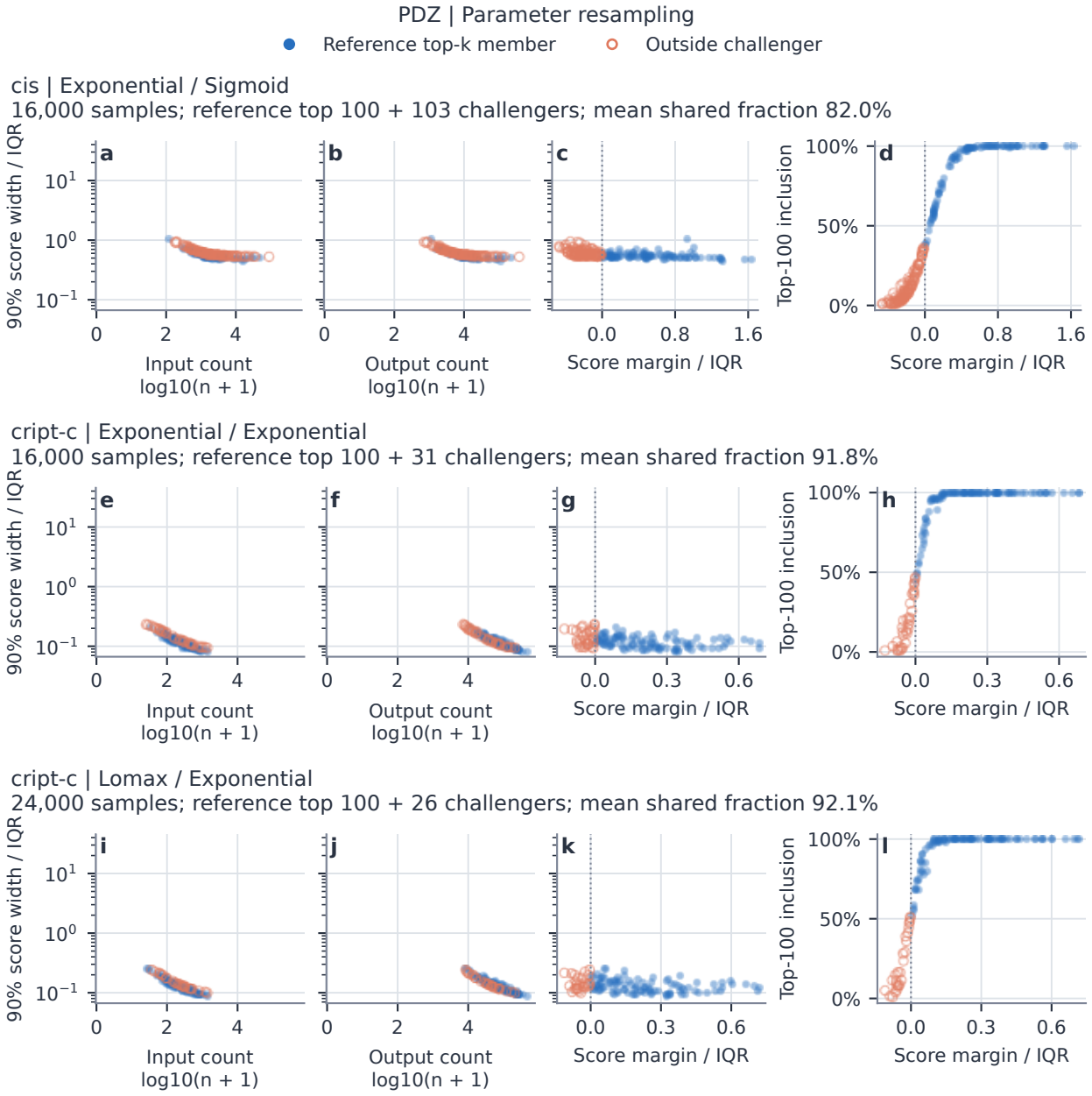


Figure S5: PDZ decision-cohort summaries: parameter resampling. Rows, top to bottom: pdz-cis, Exponential–Sigmoid; pdz-cript-c, Exponential–Exponential; pdz-cript-c, Lomax–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

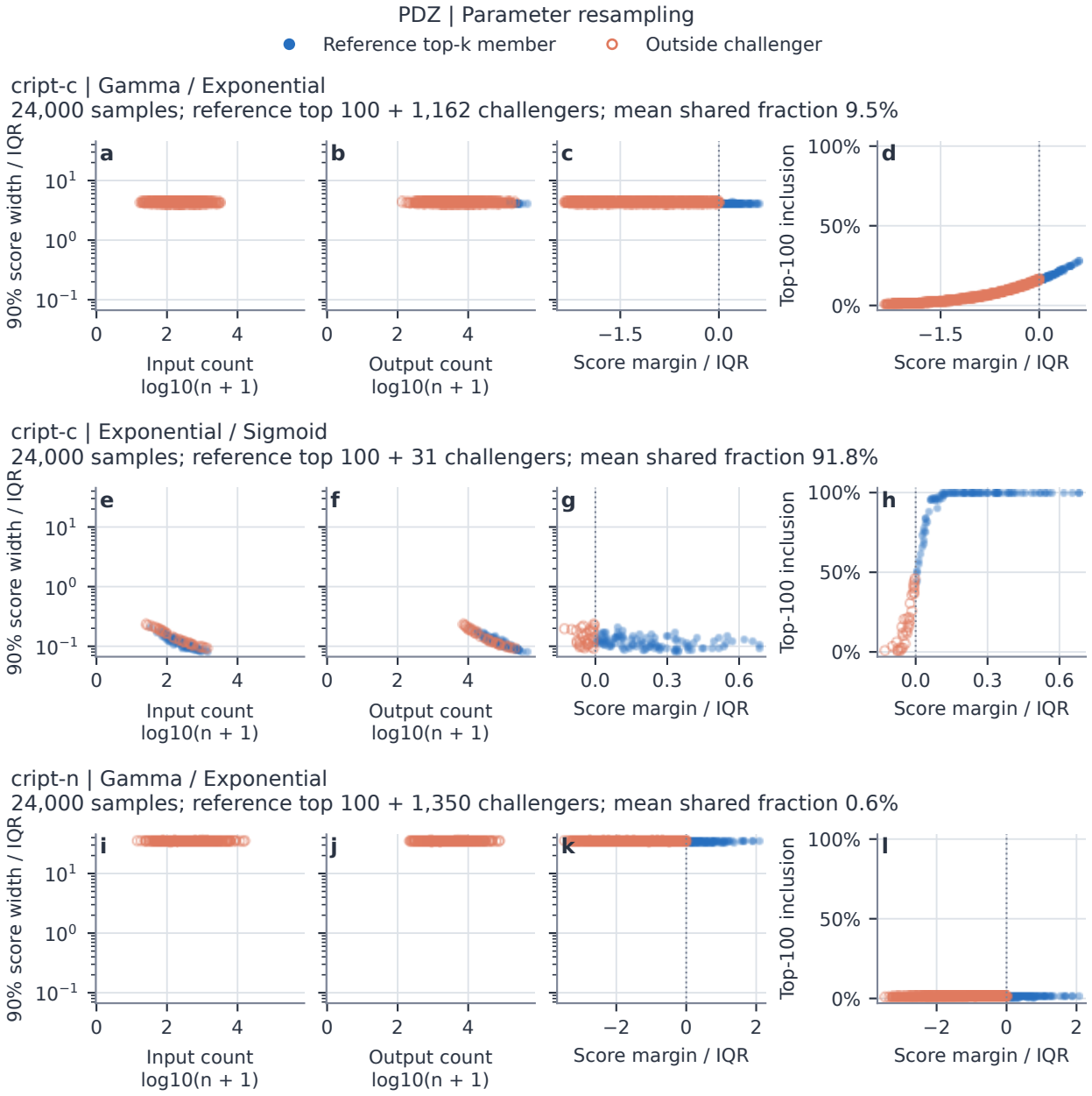


Figure S6: PDZ decision-cohort summaries: parameter resampling. Rows, top to bottom: pdz-cript-c, Gamma–Exponential; pdz-cript-c, Exponential–Sigmoid; pdz-cript-n, Gamma–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

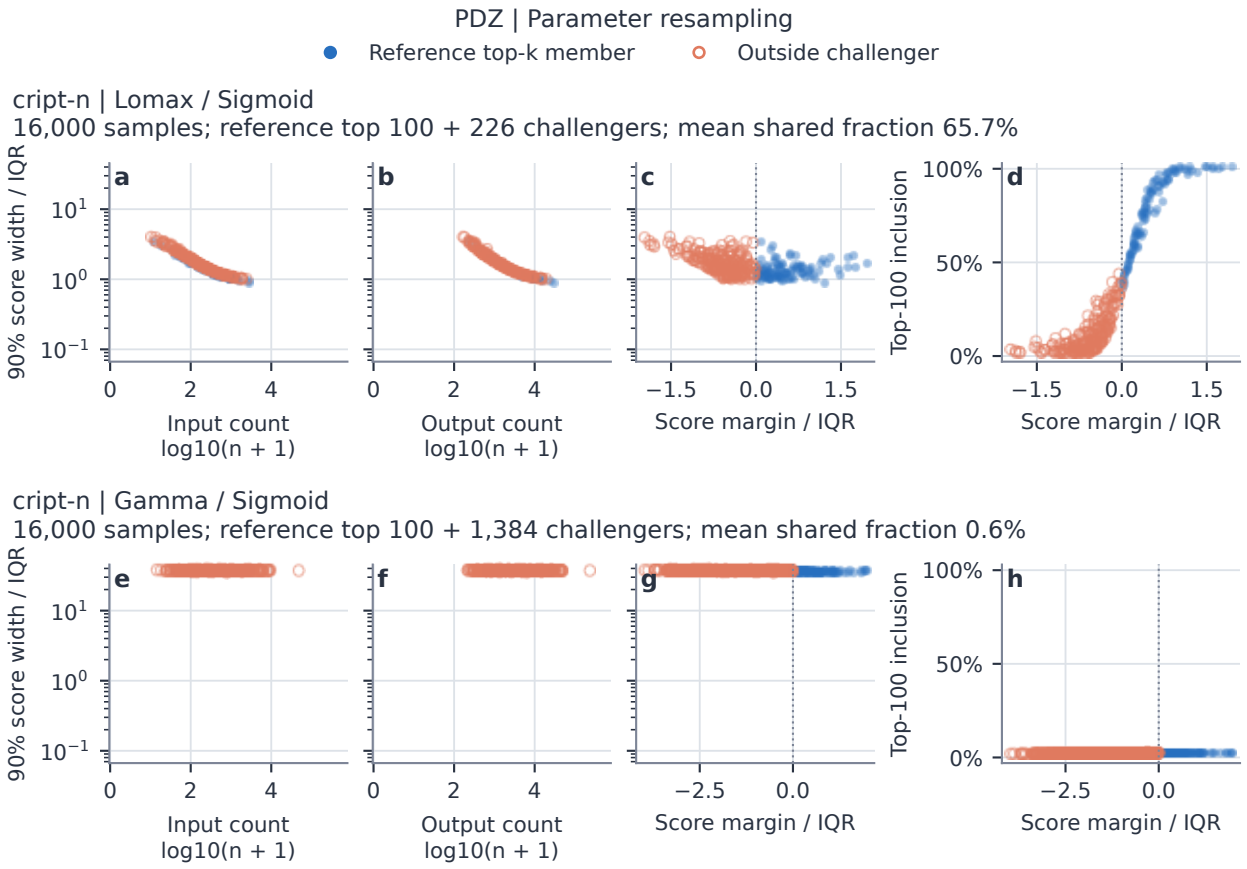


Figure S7: PDZ decision-cohort summaries: parameter resampling. Rows, top to bottom: pdz-cript-n, Lomax–Sigmoid; pdz-cript-n, Gamma–Sigmoid. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

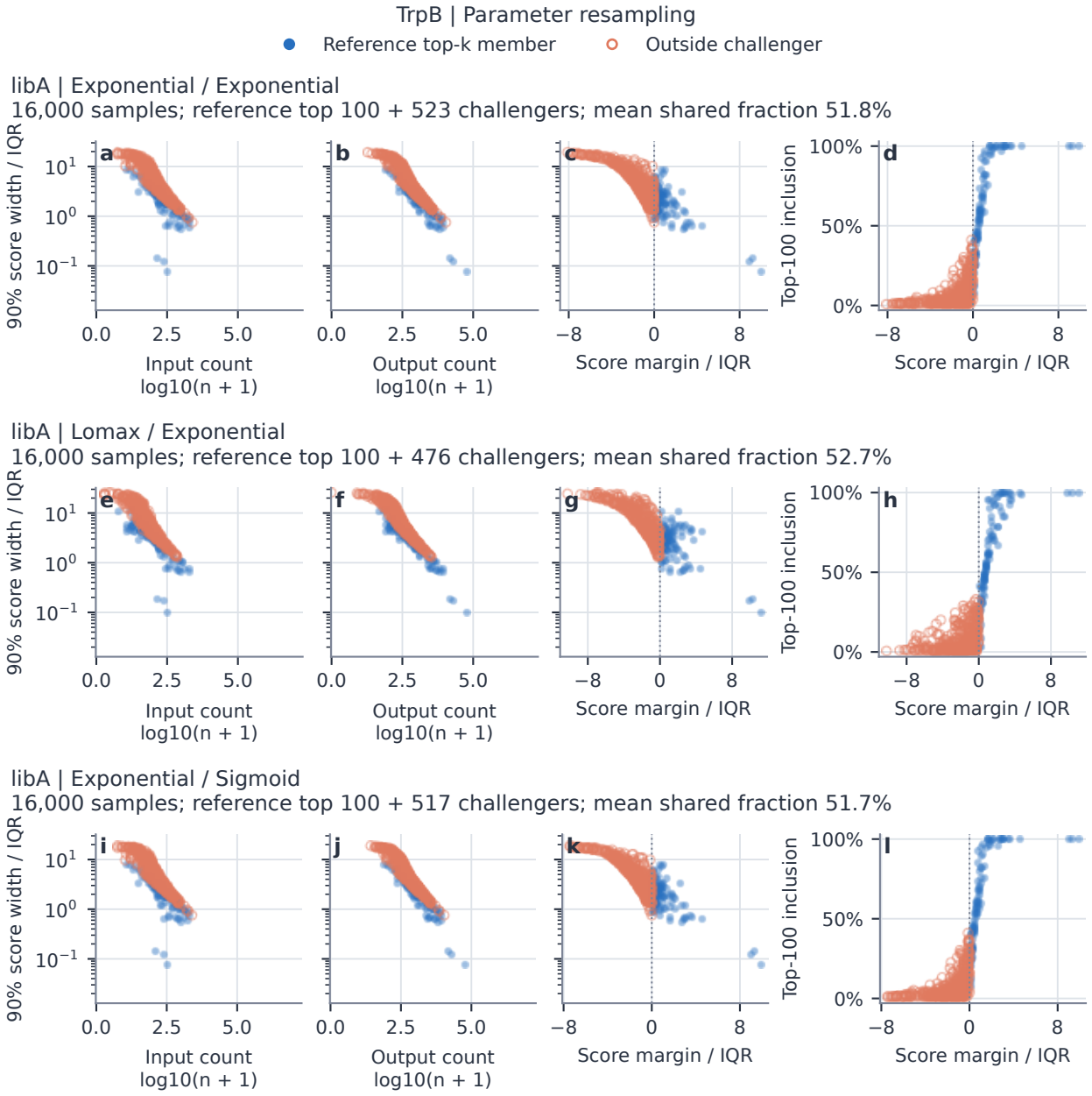


Figure S8: TrpB decision-cohort summaries: parameter resampling. Rows, top to bottom: trpb-libA, Exponential–Exponential; trpb-libA, Lomax–Exponential; trpb-libA, Exponential–Sigmoid. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

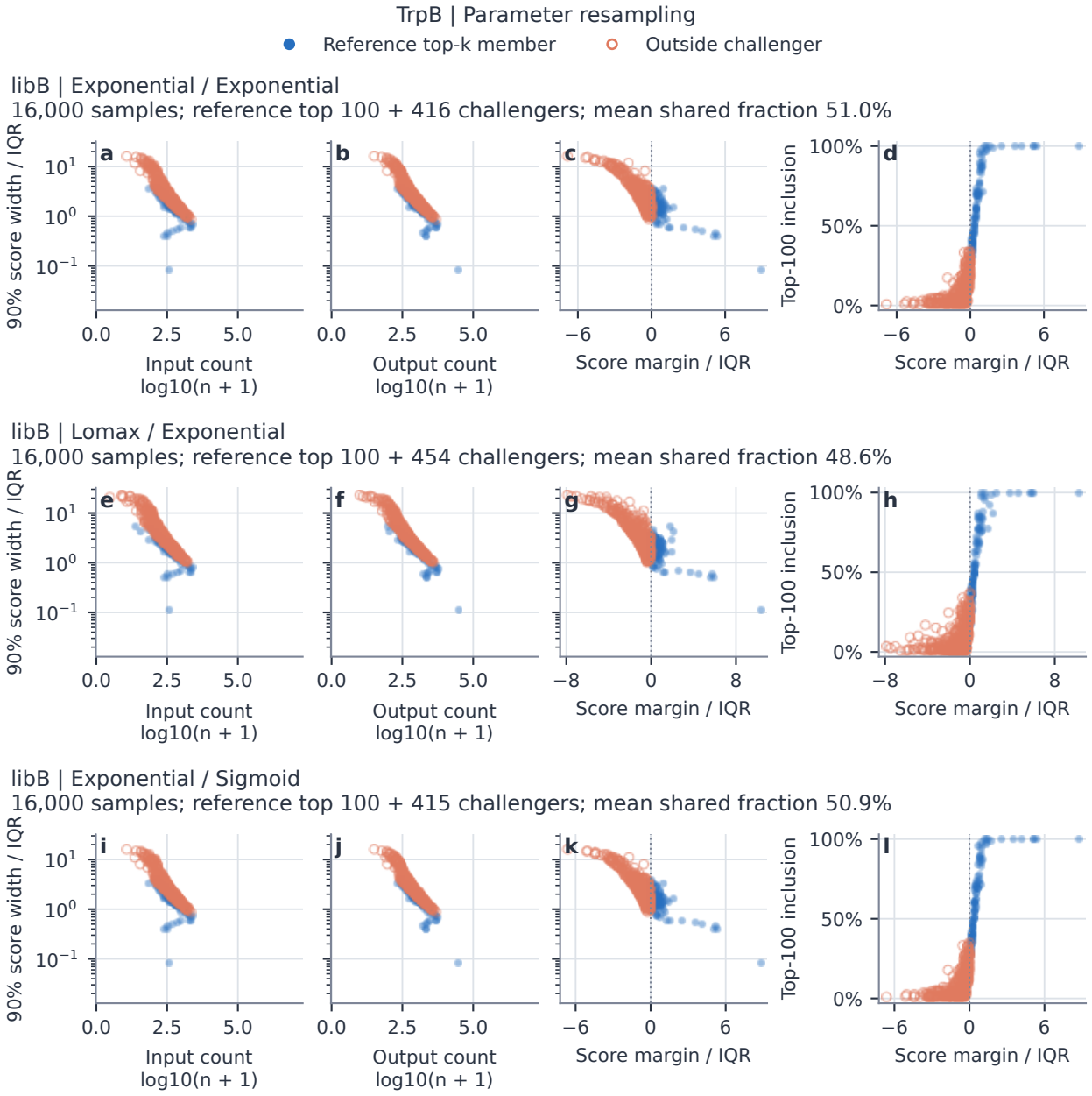


Figure S9: TrpB decision-cohort summaries: parameter resampling. Rows, top to bottom: trpb-libB, Exponential–Exponential; trpb-libB, Lomax–Exponential; trpb-libB, Exponential–Sigmoid. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

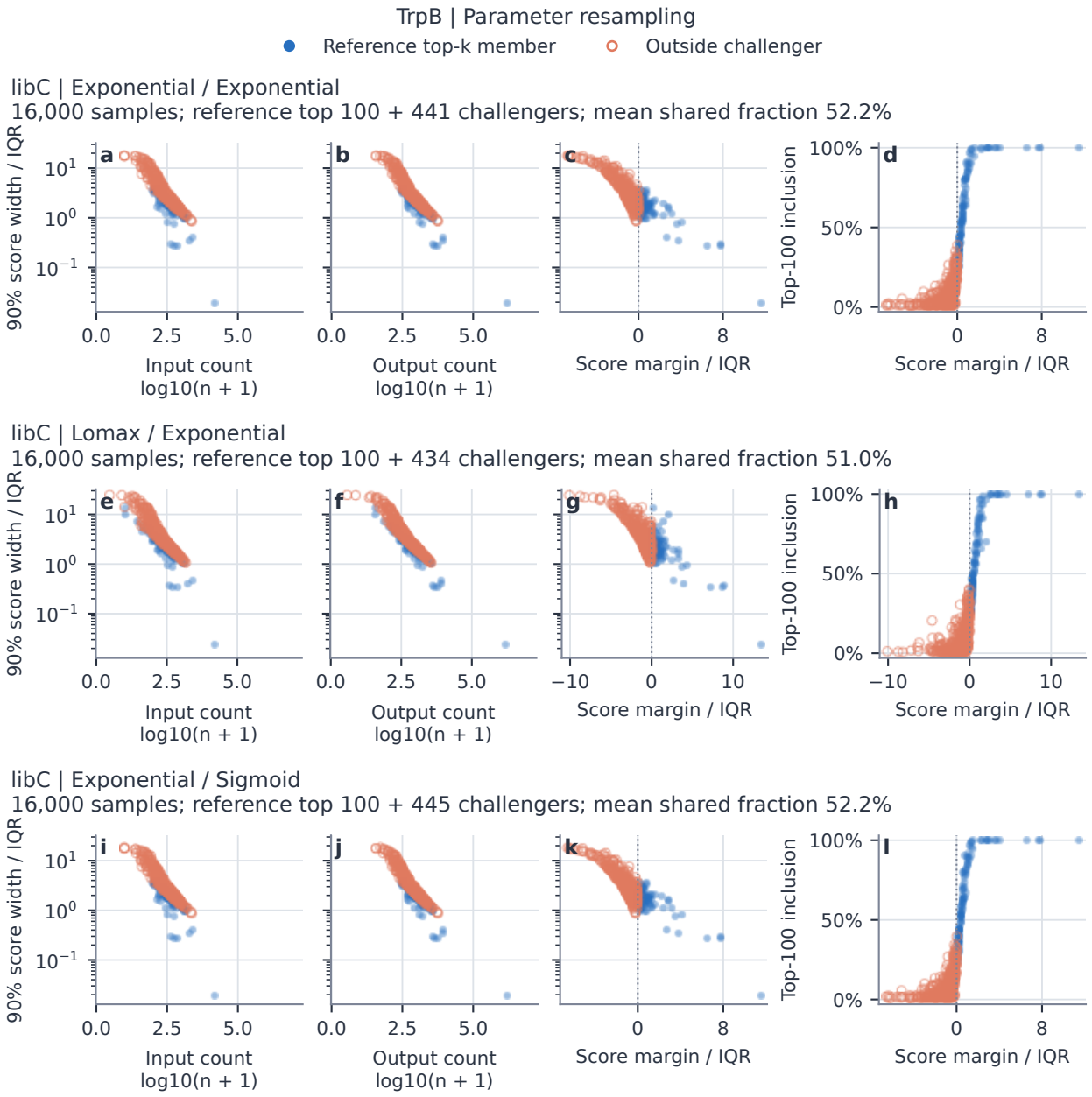


Figure S10: TrpB decision-cohort summaries: parameter resampling. Rows, top to bottom: trpb-libC, Exponential–Exponential; trpb-libC, Lomax–Exponential; trpb-libC, Exponential–Sigmoid. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

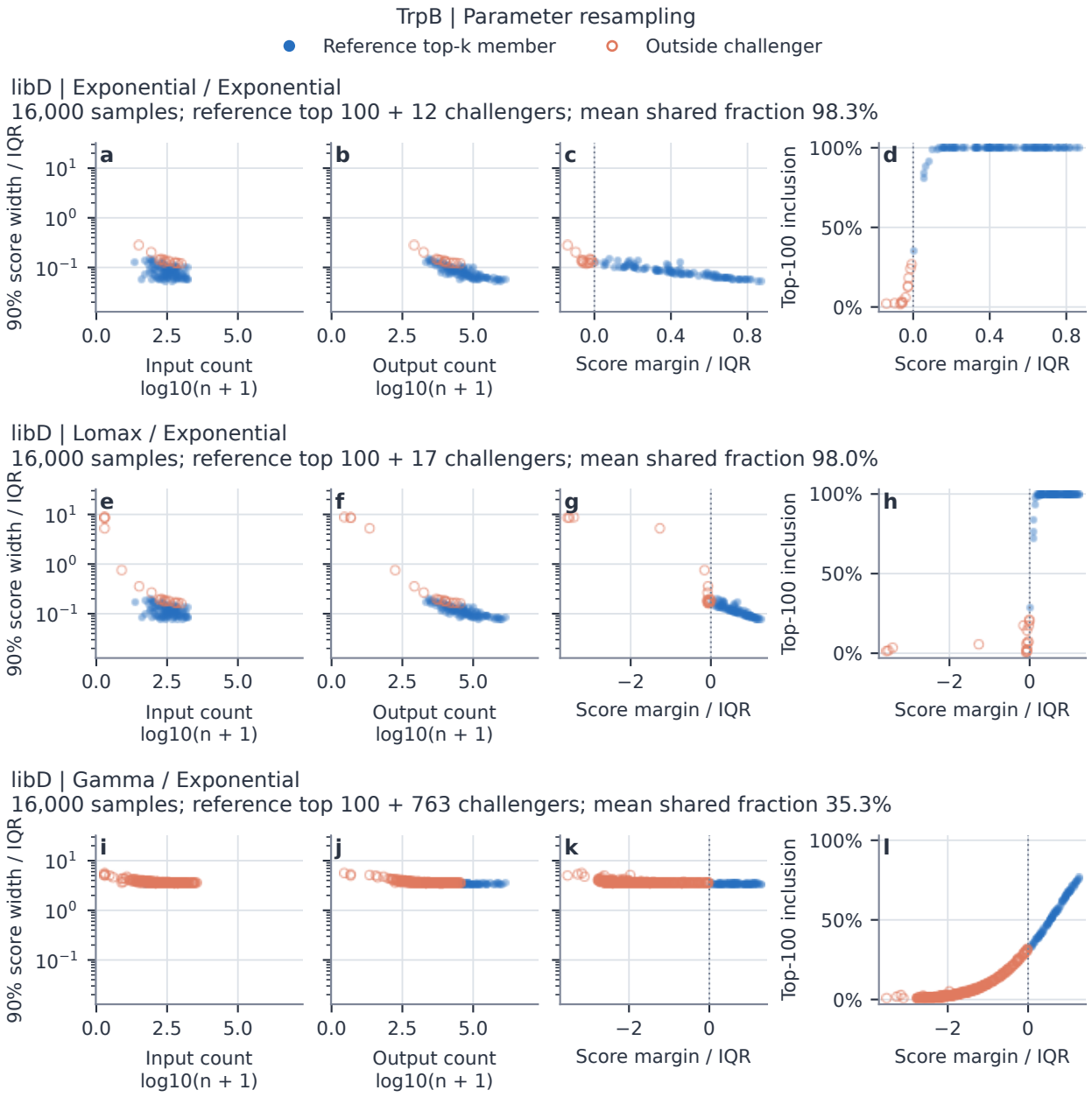


Figure S11: TrpB decision-cohort summaries: parameter resampling. Rows, top to bottom: trpb-libD, Exponential–Exponential; trpb-libD, Lomax–Exponential; trpb-libD, Gamma–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

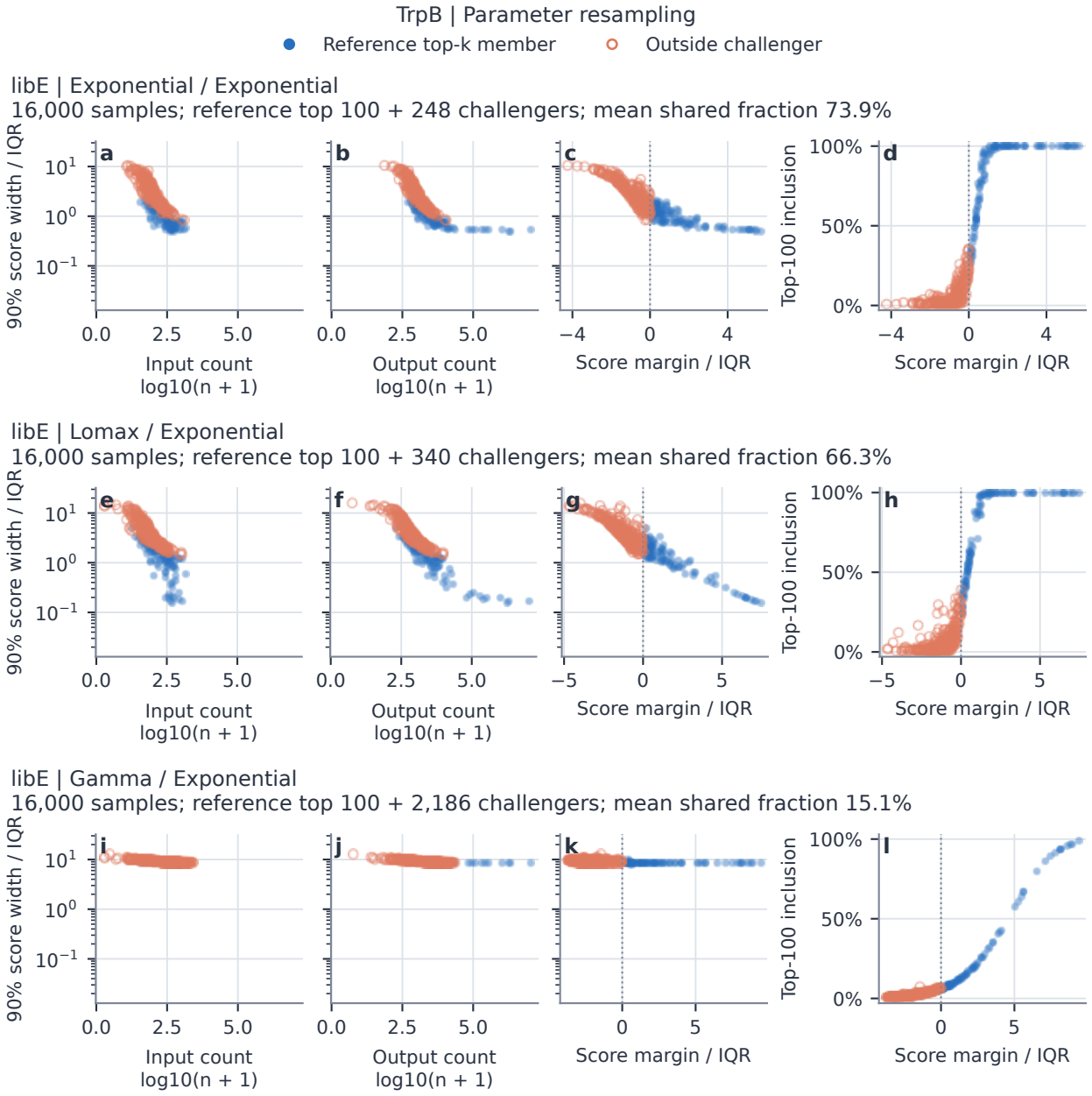


Figure S12: TrpB decision-cohort summaries: parameter resampling. Rows, top to bottom: trpb-libE, Exponential–Exponential; trpb-libE, Lomax–Exponential; trpb-libE, Gamma–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

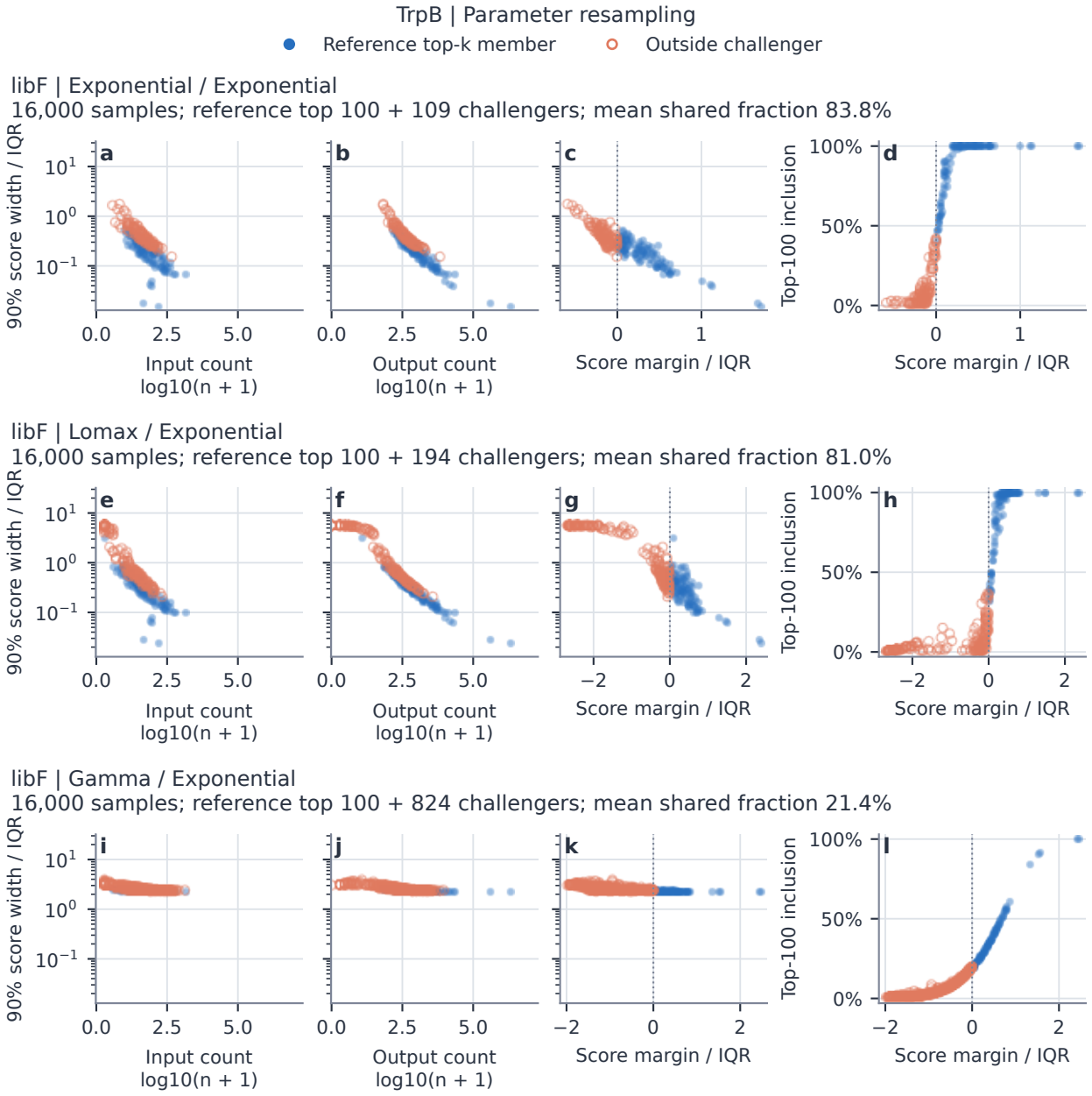


Figure S13: TrpB decision-cohort summaries: parameter resampling. Rows, top to bottom: trpb-libF, Exponential–Exponential; trpb-libF, Lomax–Exponential; trpb-libF, Gamma–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

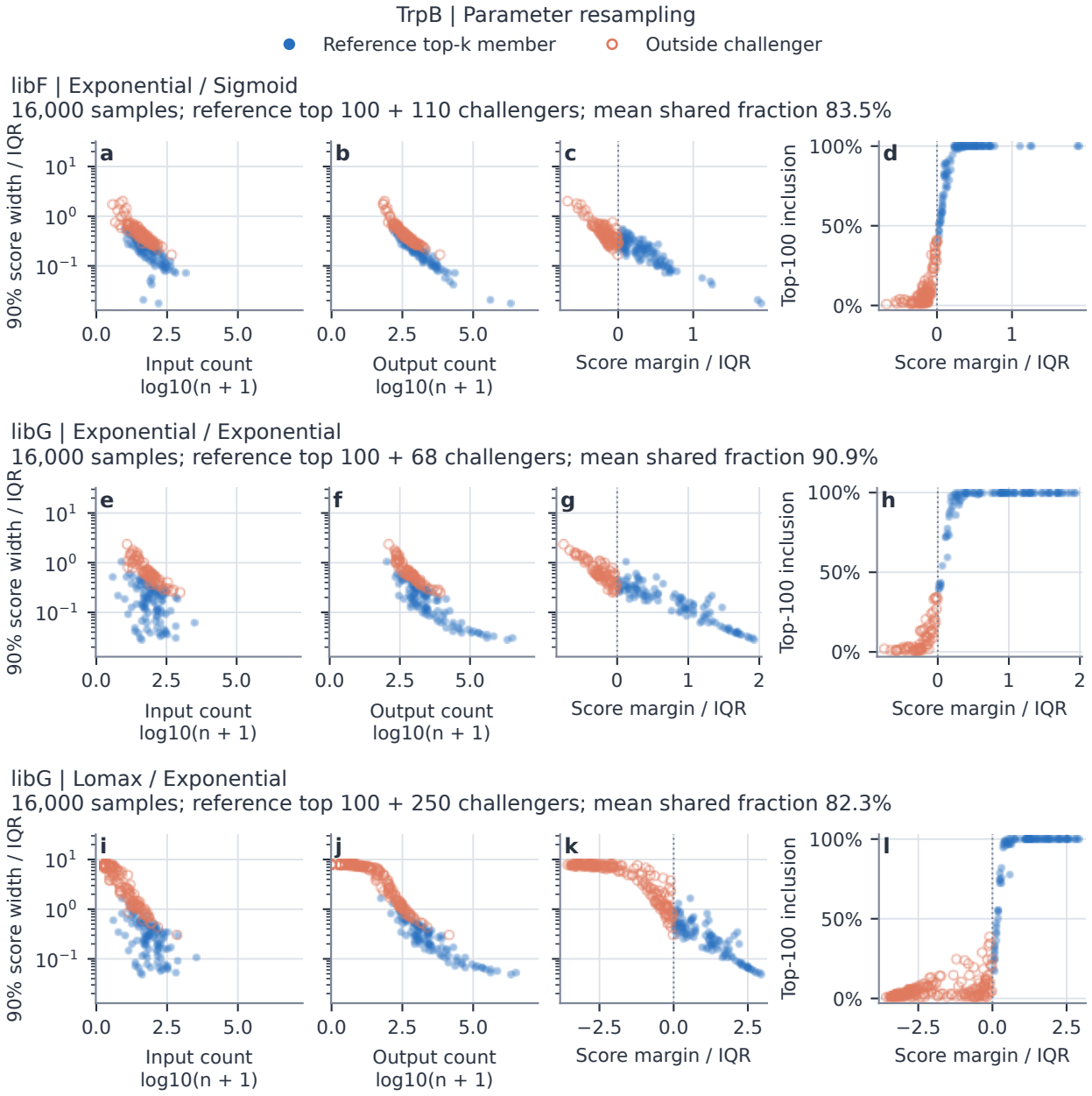


Figure S14: TrpB decision-cohort summaries: parameter resampling. Rows, top to bottom: trpb-libF, Exponential–Sigmoid; trpb-libG, Exponential–Exponential; trpb-libG, Lomax–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

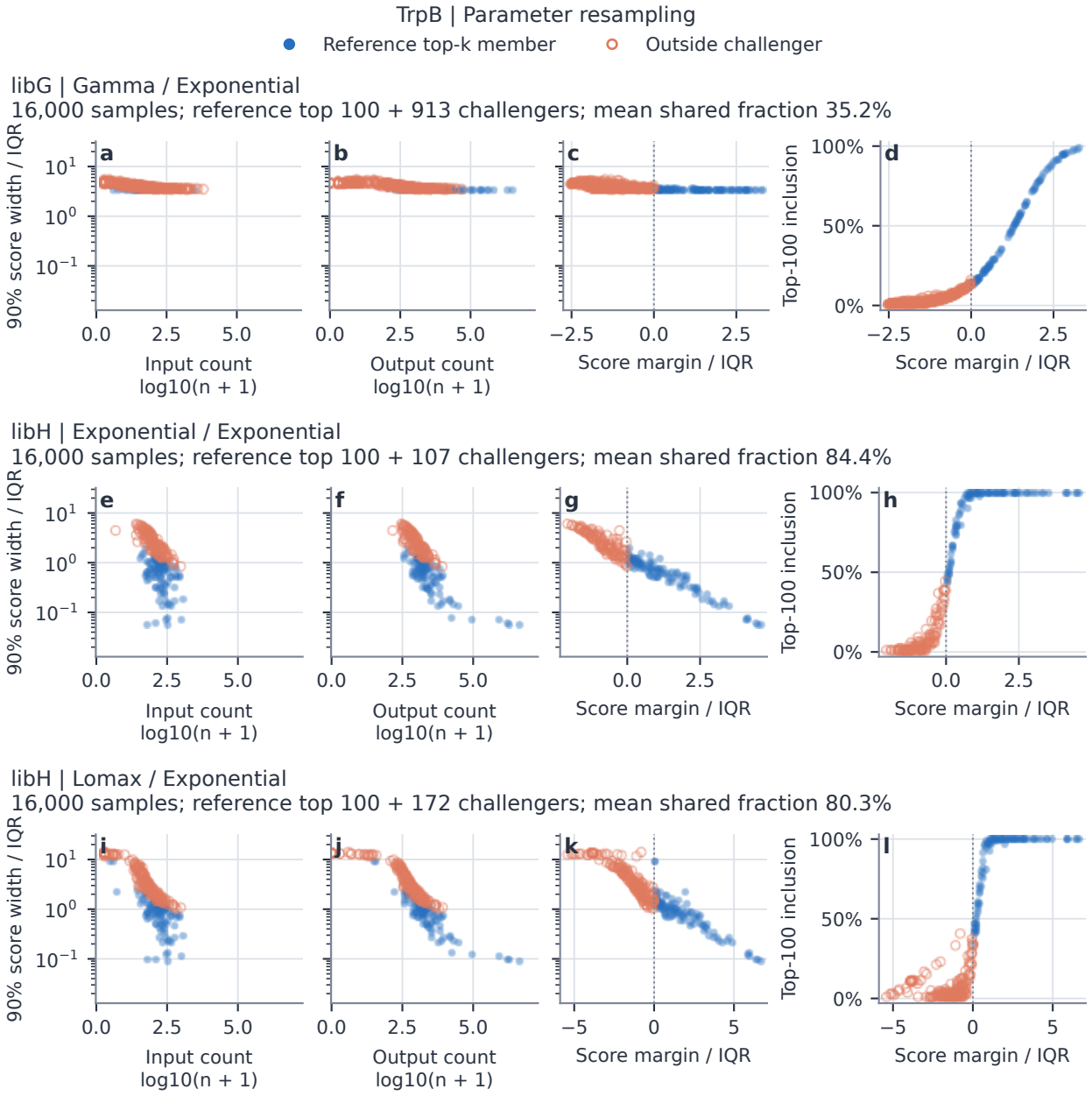


Figure S15: TrpB decision-cohort summaries: parameter resampling. Rows, top to bottom: trpb-libG, Gamma–Exponential; trpb-libH, Exponential–Exponential; trpb-libH, Lomax–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

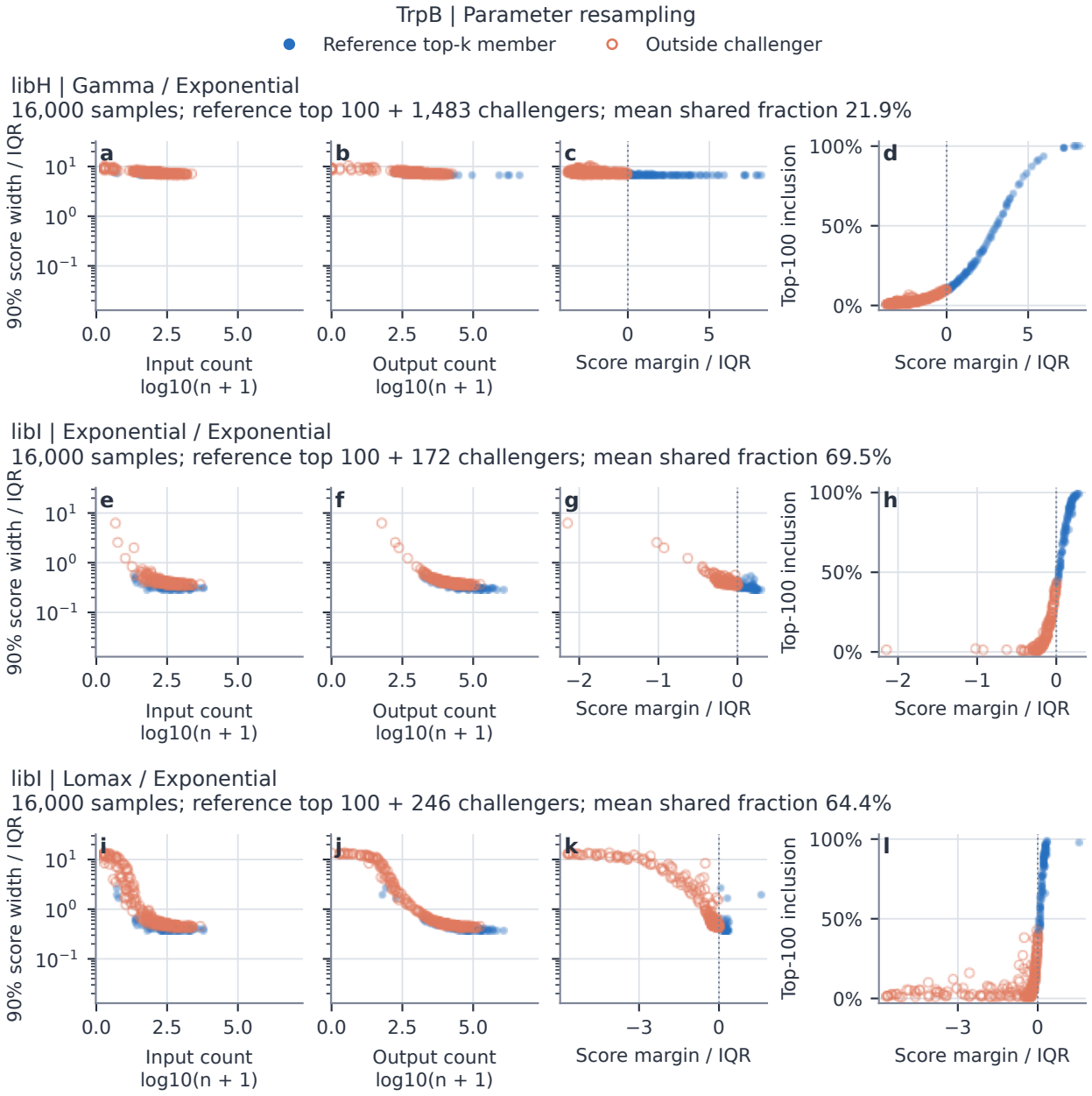


Figure S16: TrpB decision-cohort summaries: parameter resampling. Rows, top to bottom: trpb-libH, Gamma–Exponential; trpb-libI, Exponential–Exponential; trpb-libI, Lomax–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

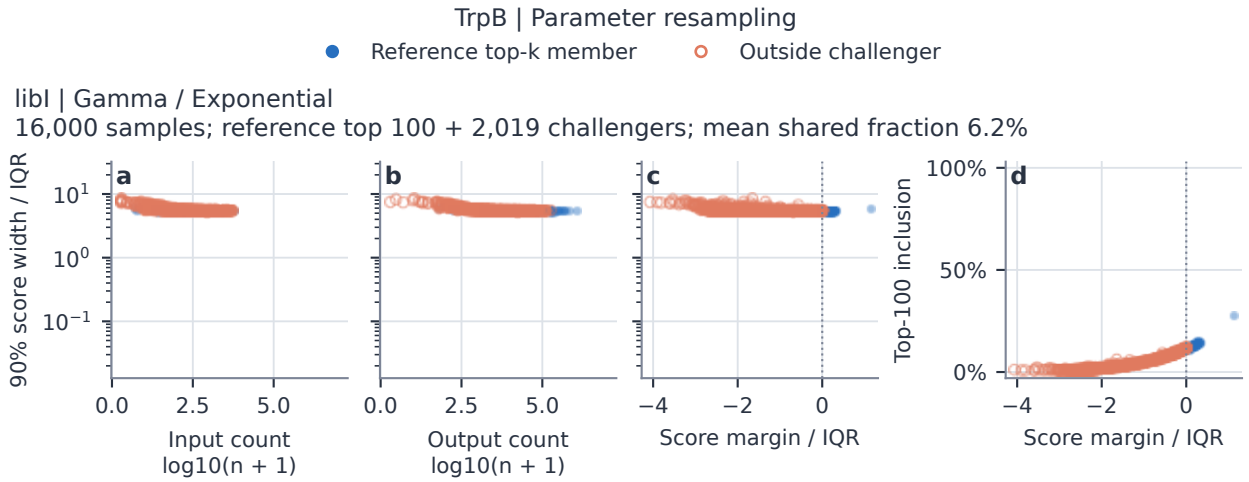


Figure S17: TrpB decision-cohort summaries: parameter resampling. Rows, top to bottom: trpb-libI, Gamma–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

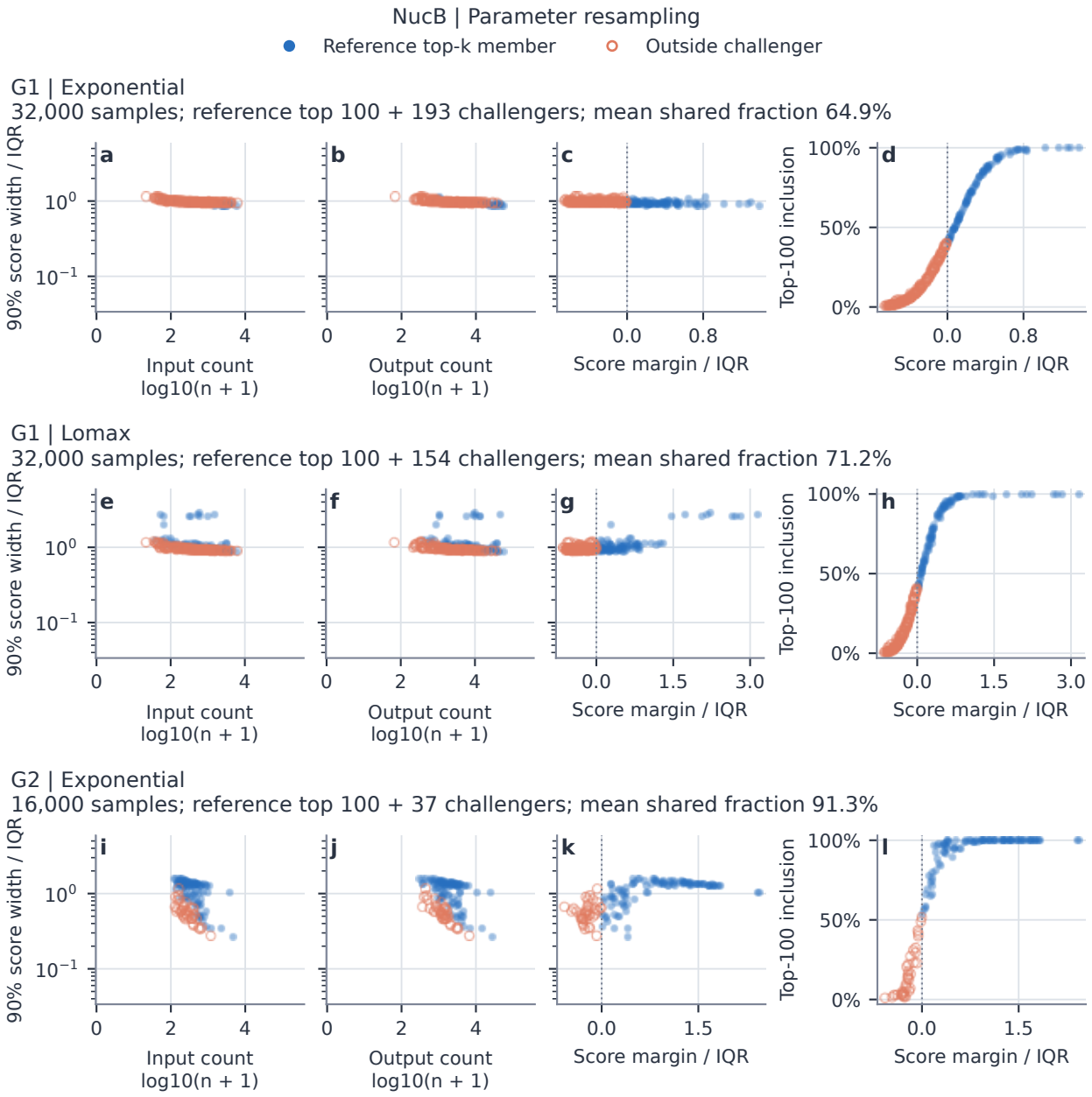


Figure S18: NucB decision-cohort summaries: parameter resampling. Rows, top to bottom: nuc-G1, Exponential; nuc-G1, Lomax; nuc-G2, Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

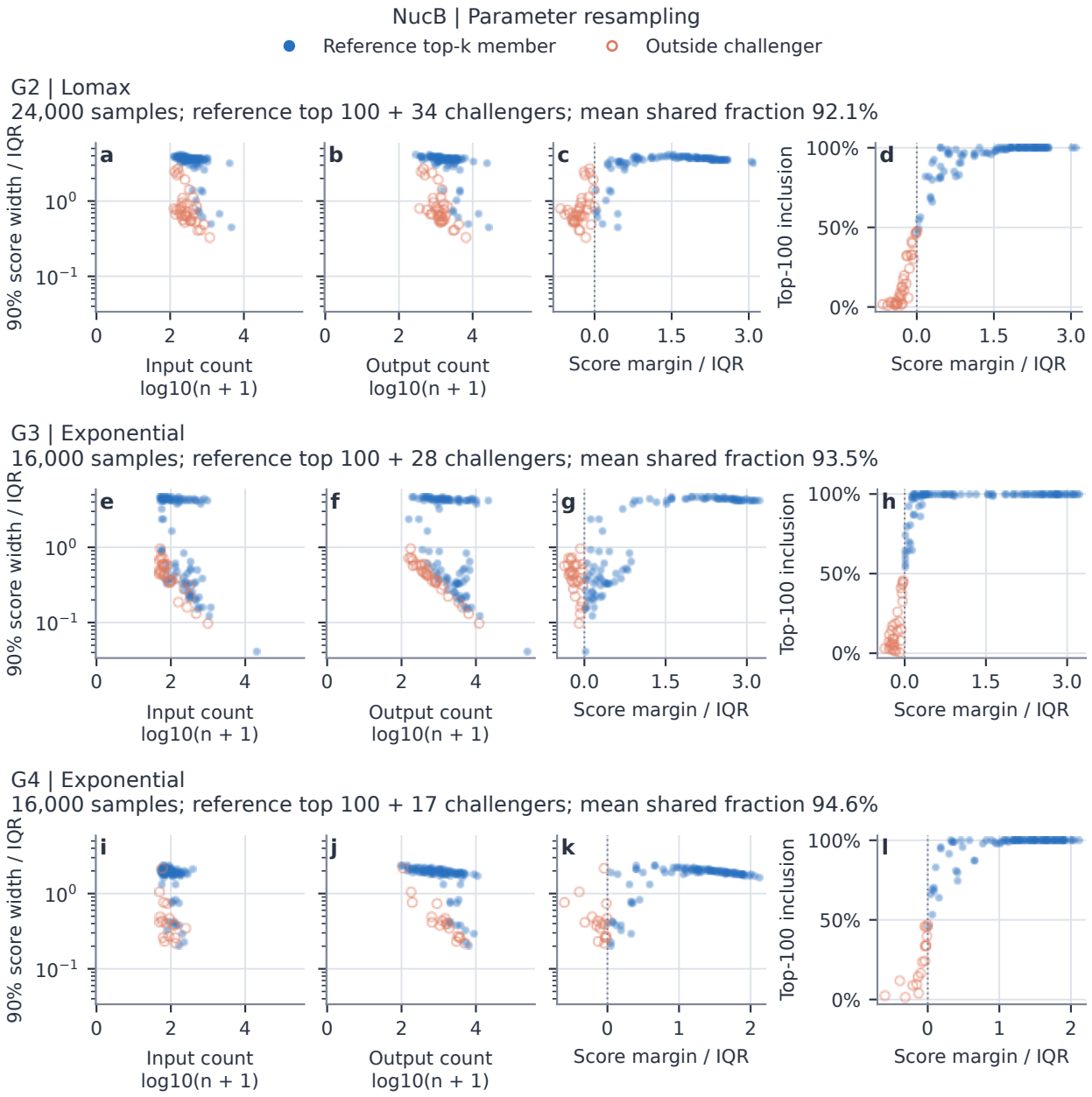


Figure S19: NucB decision-cohort summaries: parameter resampling. Rows, top to bottom: nuc-G2, Lomax; nuc-G3, Exponential; nuc-G4, Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

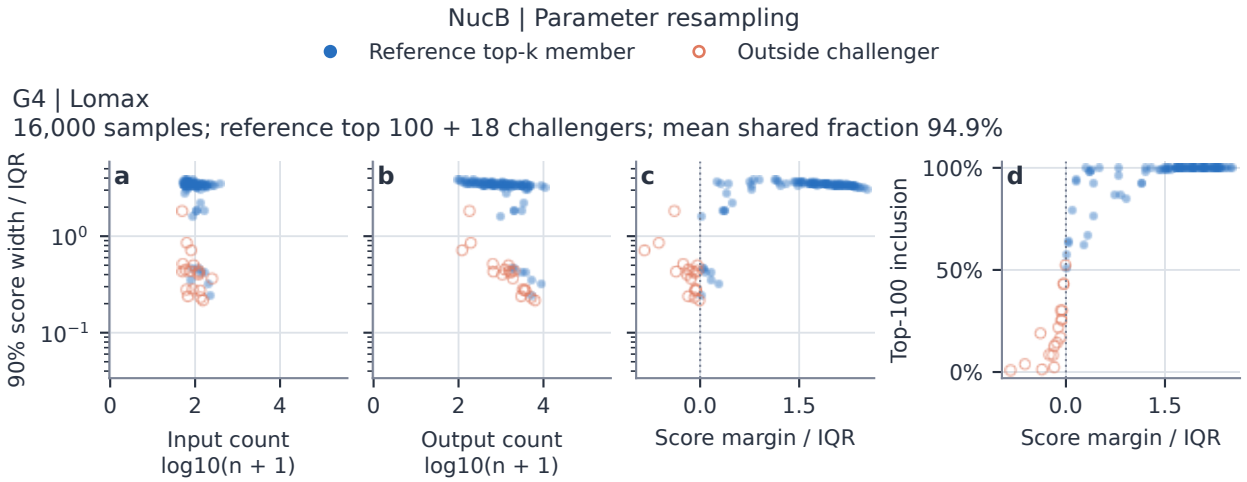


Figure S20: NucB decision-cohort summaries: parameter resampling. Rows, top to bottom: nuc-G4, Lomax. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

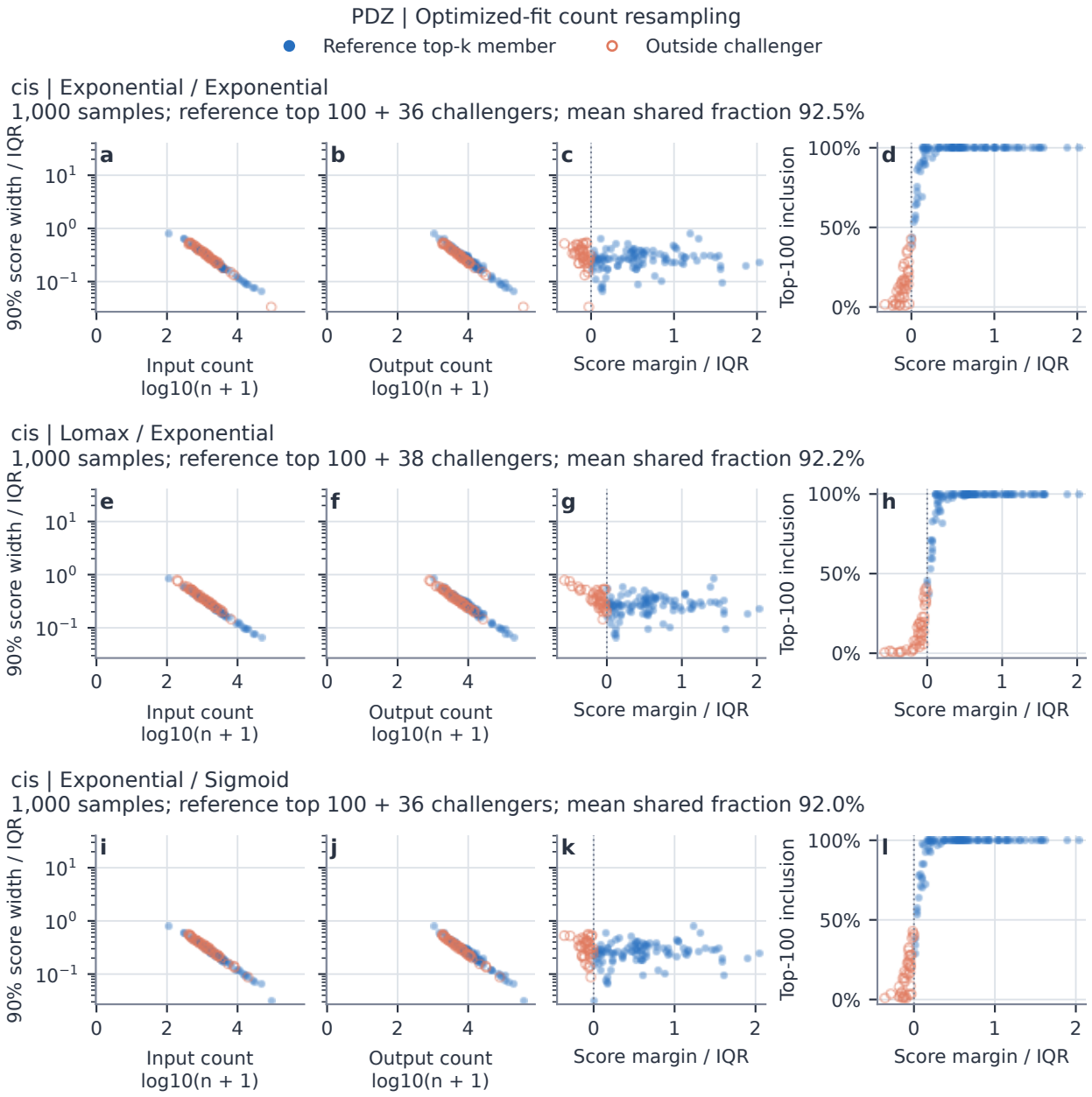


Figure S21: PDZ decision-cohort summaries: optimized-fit count resampling. Rows, top to bottom: pdz-cis, Exponential–Exponential; pdz-cis, Lomax–Exponential; pdz-cis, Exponential–Sigmoid. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

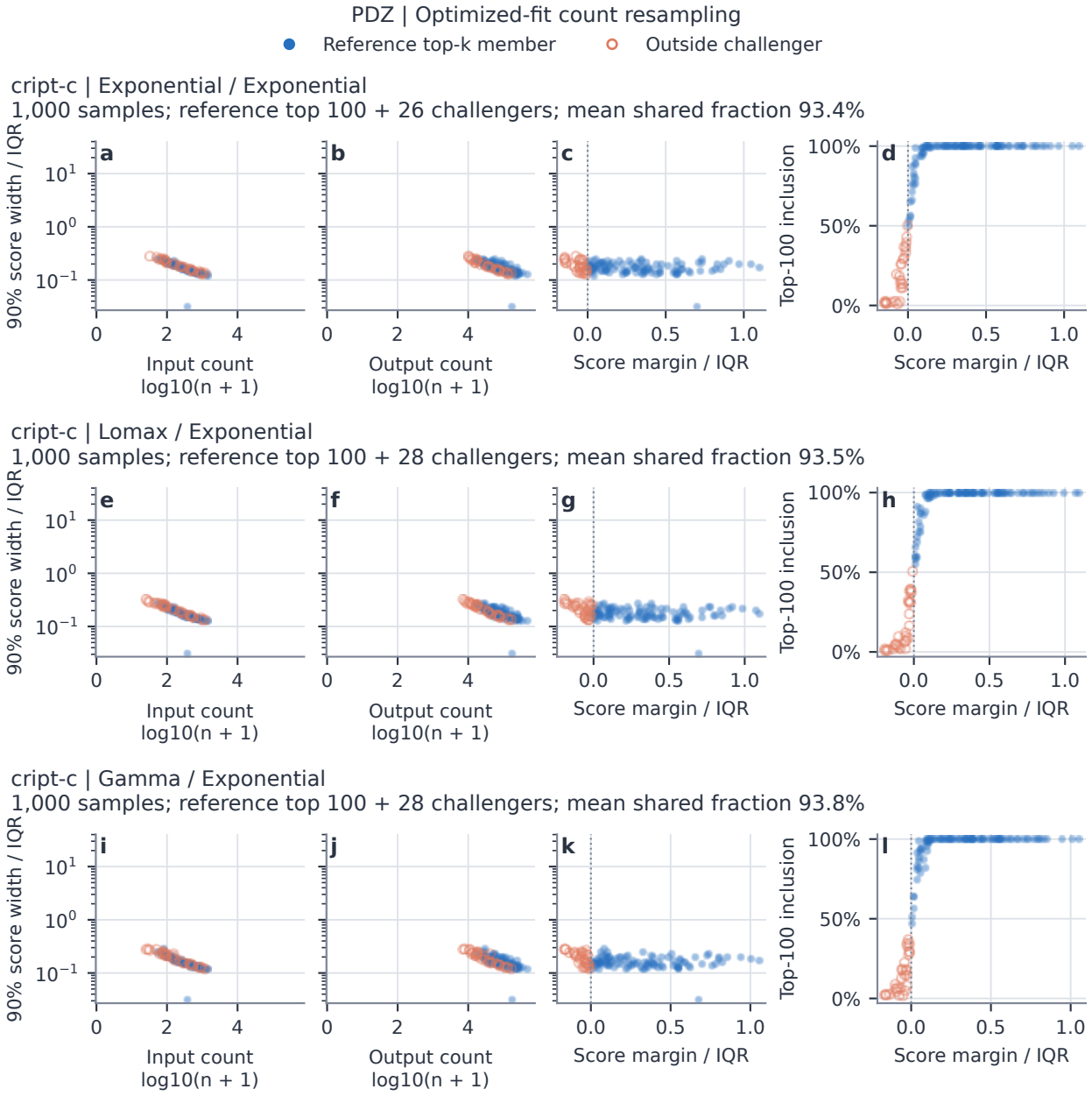


Figure S22: PDZ decision-cohort summaries: optimized-fit count resampling. Rows, top to bottom: pdz-cript-c, Exponential–Exponential; pdz-cript-c, Lomax–Exponential; pdz-cript-c, Gamma–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

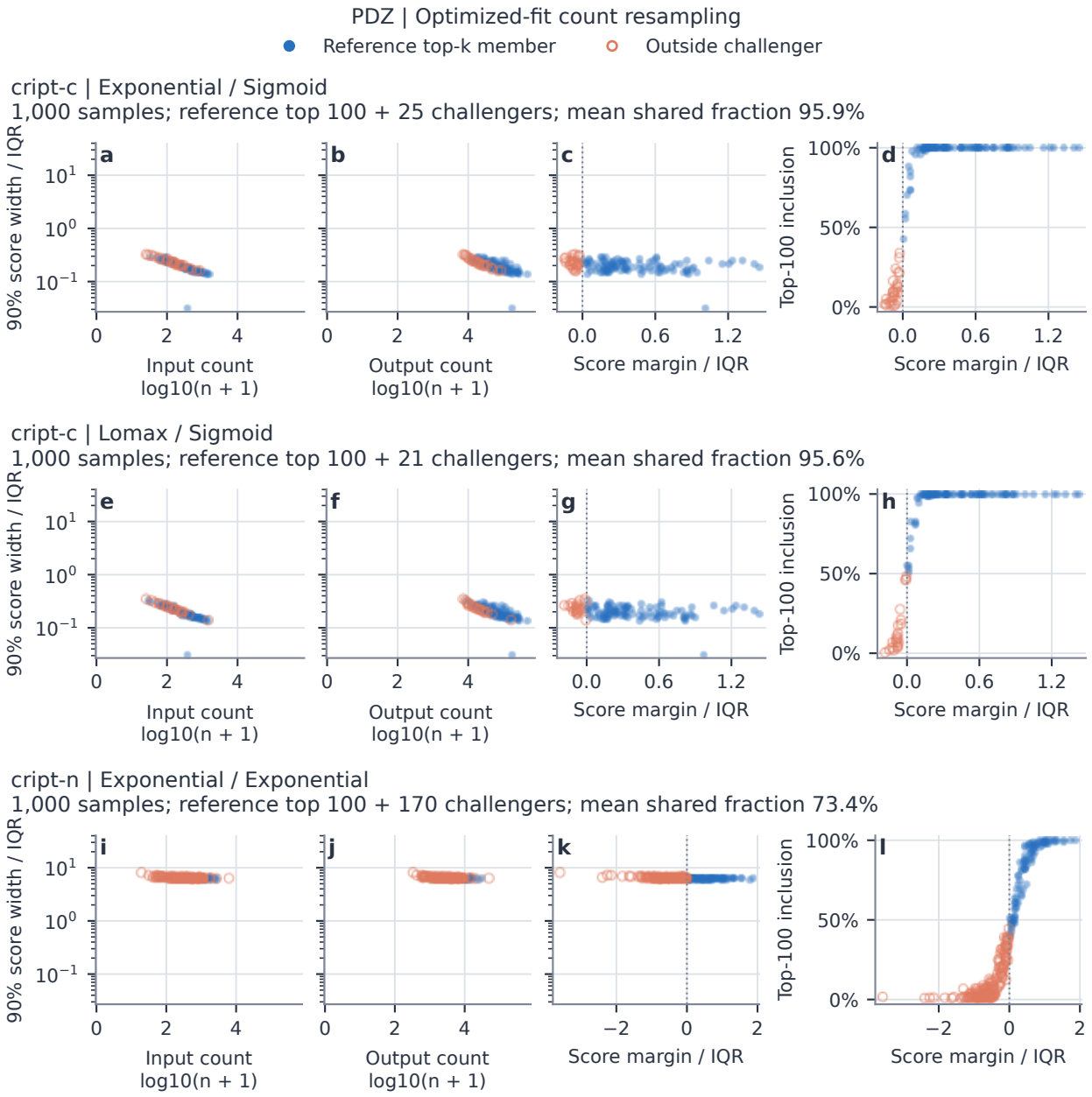


Figure S23: PDZ decision-cohort summaries: optimized-fit count resampling. Rows, top to bottom: pdz-cript-c, Exponential–Sigmoid; pdz-cript-c, Lomax–Sigmoid; pdz-cript-n, Exponential–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

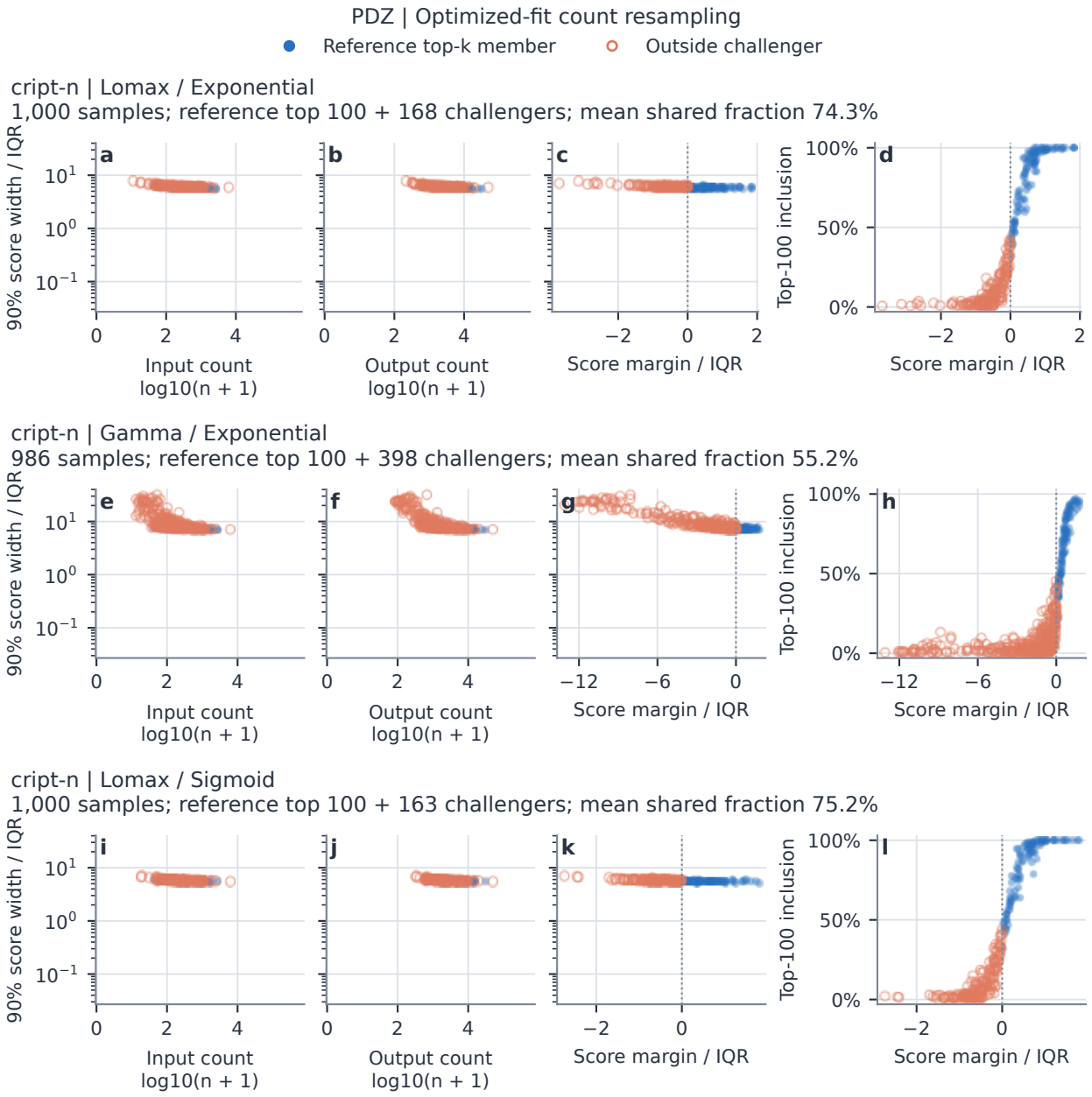


Figure S24: PDZ decision-cohort summaries: optimized-fit count resampling. Rows, top to bottom: pdz-cript-n, Lomax–Exponential; pdz-cript-n, Gamma–Exponential; pdz-cript-n, Lomax–Sigmoid. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

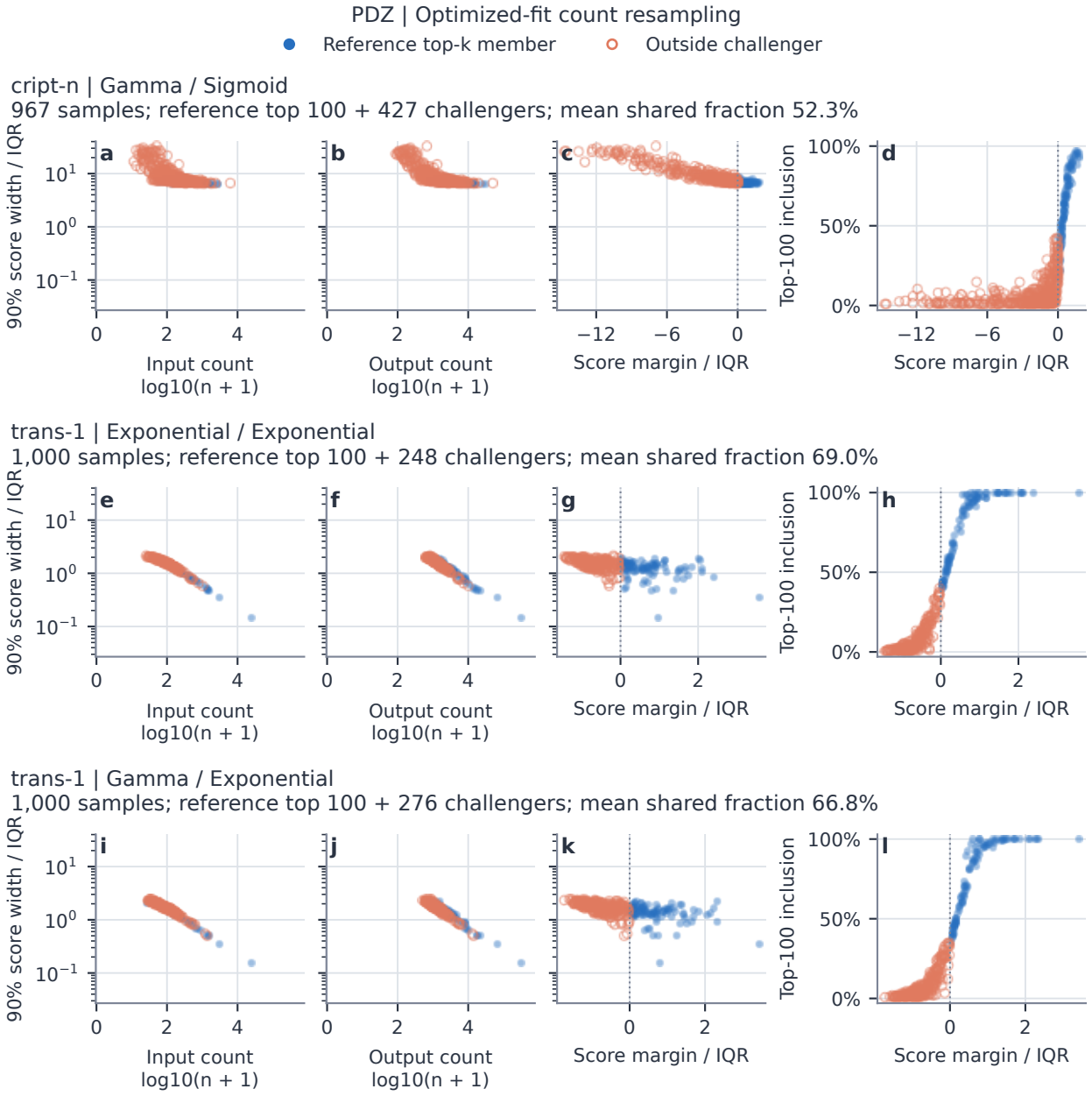


Figure S25: PDZ decision-cohort summaries: optimized-fit count resampling. Rows, top to bottom: pdz-cript-n, Gamma–Sigmoid; pdz-trans-1, Exponential–Exponential; pdz-trans-1, Gamma–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

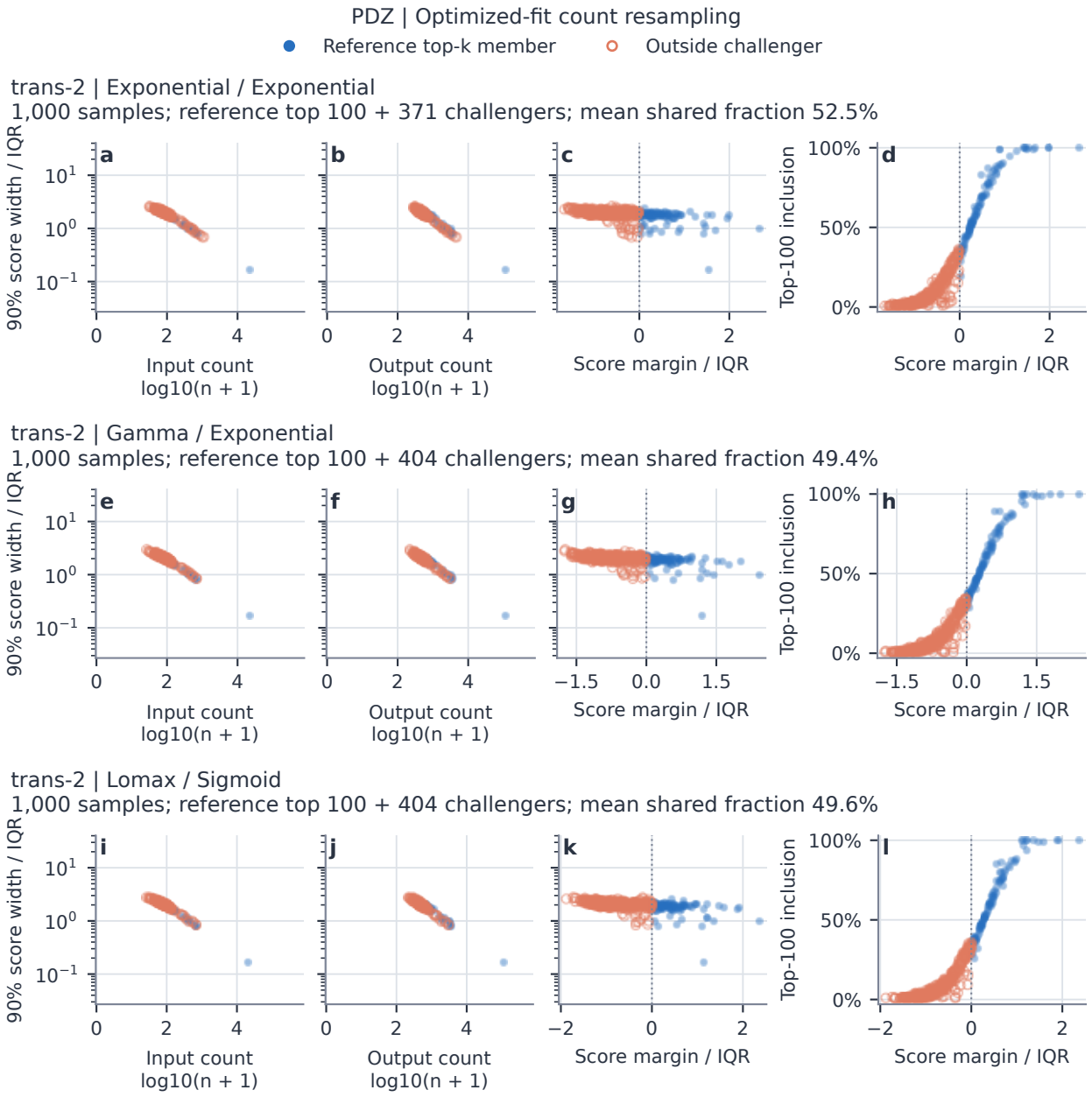


Figure S26: PDZ decision-cohort summaries: optimized-fit count resampling. Rows, top to bottom: pdz-trans-2, Exponential–Exponential; pdz-trans-2, Gamma–Exponential; pdz-trans-2, Lomax–Sigmoid. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

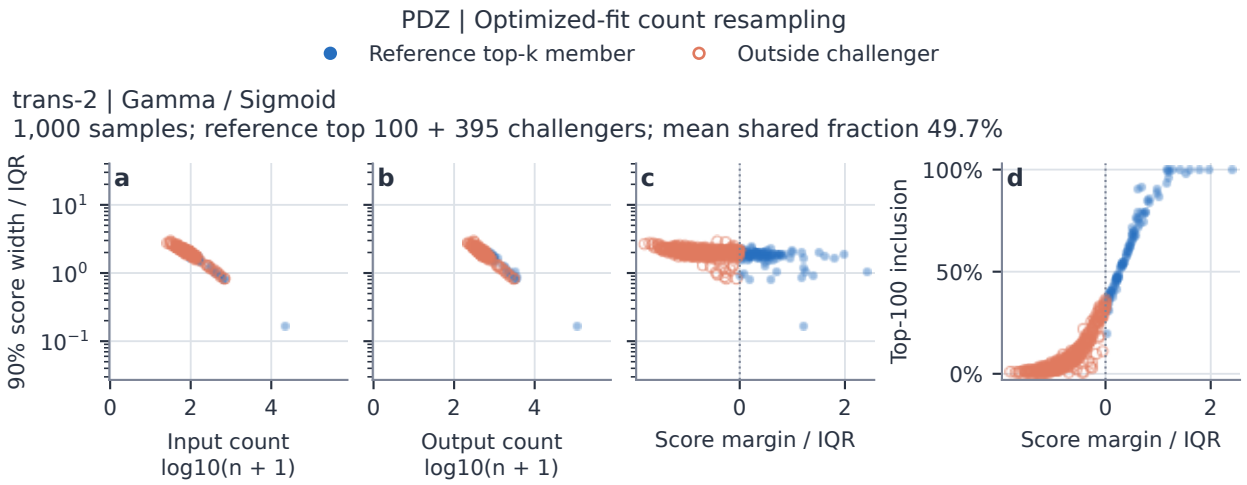


Figure S27: PDZ decision-cohort summaries: optimized-fit count resampling. Rows, top to bottom: pdz-trans-2, Gamma–Sigmoid. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

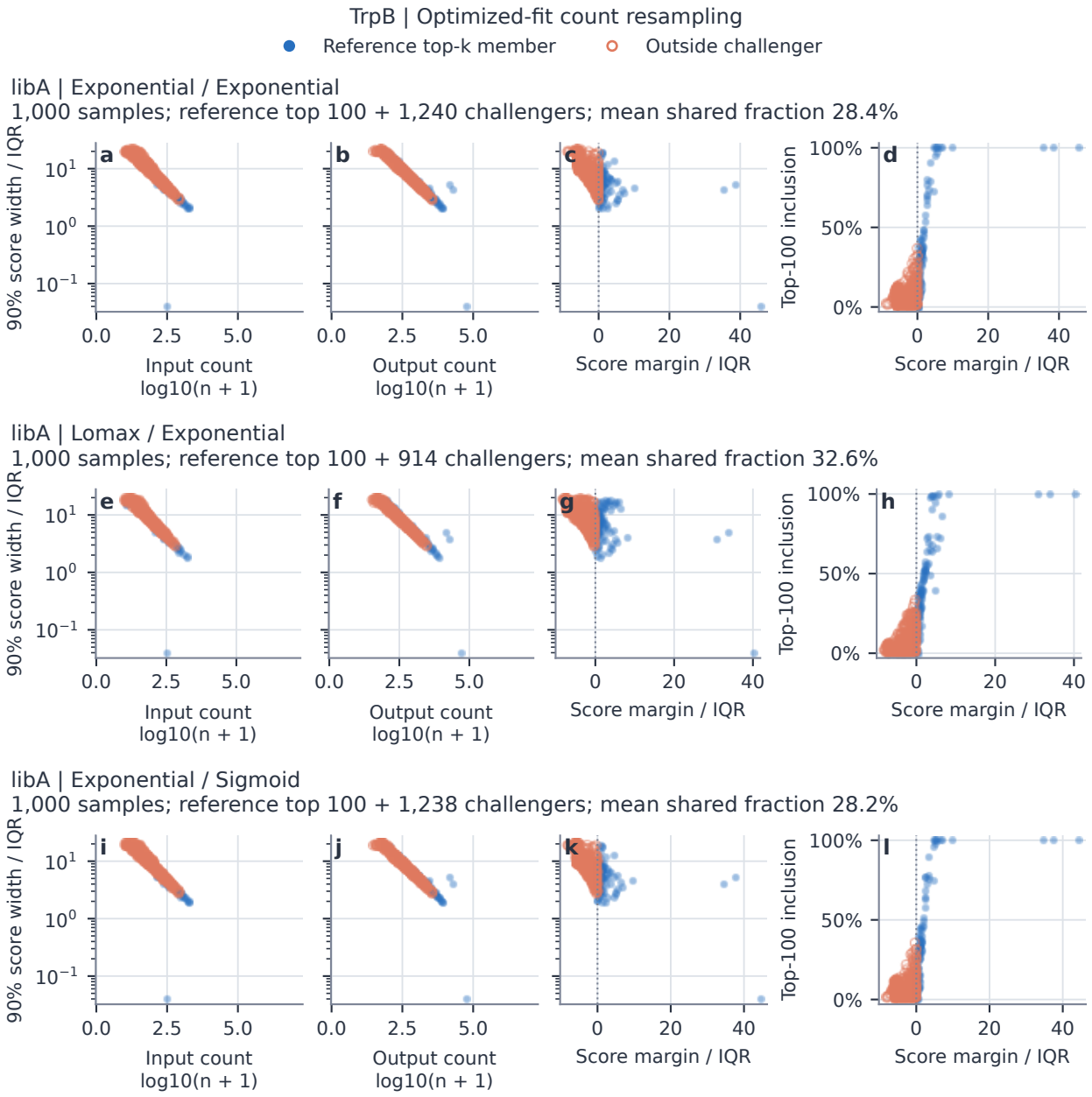


Figure S28: TrpB decision-cohort summaries: optimized-fit count resampling. Rows, top to bottom: trpb-libA, Exponential–Exponential; trpb-libA, Lomax–Exponential; trpb-libA, Exponential–Sigmoid. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

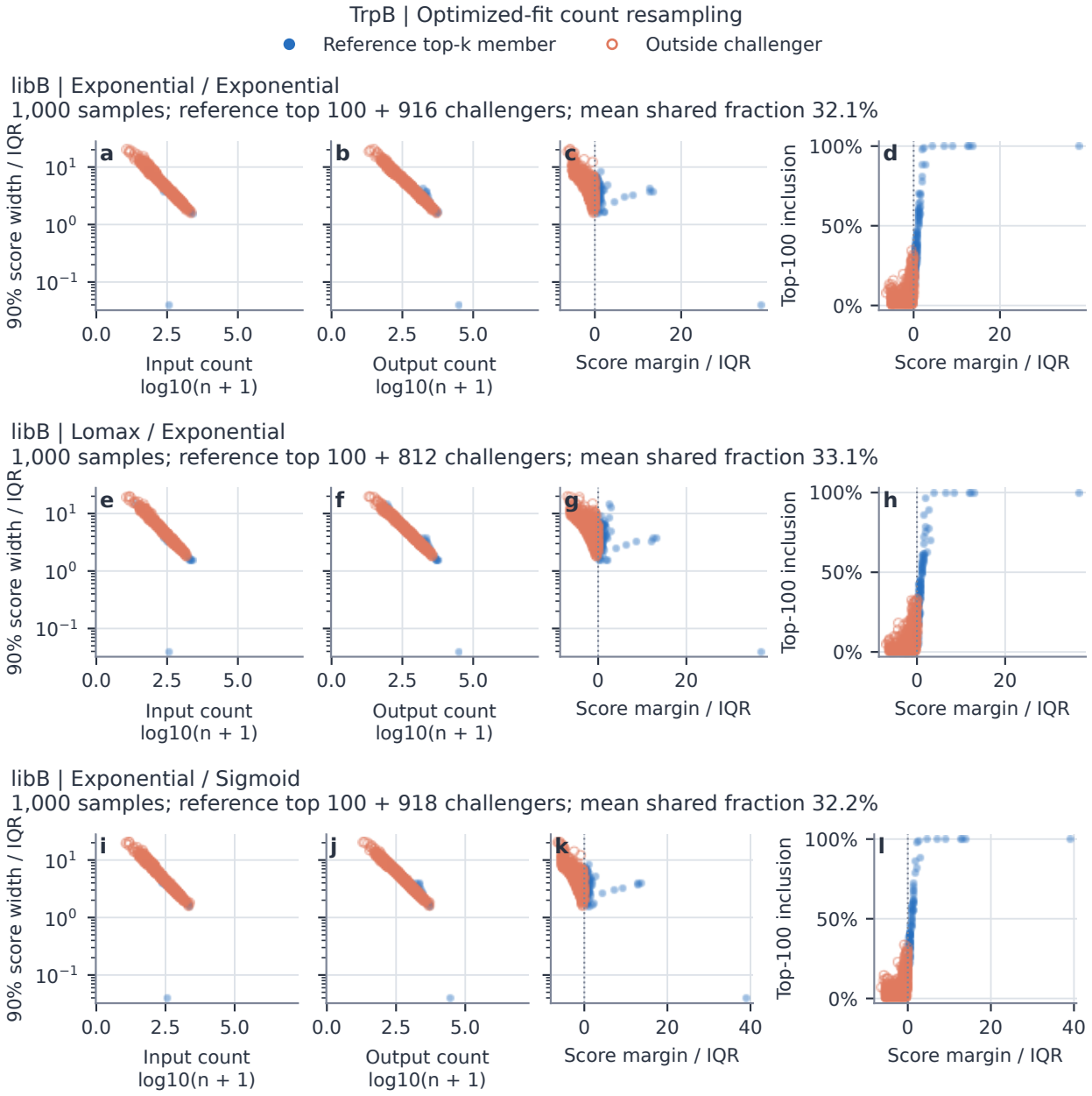


Figure S29: TrpB decision-cohort summaries: optimized-fit count resampling. Rows, top to bottom: trpb-libB, Exponential–Exponential; trpb-libB, Lomax–Exponential; trpb-libB, Exponential–Sigmoid. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

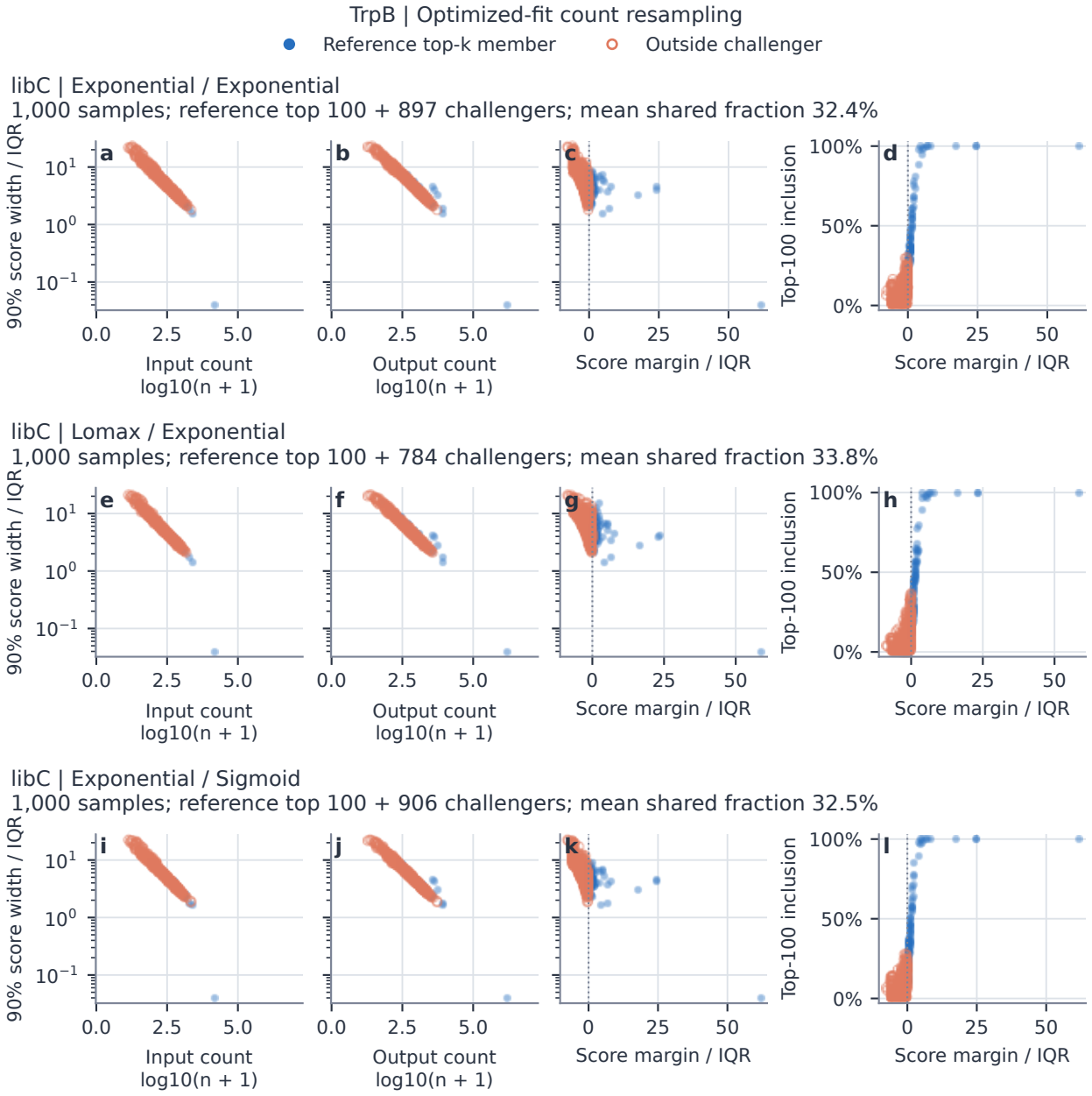


Figure S30: TrpB decision-cohort summaries: optimized-fit count resampling. Rows, top to bottom: trpb-libC, Exponential–Exponential; trpb-libC, Lomax–Exponential; trpb-libC, Exponential–Sigmoid. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

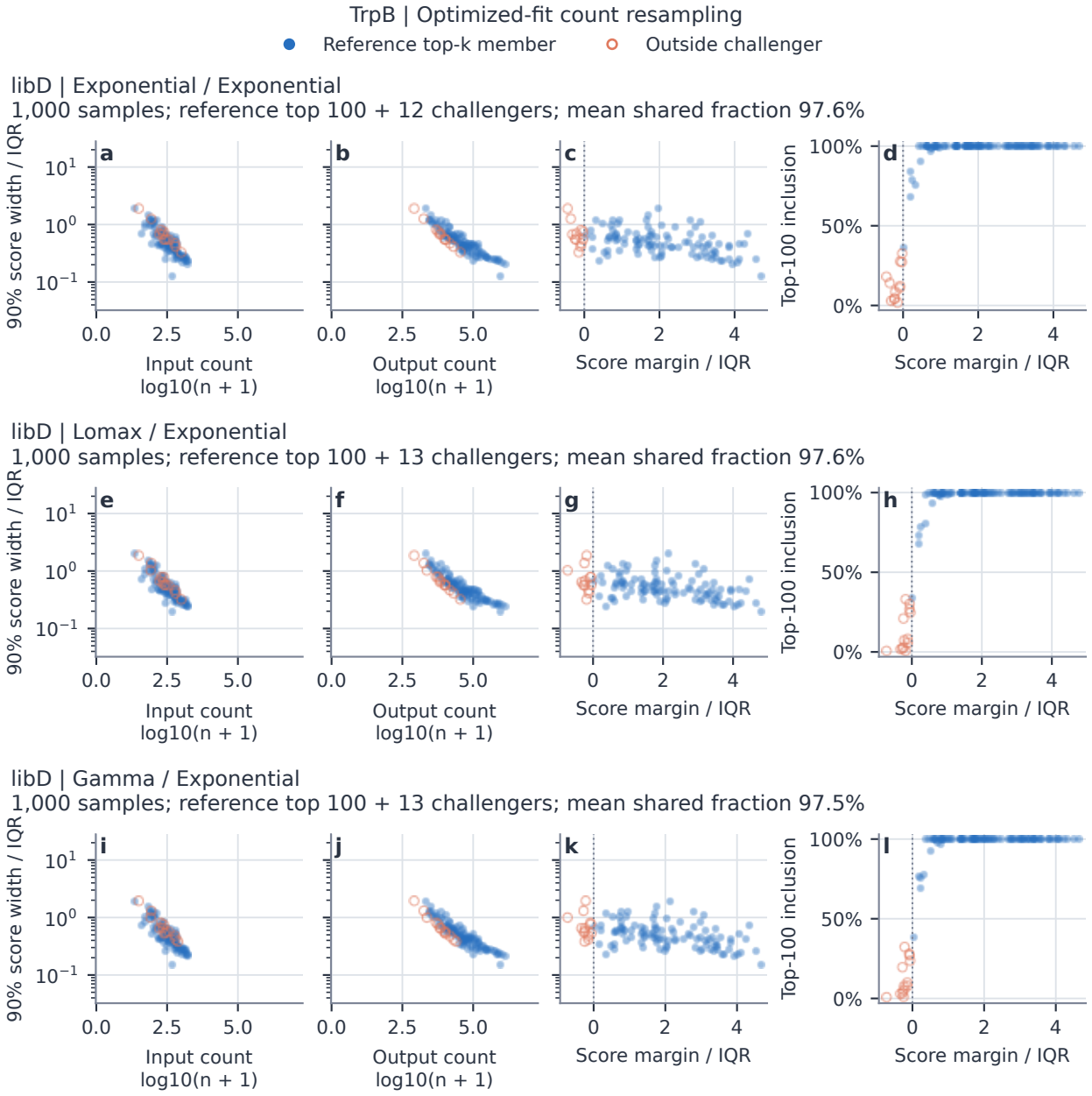


Figure S31: TrpB decision-cohort summaries: optimized-fit count resampling. Rows, top to bottom: trpb-libD, Exponential–Exponential; trpb-libD, Lomax–Exponential; trpb-libD, Gamma–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

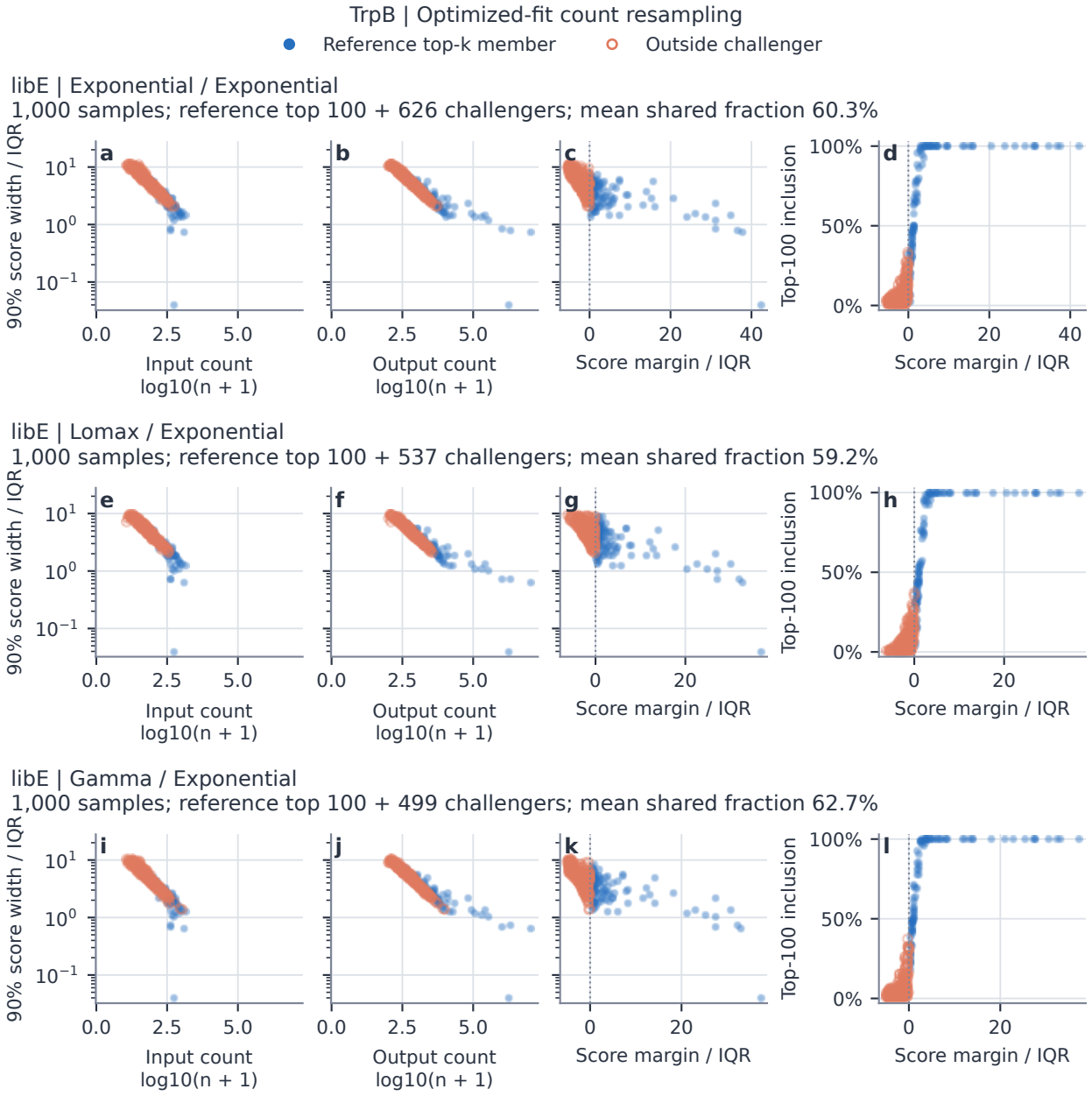


Figure S32: TrpB decision-cohort summaries: optimized-fit count resampling. Rows, top to bottom: trpb-libE, Exponential–Exponential; trpb-libE, Lomax–Exponential; trpb-libE, Gamma–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

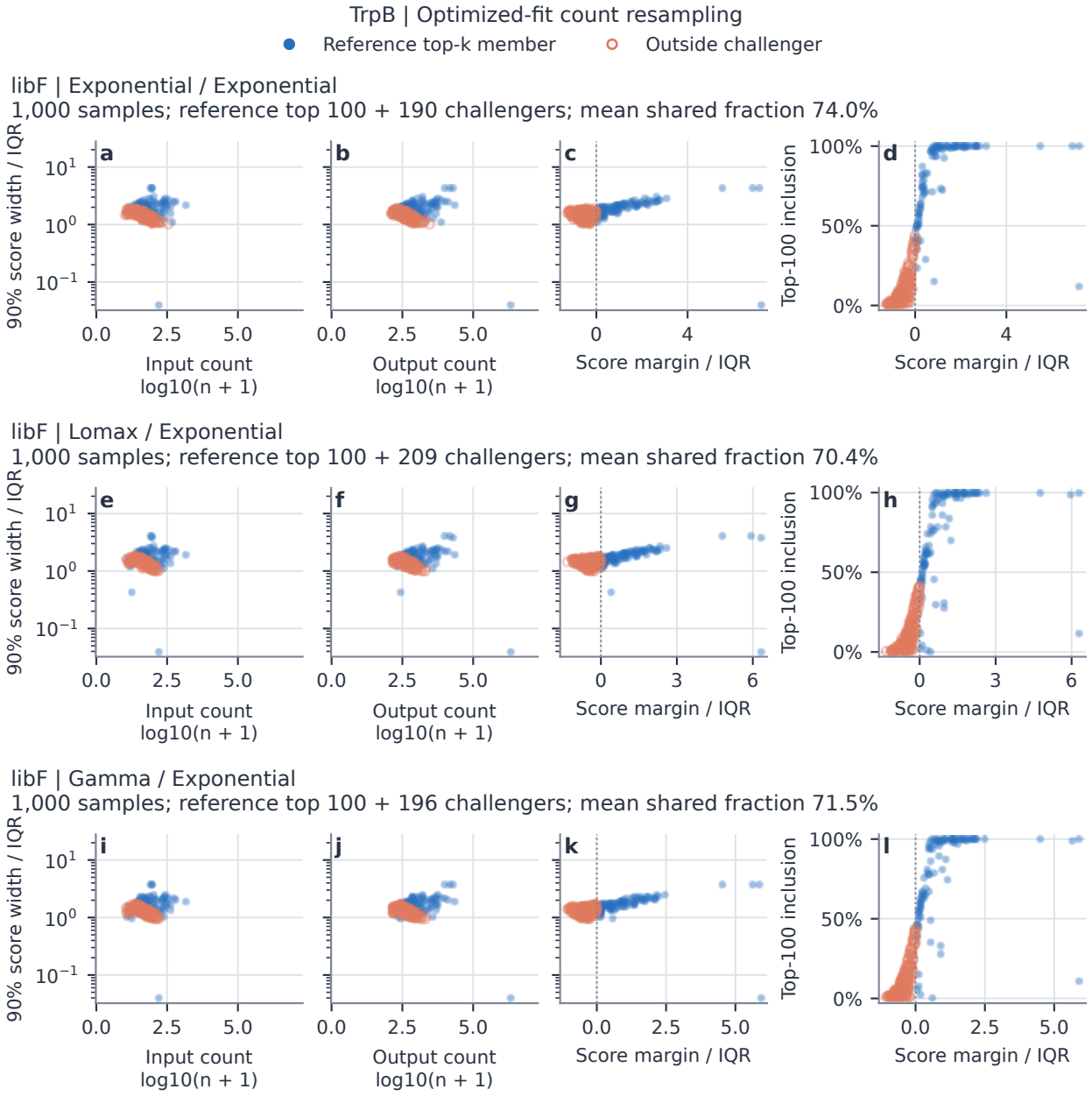


Figure S33: TrpB decision-cohort summaries: optimized-fit count resampling. Rows, top to bottom: trpb-libF, Exponential–Exponential; trpb-libF, Lomax–Exponential; trpb-libF, Gamma–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

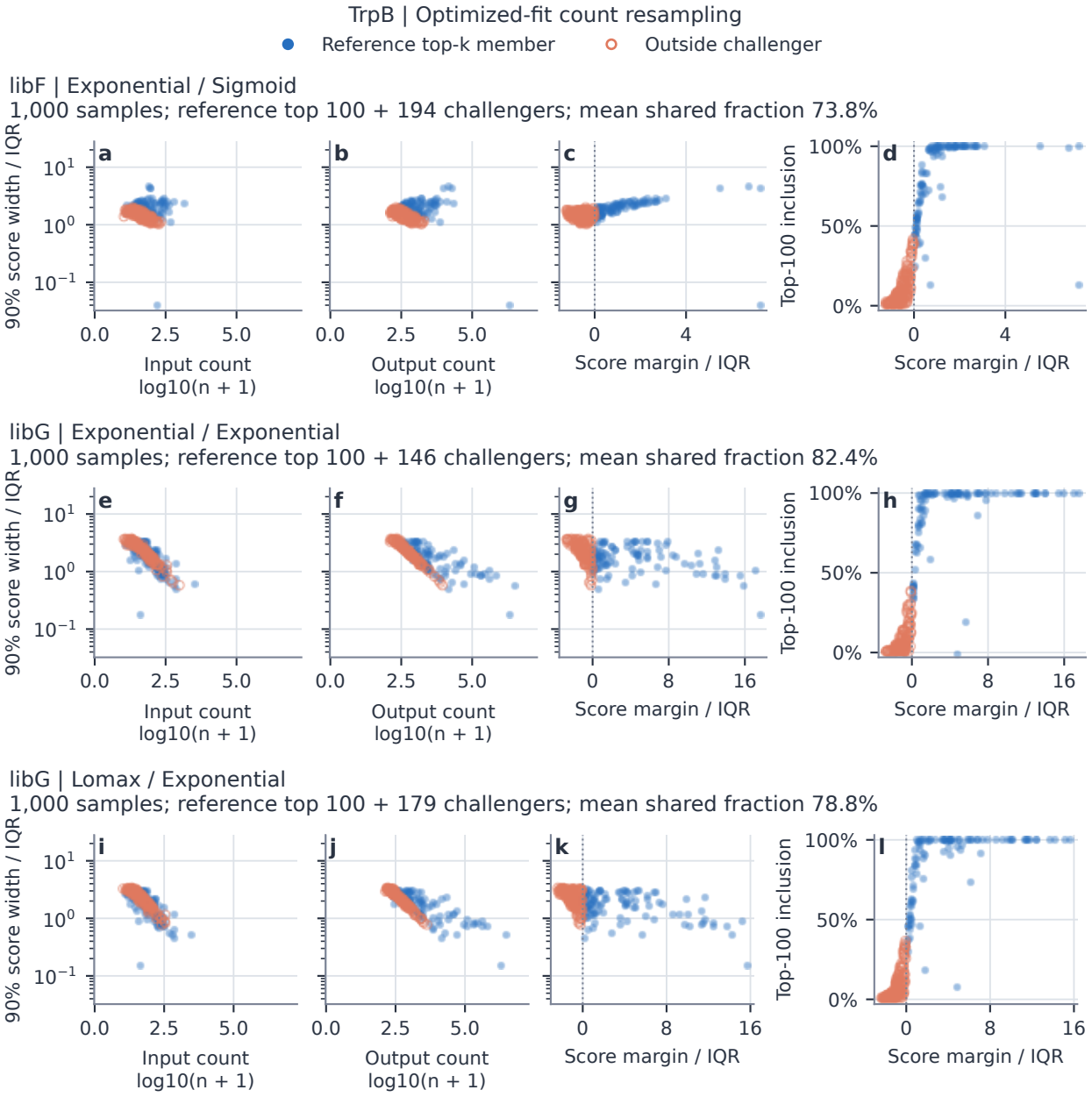


Figure S34: TrpB decision-cohort summaries: optimized-fit count resampling. Rows, top to bottom: trpb-libF, Exponential–Sigmoid; trpb-libG, Exponential–Exponential; trpb-libG, Lomax–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

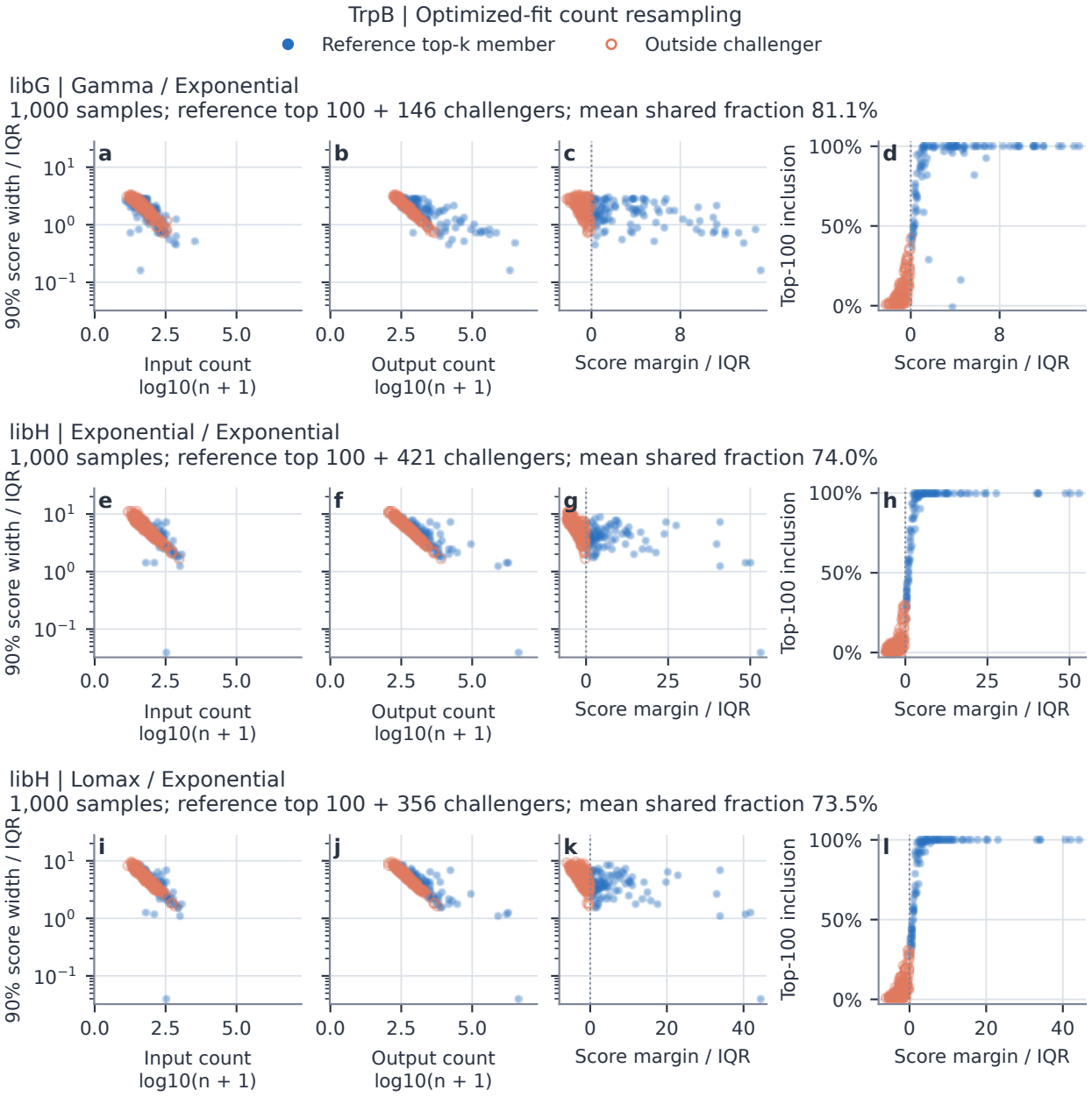


Figure S35: TrpB decision-cohort summaries: optimized-fit count resampling. Rows, top to bottom: trpb-libG, Gamma–Exponential; trpb-libH, Exponential–Exponential; trpb-libH, Lomax–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

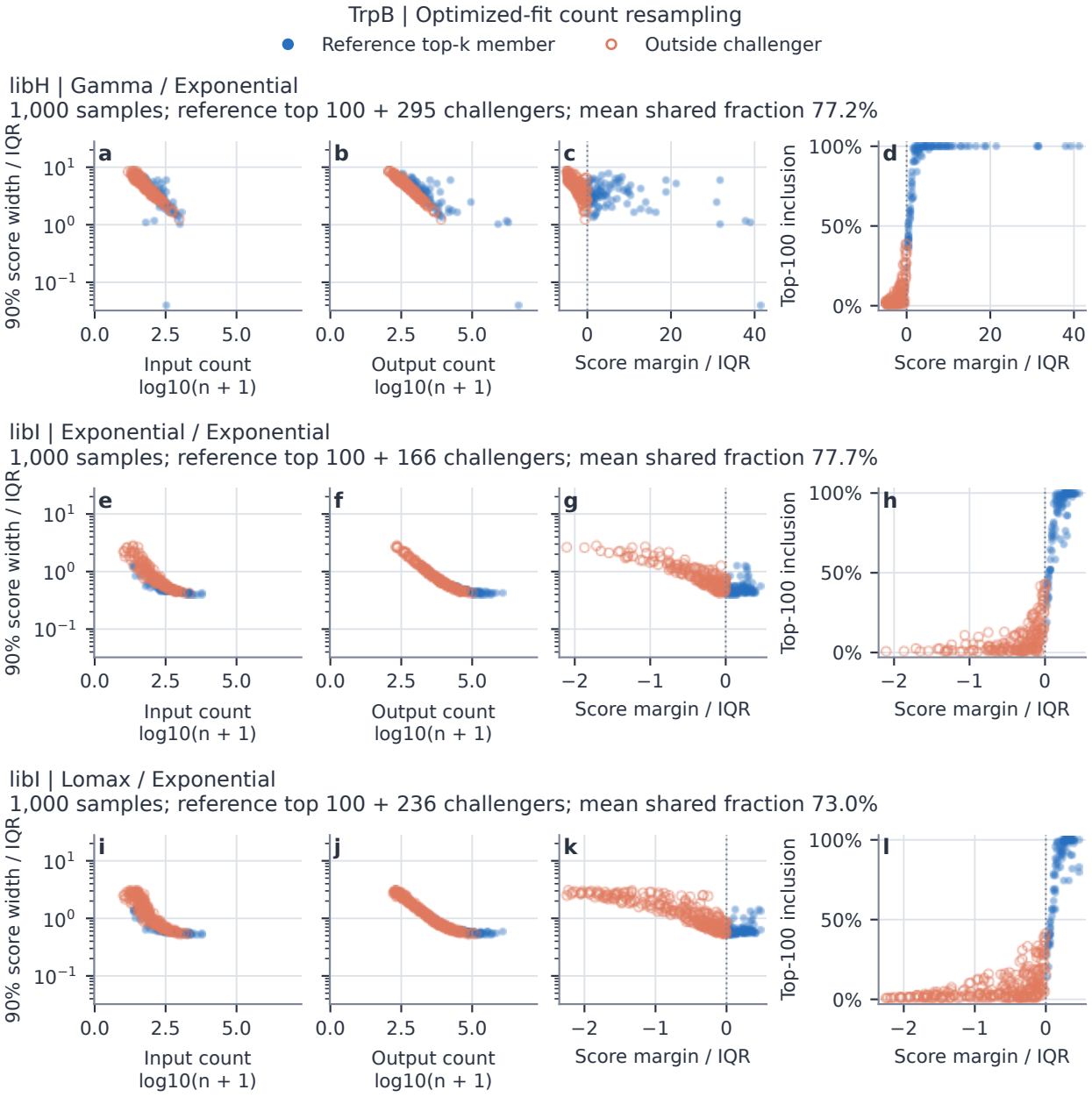


Figure S36: TrpB decision-cohort summaries: optimized-fit count resampling. Rows, top to bottom: trpb-libH, Gamma–Exponential; trpb-libI, Exponential–Exponential; trpb-libI, Lomax–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

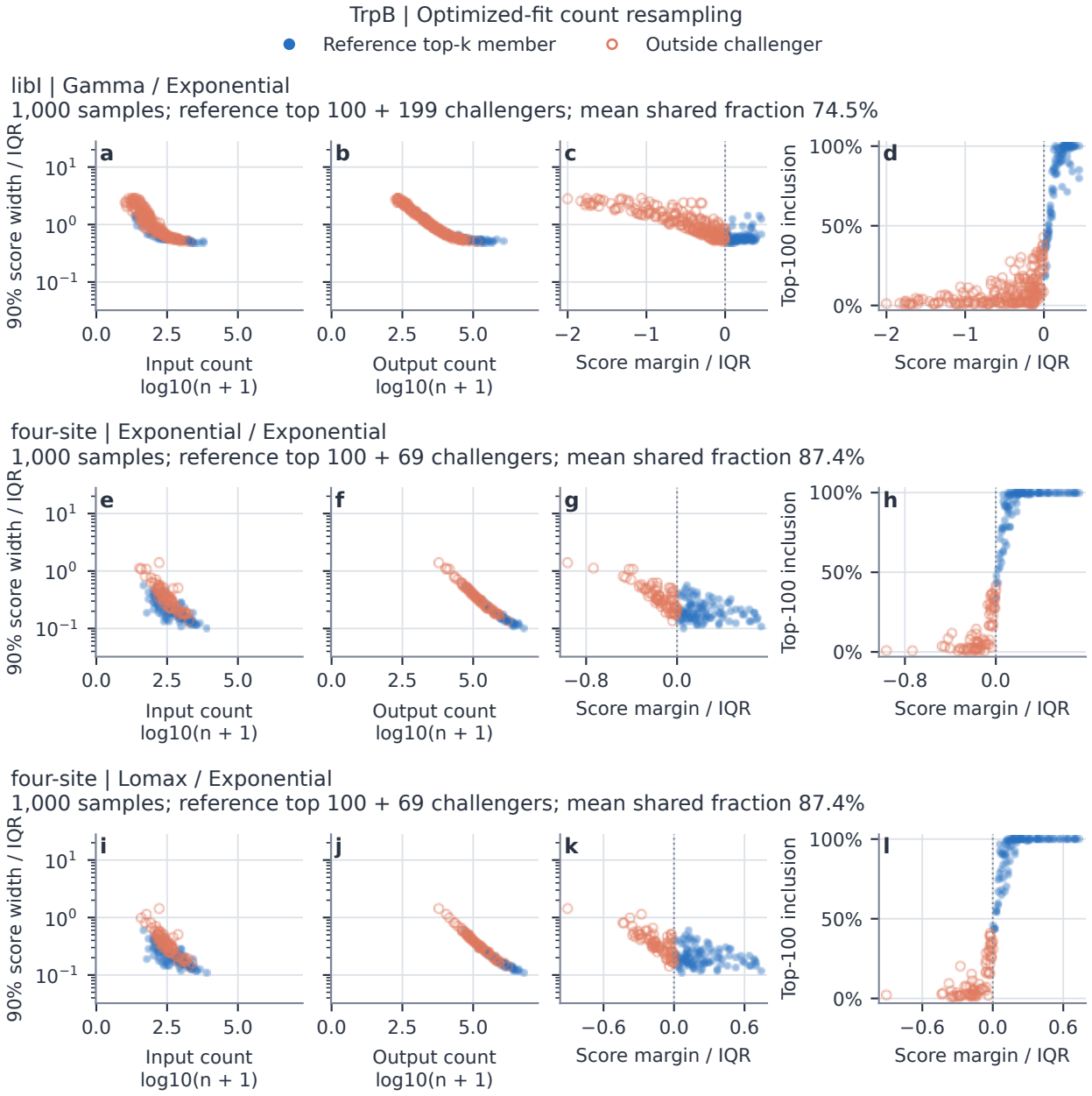


Figure S37: TrpB decision-cohort summaries: optimized-fit count resampling. Rows, top to bottom: trpb-libI, Gamma–Exponential; trpb-four-site, Exponential–Exponential; trpb-four-site, Lomax–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

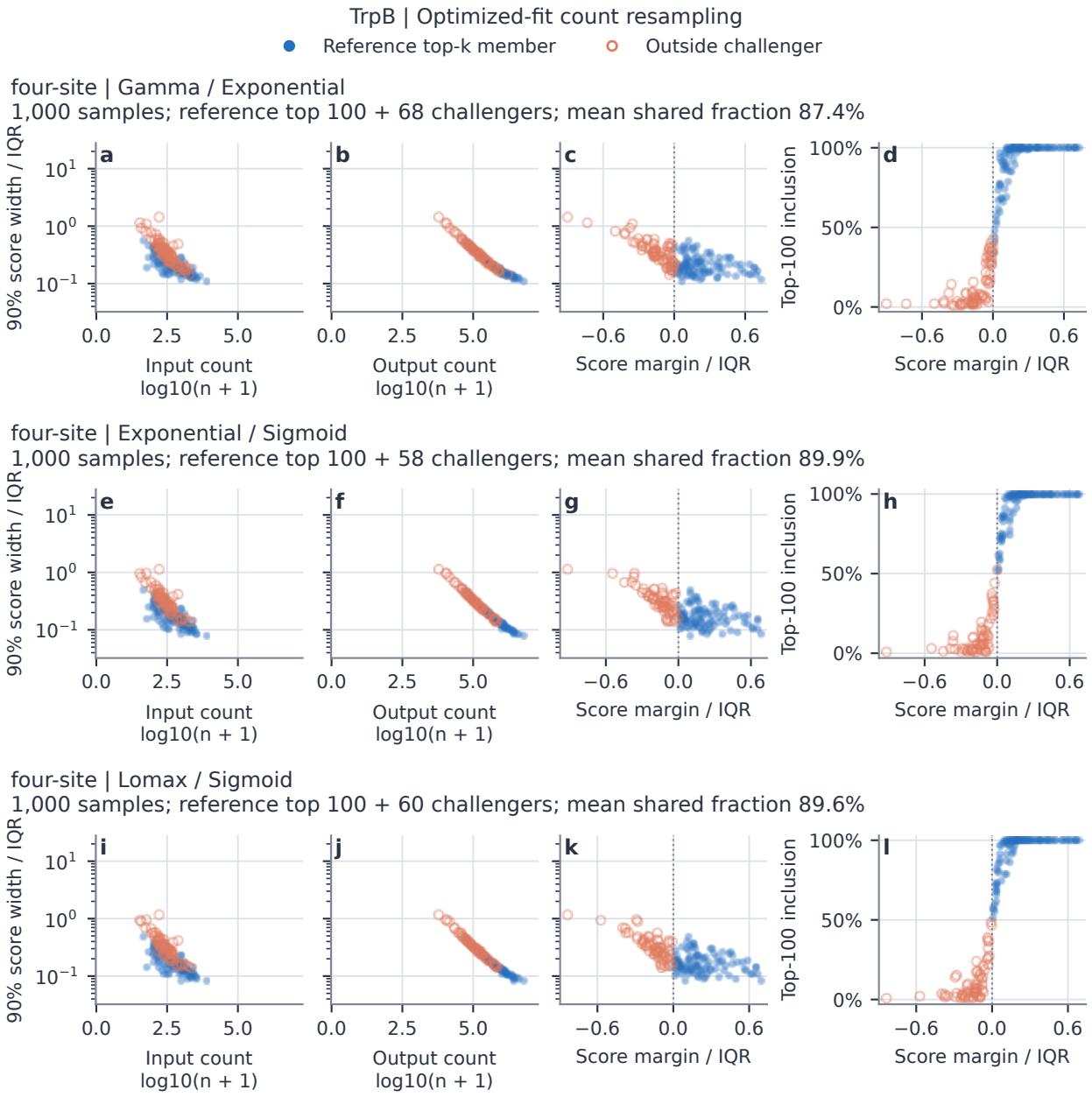


Figure S38: TrpB decision-cohort summaries: optimized-fit count resampling. Rows, top to bottom: trpb-four-site, Gamma–Exponential; trpb-four-site, Exponential–Sigmoid; trpb-four-site, Lomax–Sigmoid. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

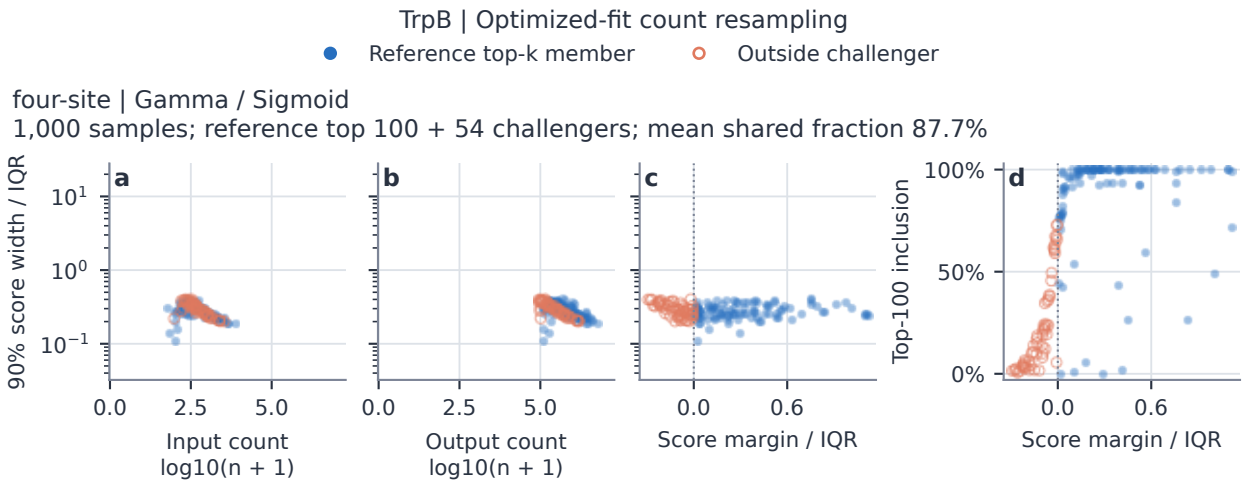


Figure S39: TrpB decision-cohort summaries: optimized-fit count resampling. Rows, top to bottom: trpb-four-site, Gamma–Sigmoid. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

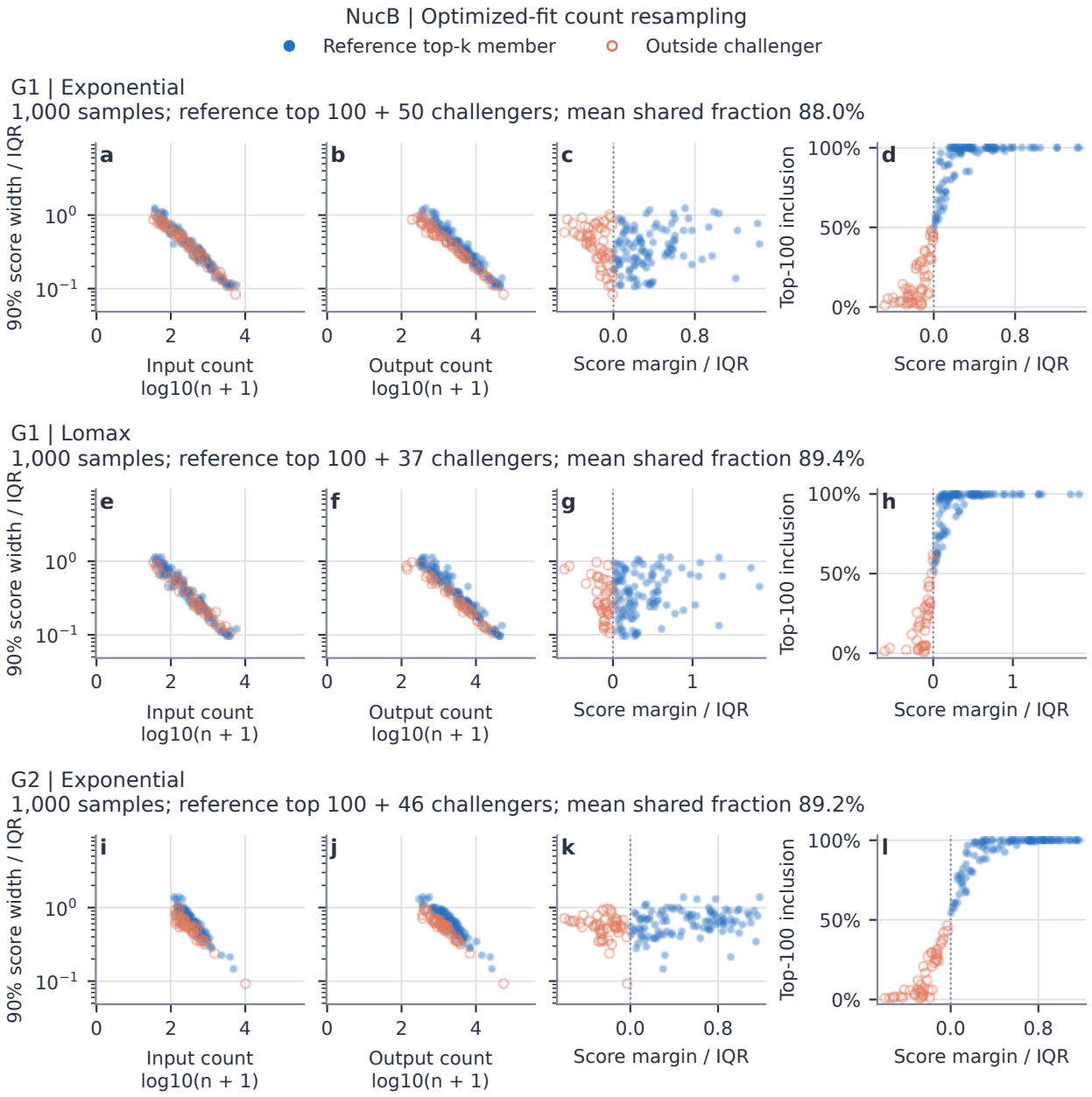


Figure S40: NucB decision-cohort summaries: optimized-fit count resampling. Rows, top to bottom: nuc-G1, Exponential; nuc-G1, Lomax; nuc-G2, Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

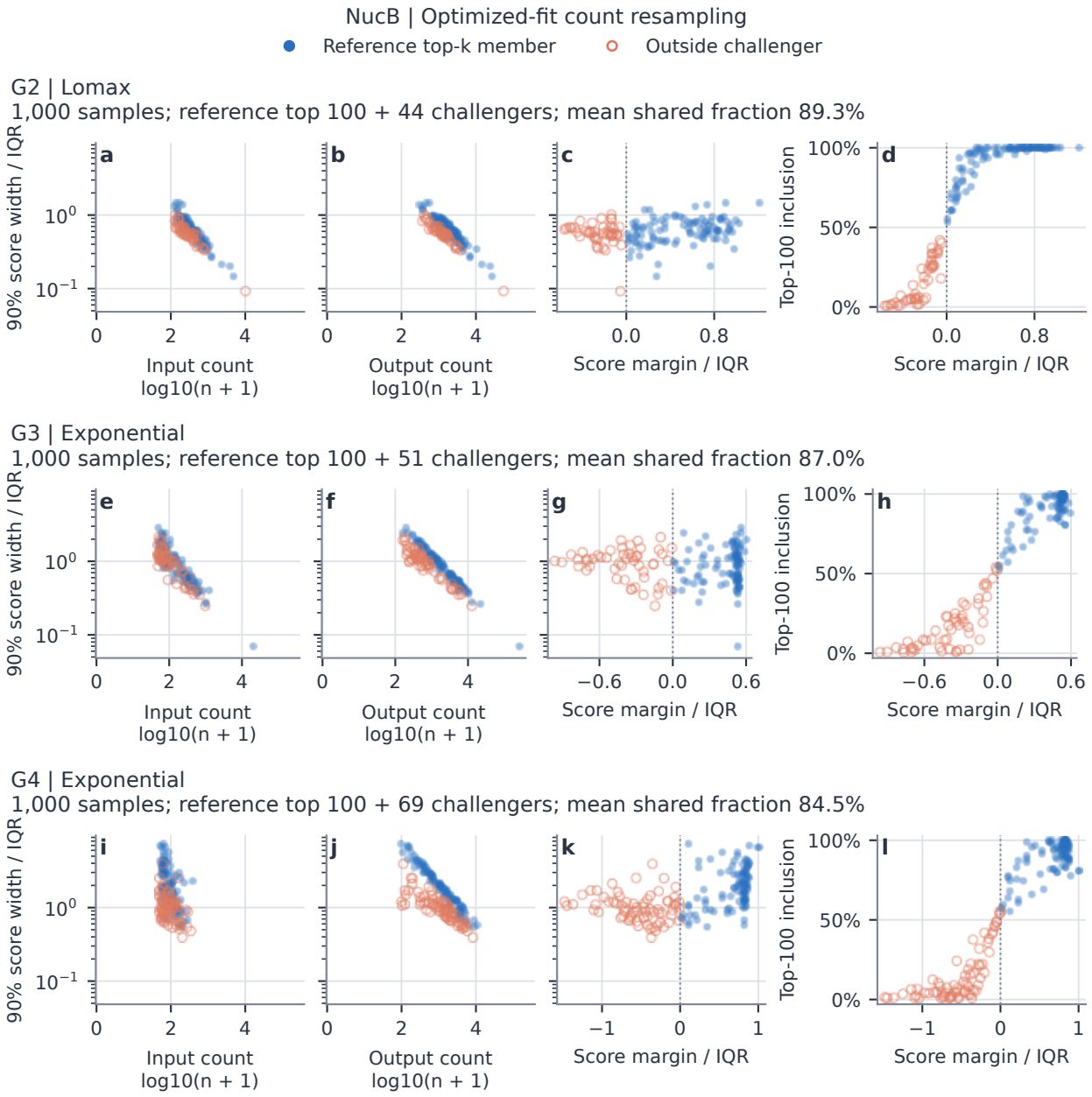


Figure S41: NucB decision-cohort summaries: optimized-fit count resampling. Rows, top to bottom: nuc-G2, Lomax; nuc-G3, Exponential; nuc-G4, Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

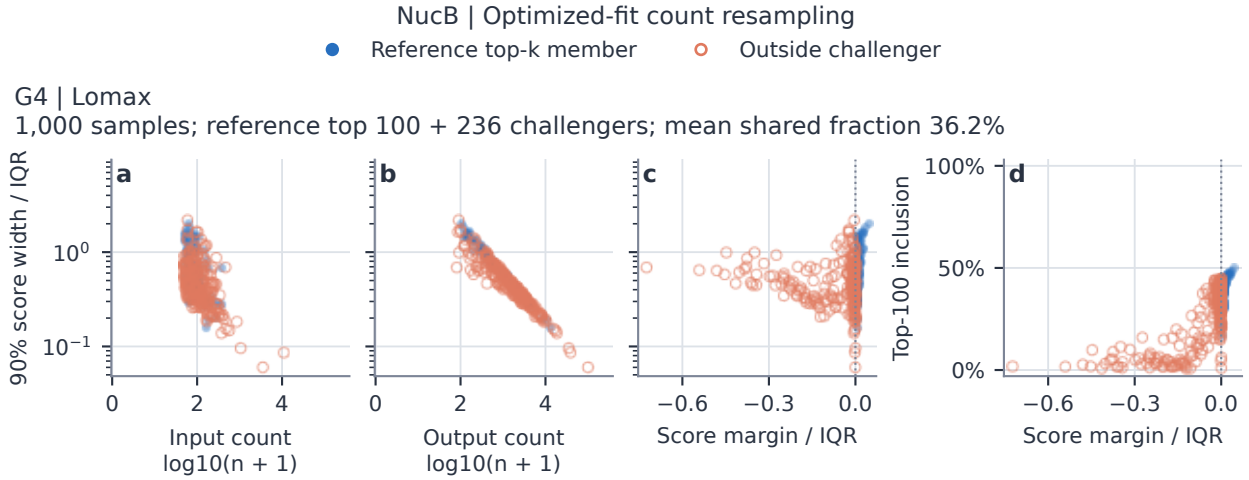


Figure S42: NucB decision-cohort summaries: optimized-fit count resampling. Rows, top to bottom: nuc-G4, Lomax. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

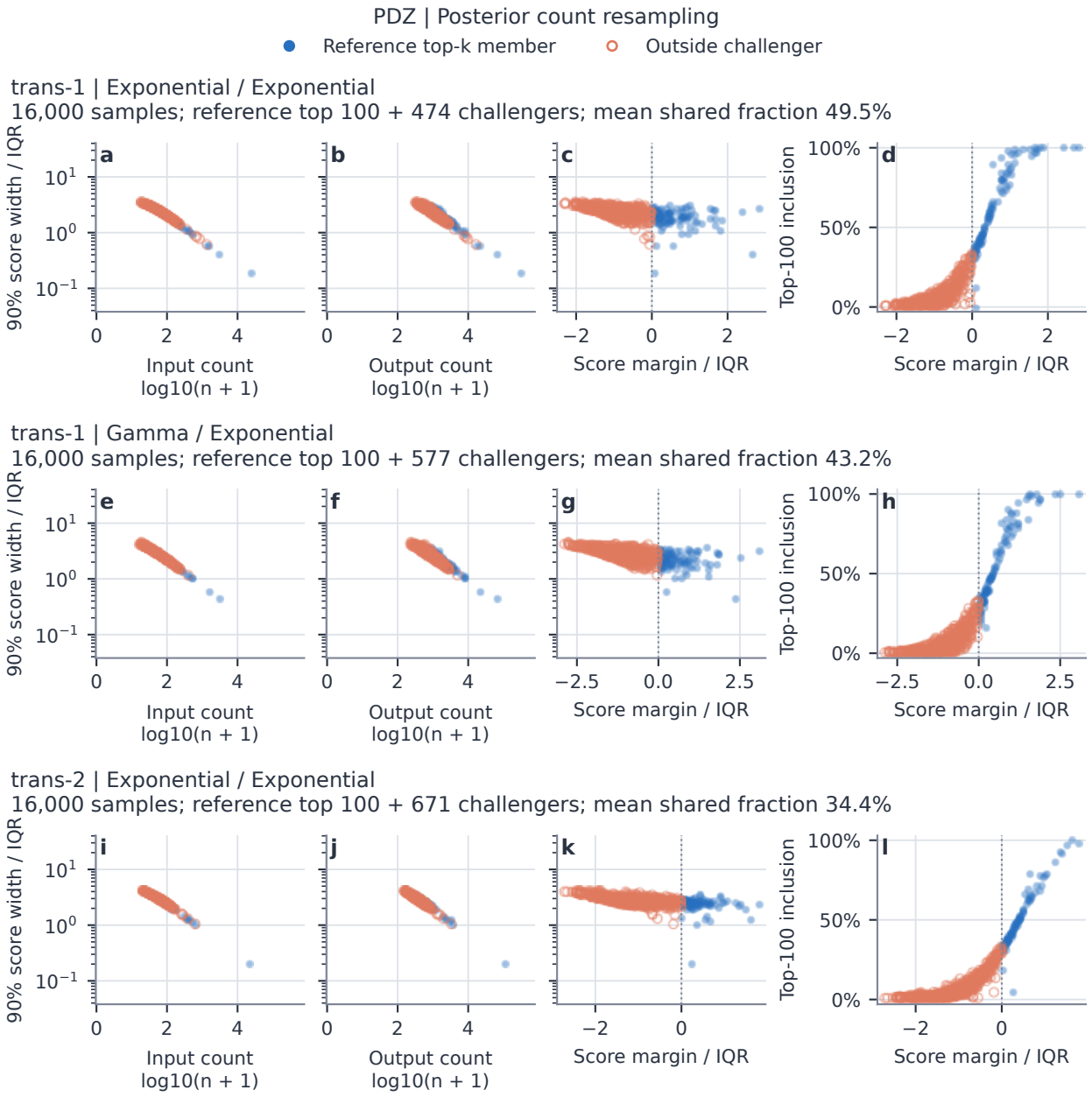


Figure S43: PDZ decision-cohort summaries: posterior count resampling. Rows, top to bottom: pdz-trans-1, Exponential–Exponential; pdz-trans-1, Gamma–Exponential; pdz-trans-2, Exponential–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

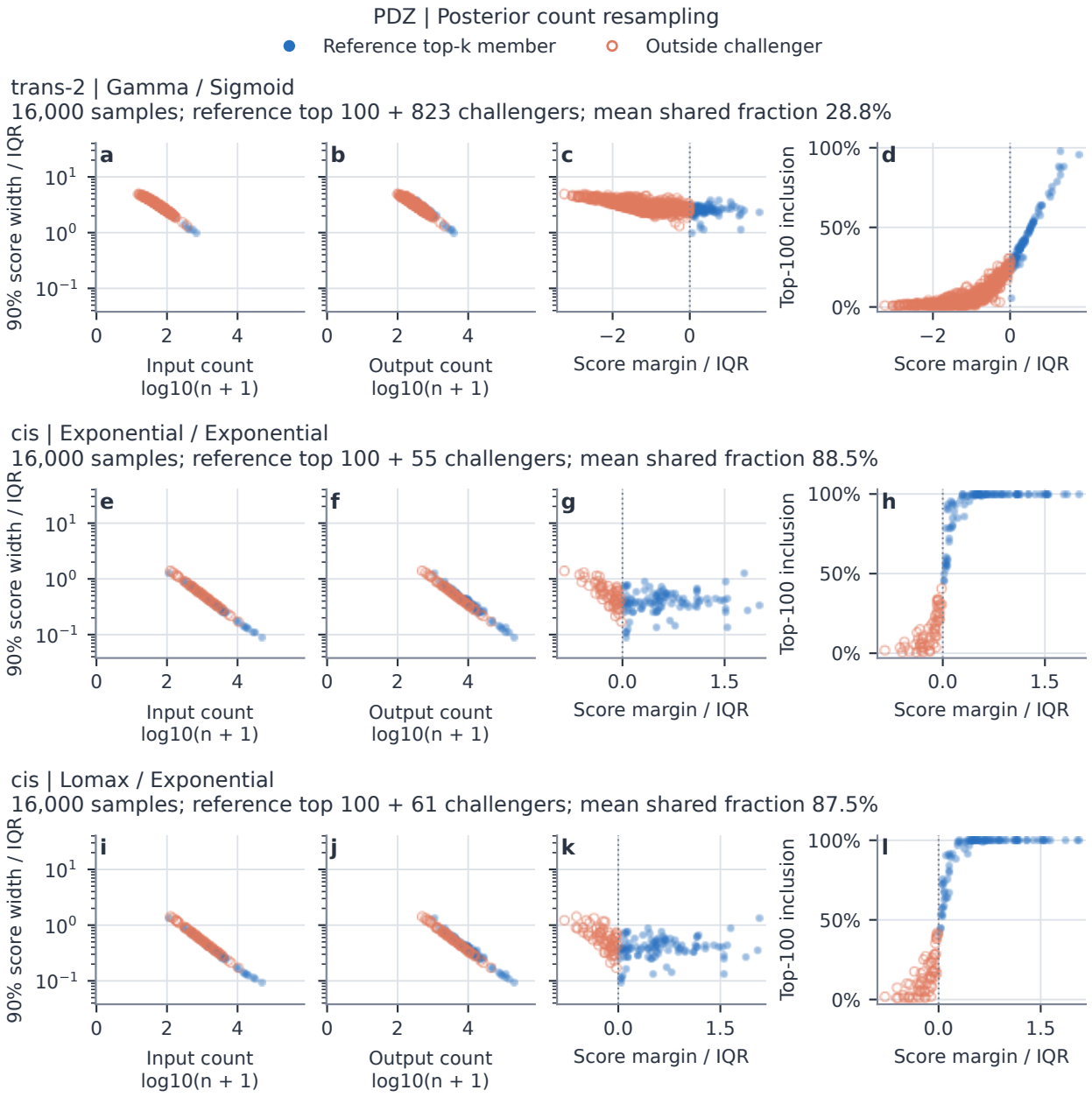


Figure S44: PDZ decision-cohort summaries: posterior count resampling. Rows, top to bottom: pdz-trans-2, Gamma–Sigmoid; pdz-cis, Exponential–Exponential; pdz-cis, Lomax–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

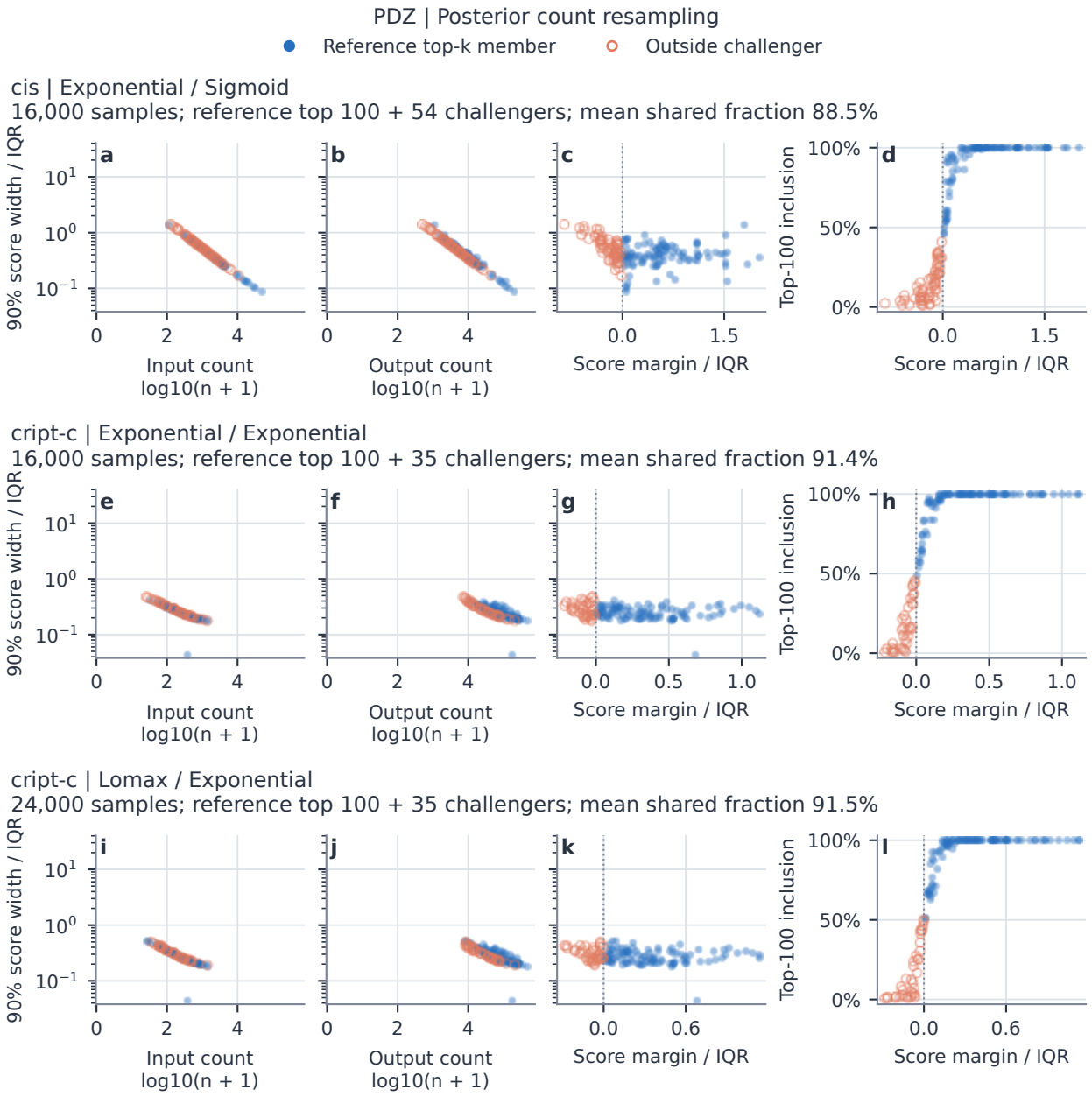


Figure S45: PDZ decision-cohort summaries: posterior count resampling. Rows, top to bottom: pdz-cis, Exponential–Sigmoid; pdz-cript-c, Exponential–Exponential; pdz-cript-c, Lomax–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

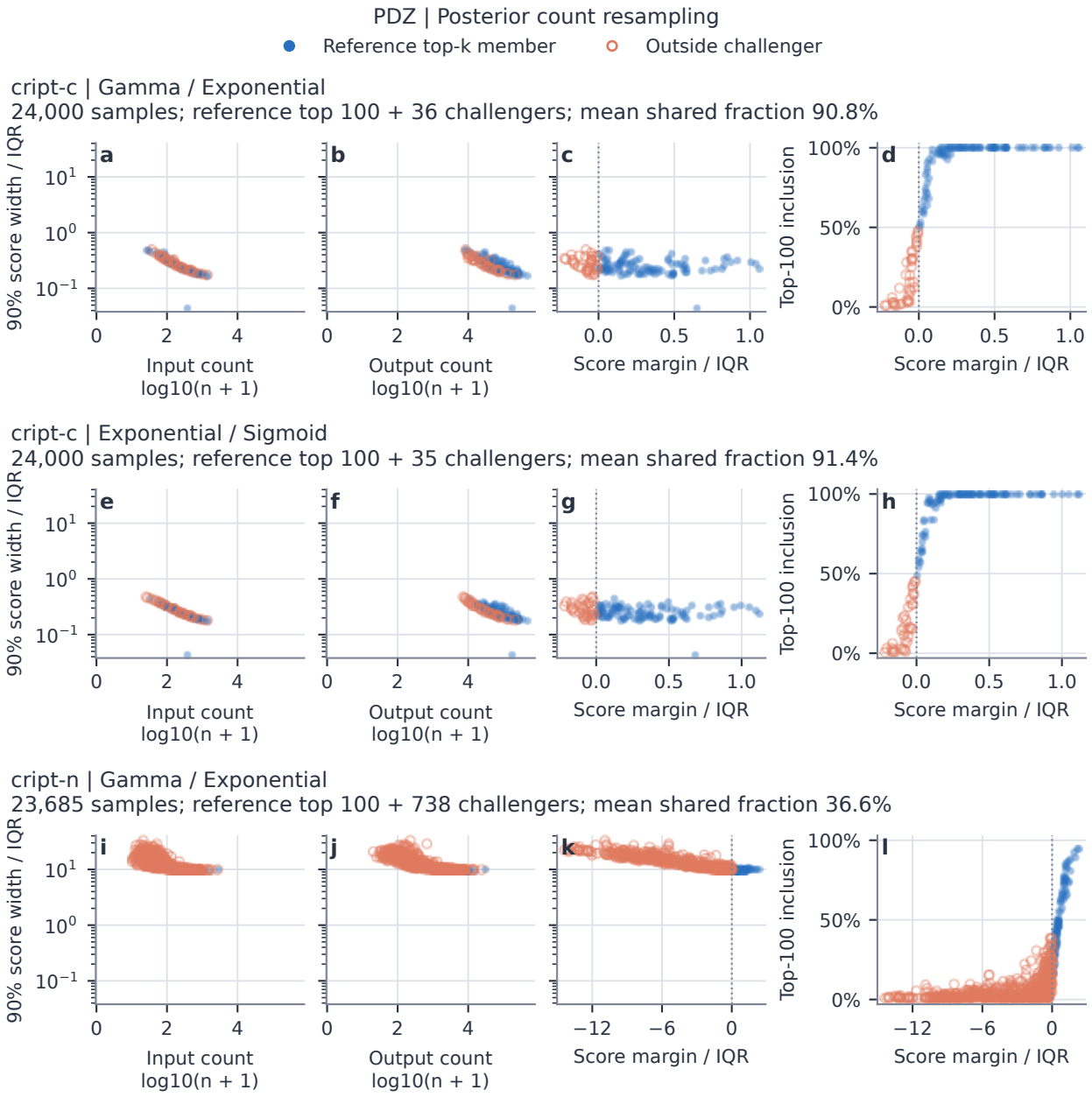


Figure S46: PDZ decision-cohort summaries: posterior count resampling. Rows, top to bottom: pdz-cript-c, Gamma–Exponential; pdz-cript-c, Exponential–Sigmoid; pdz-cript-n, Gamma–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

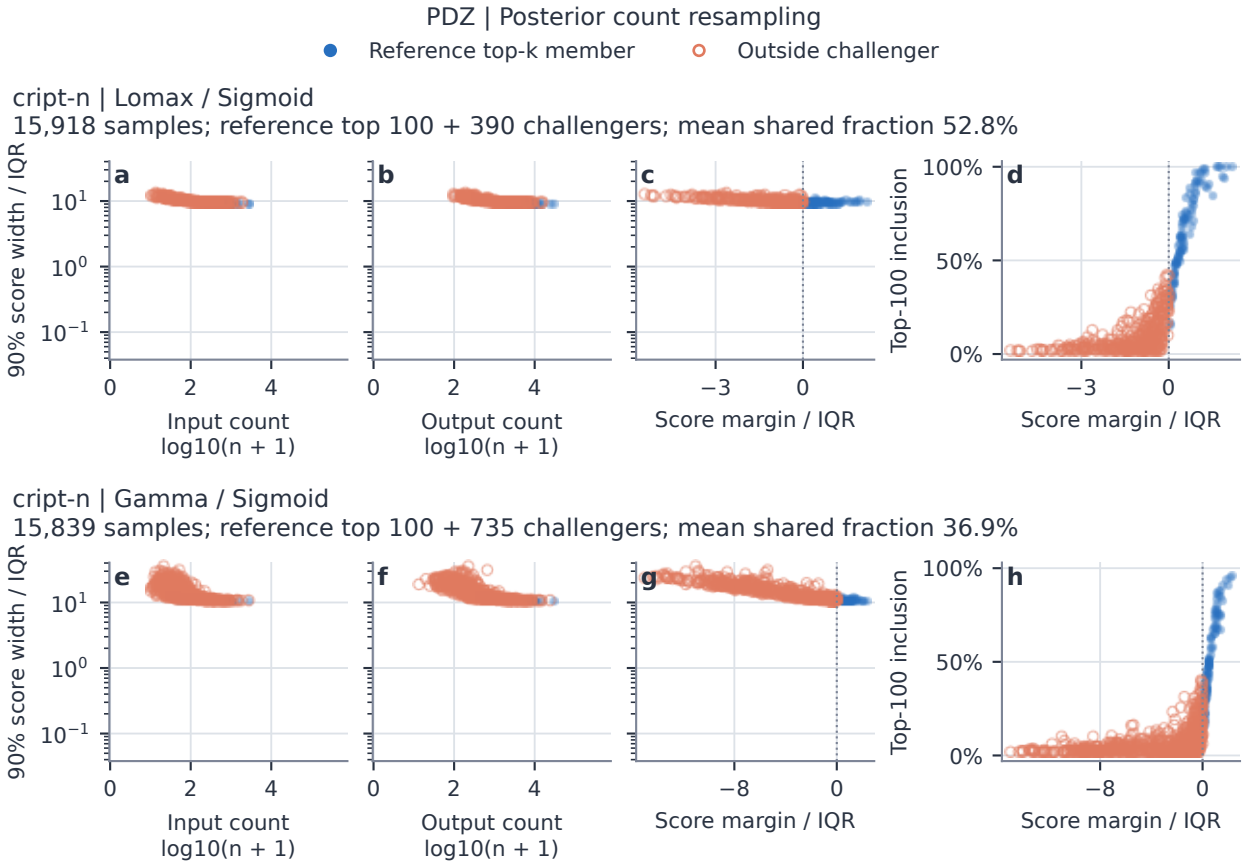


Figure S47: PDZ decision-cohort summaries: posterior count resampling. Rows, top to bottom: pdz-cript-n, Lomax–Sigmoid; pdz-cript-n, Gamma–Sigmoid. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

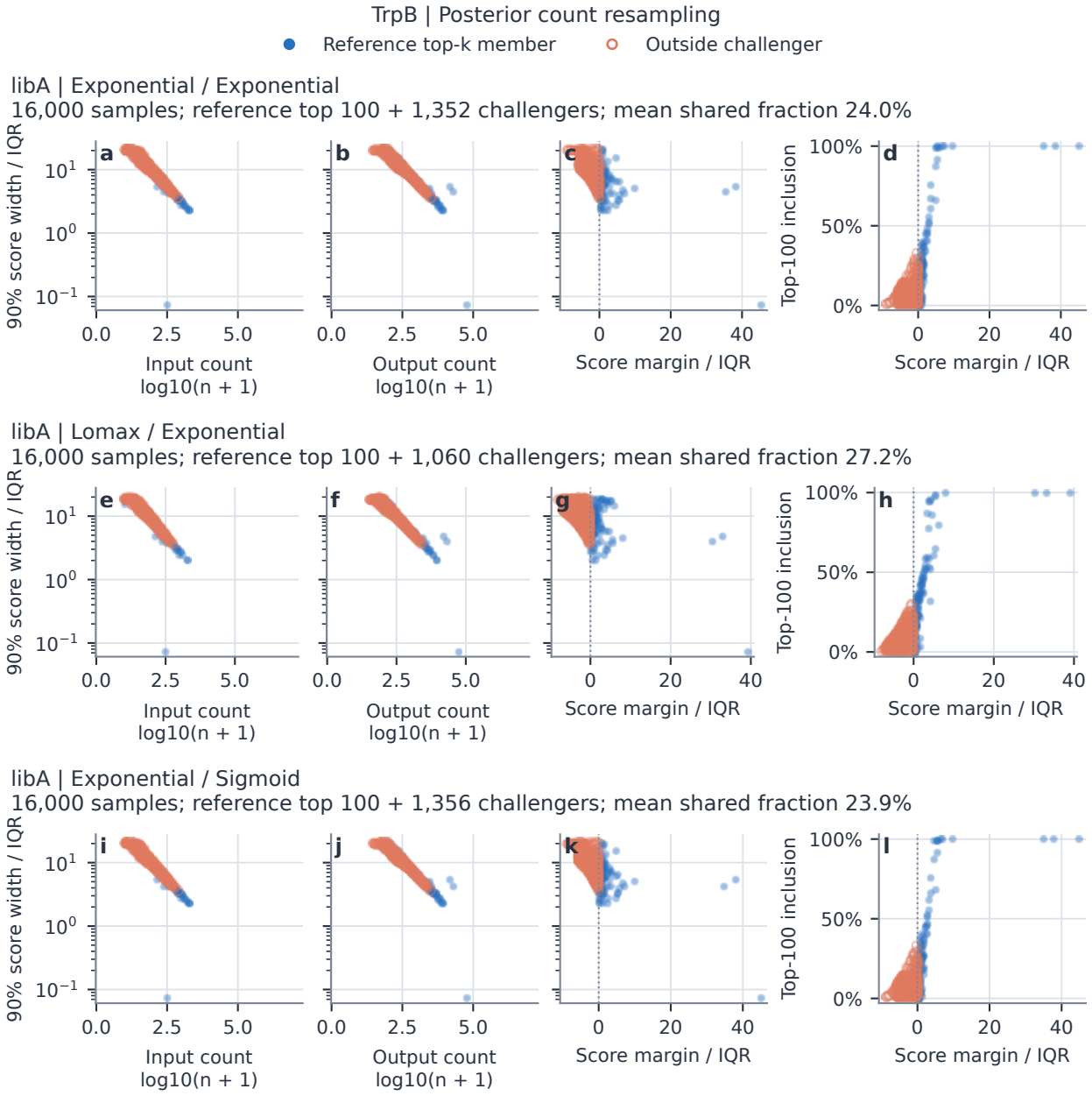


Figure S48: TrpB decision-cohort summaries: posterior count resampling. Rows, top to bottom: trpb-libA, Exponential–Exponential; trpb-libA, Lomax–Exponential; trpb-libA, Exponential–Sigmoid. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

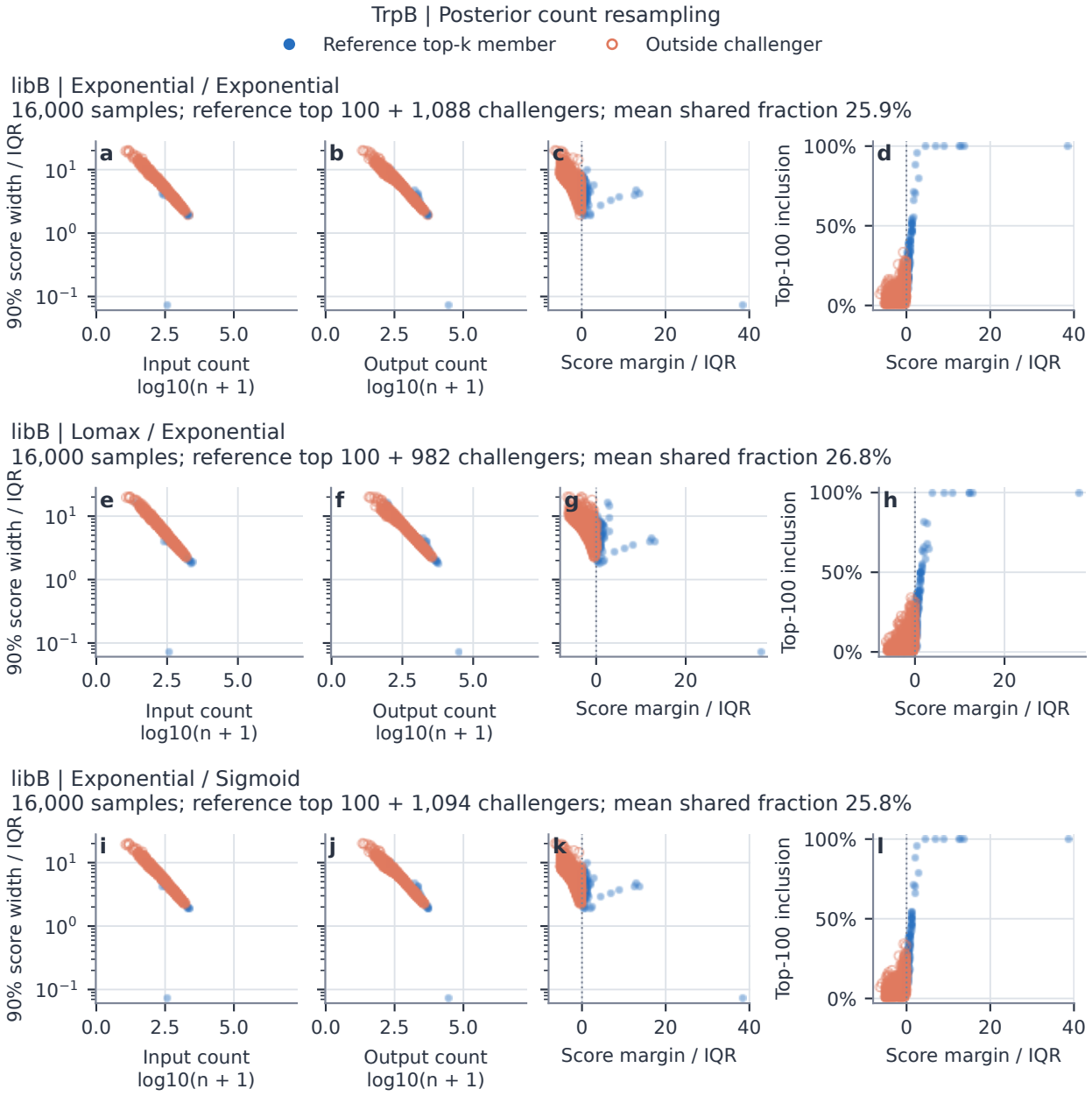


Figure S49: TrpB decision-cohort summaries: posterior count resampling. Rows, top to bottom: trpb-libB, Exponential–Exponential; trpb-libB, Lomax–Exponential; trpb-libB, Exponential–Sigmoid. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

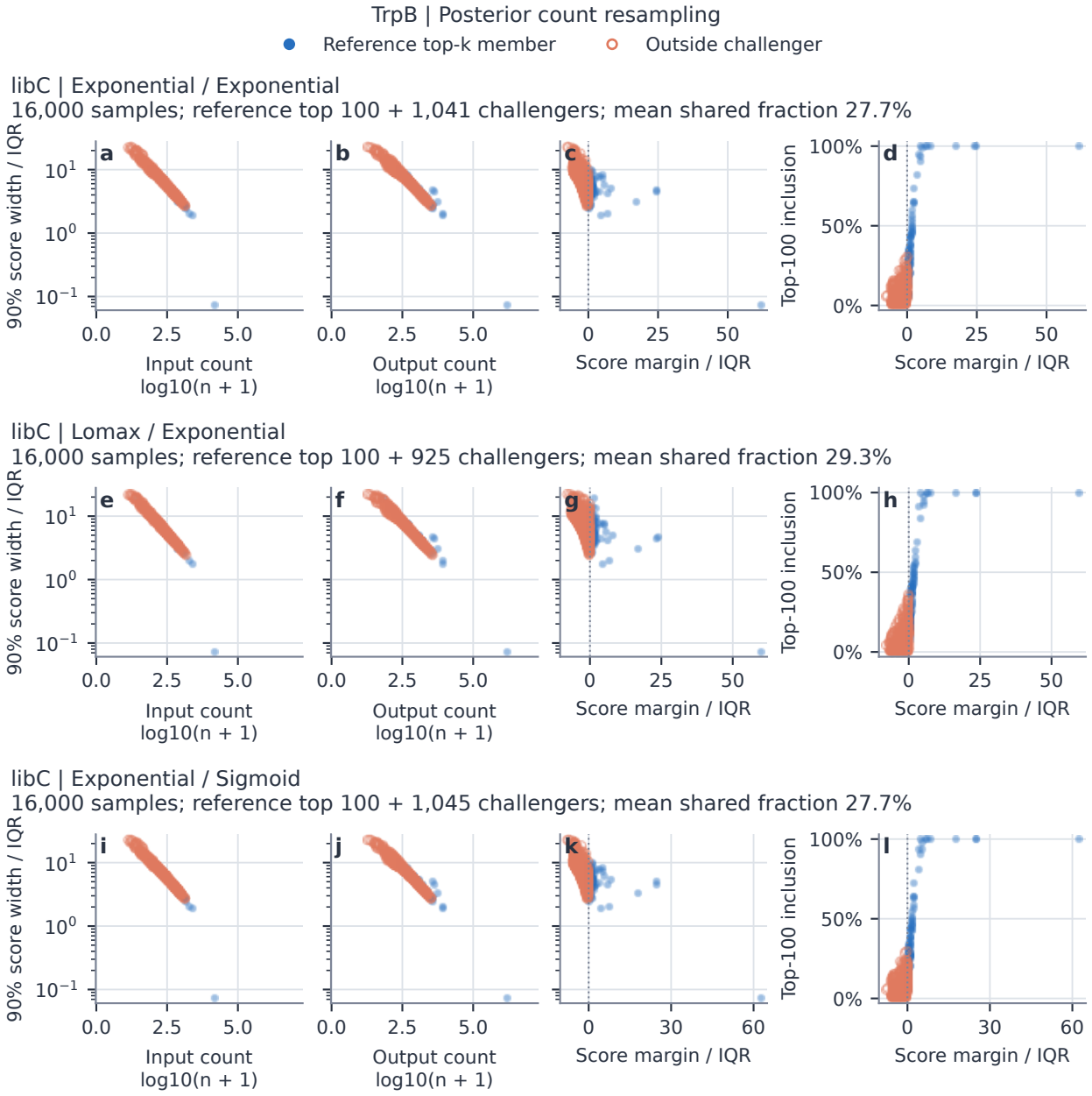


Figure S50: TrpB decision-cohort summaries: posterior count resampling. Rows, top to bottom: trpb-libC, Exponential–Exponential; trpb-libC, Lomax–Exponential; trpb-libC, Exponential–Sigmoid. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

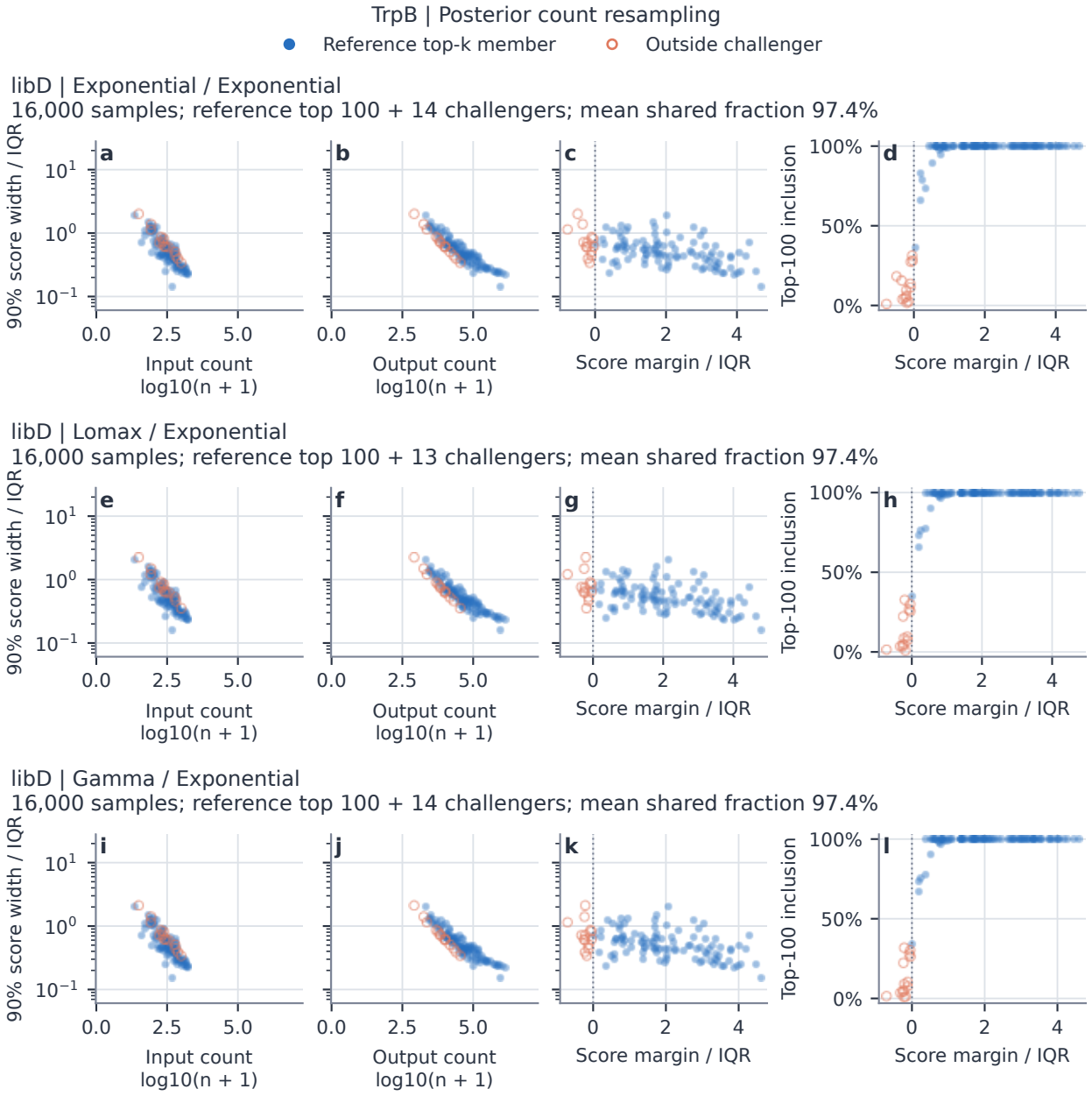


Figure S51: TrpB decision-cohort summaries: posterior count resampling. Rows, top to bottom: trpb-libD, Exponential–Exponential; trpb-libD, Lomax–Exponential; trpb-libD, Gamma–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

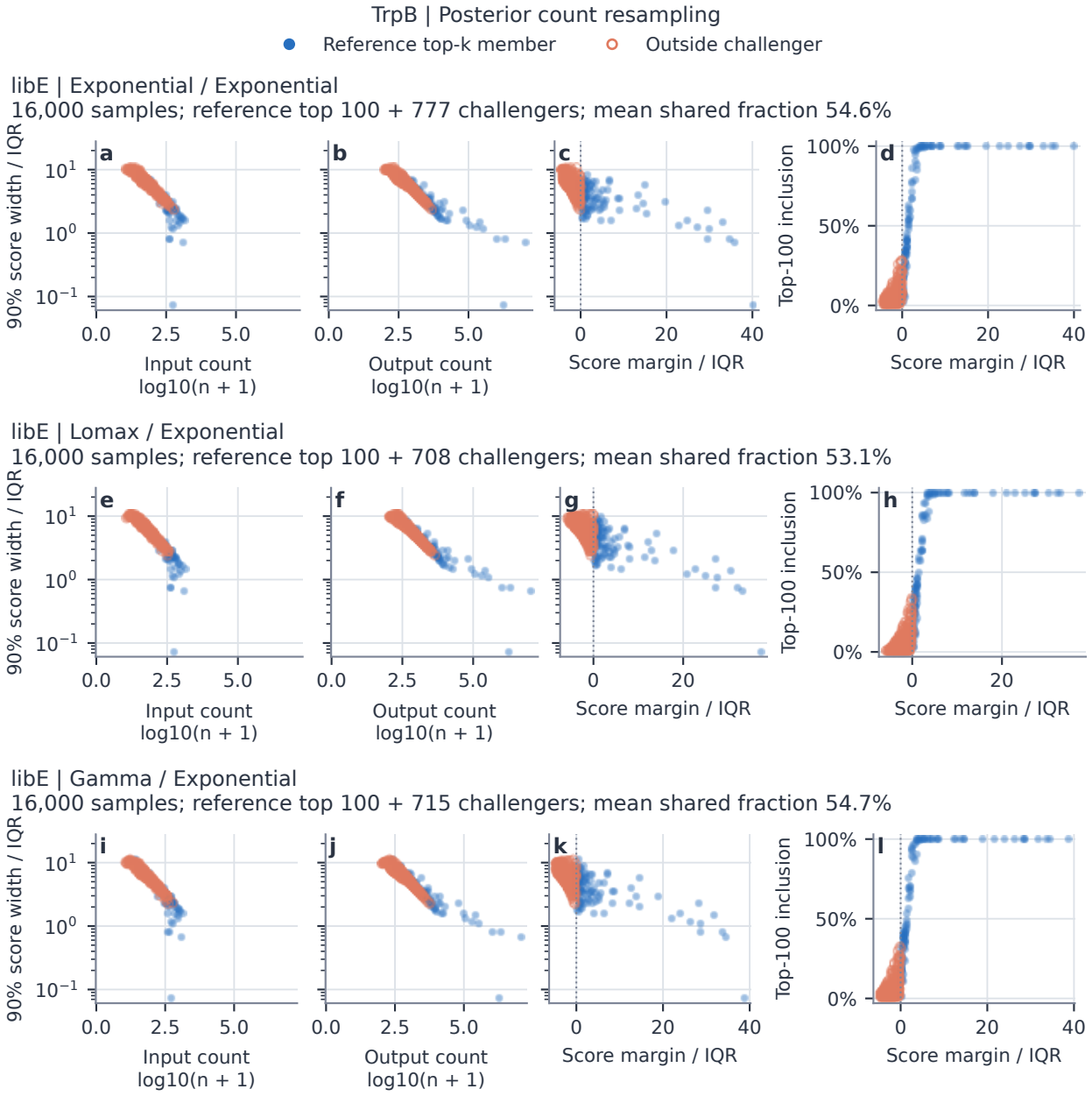


Figure S52: TrpB decision-cohort summaries: posterior count resampling. Rows, top to bottom: trpb-libE, Exponential–Exponential; trpb-libE, Lomax–Exponential; trpb-libE, Gamma–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

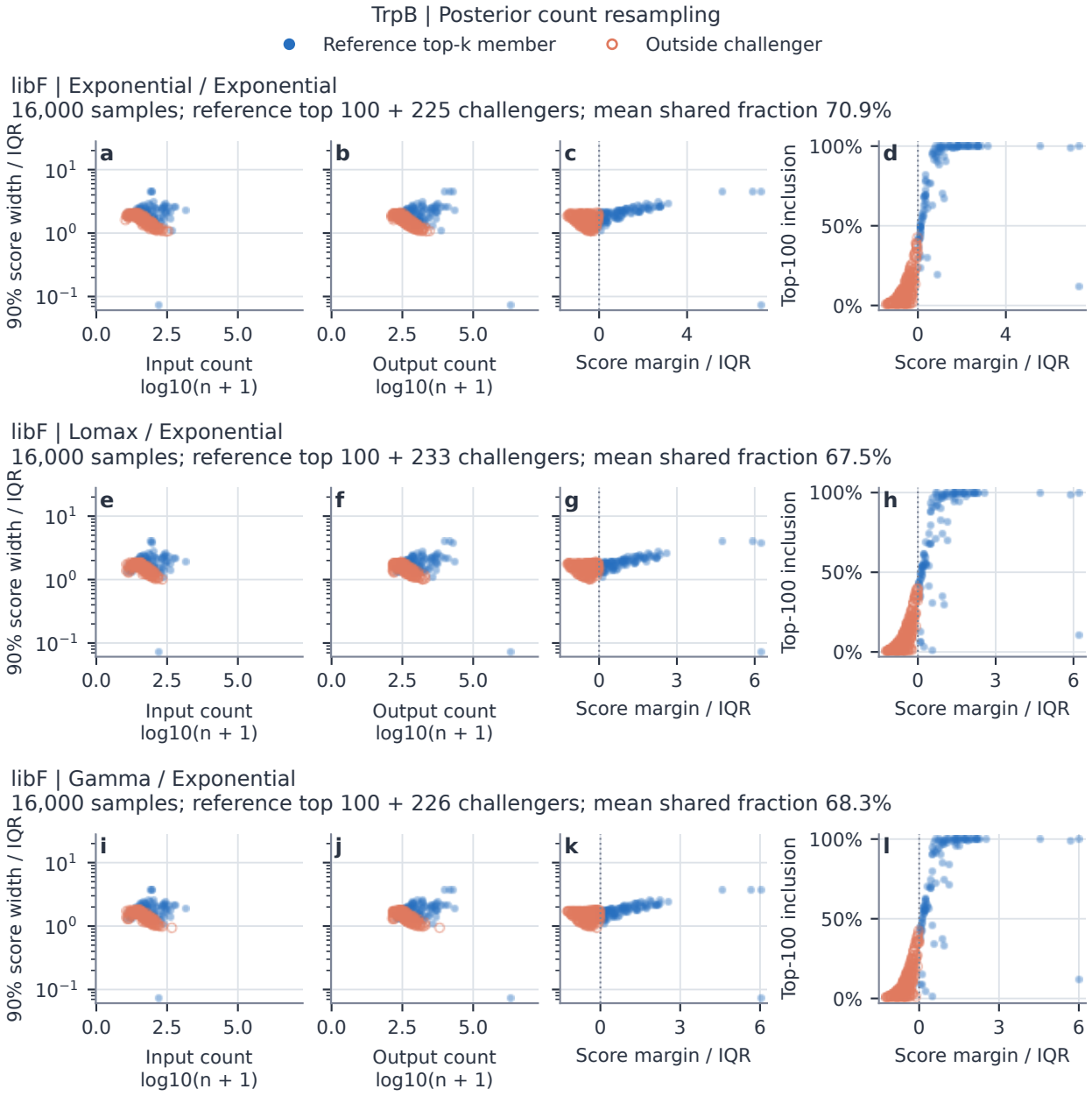


Figure S53: TrpB decision-cohort summaries: posterior count resampling. Rows, top to bottom: trpb-libF, Exponential–Exponential; trpb-libF, Lomax–Exponential; trpb-libF, Gamma–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

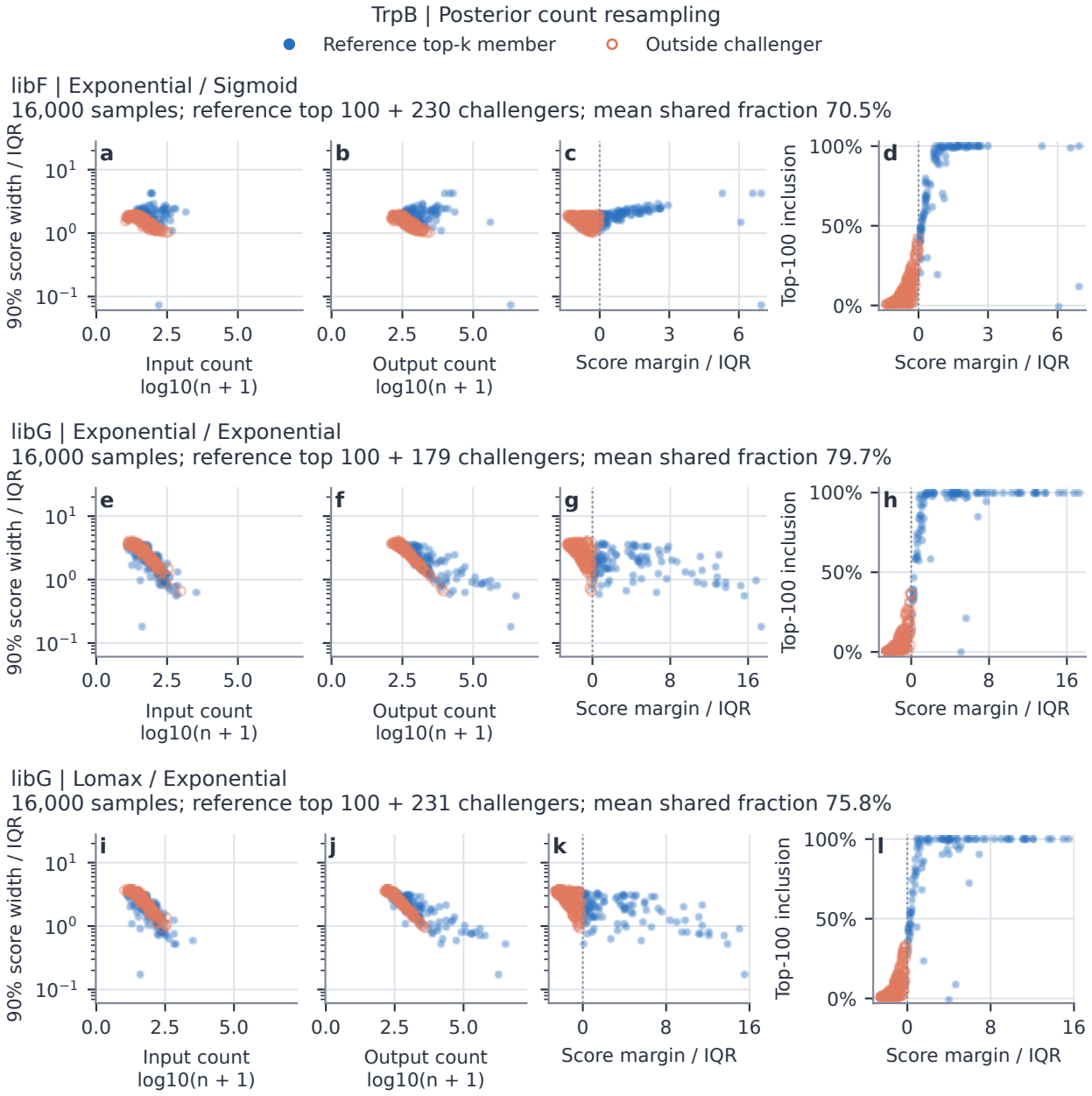


Figure S54: TrpB decision-cohort summaries: posterior count resampling. Rows, top to bottom: trpb-libF, Exponential–Sigmoid; trpb-libG, Exponential–Exponential; trpb-libG, Lomax–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

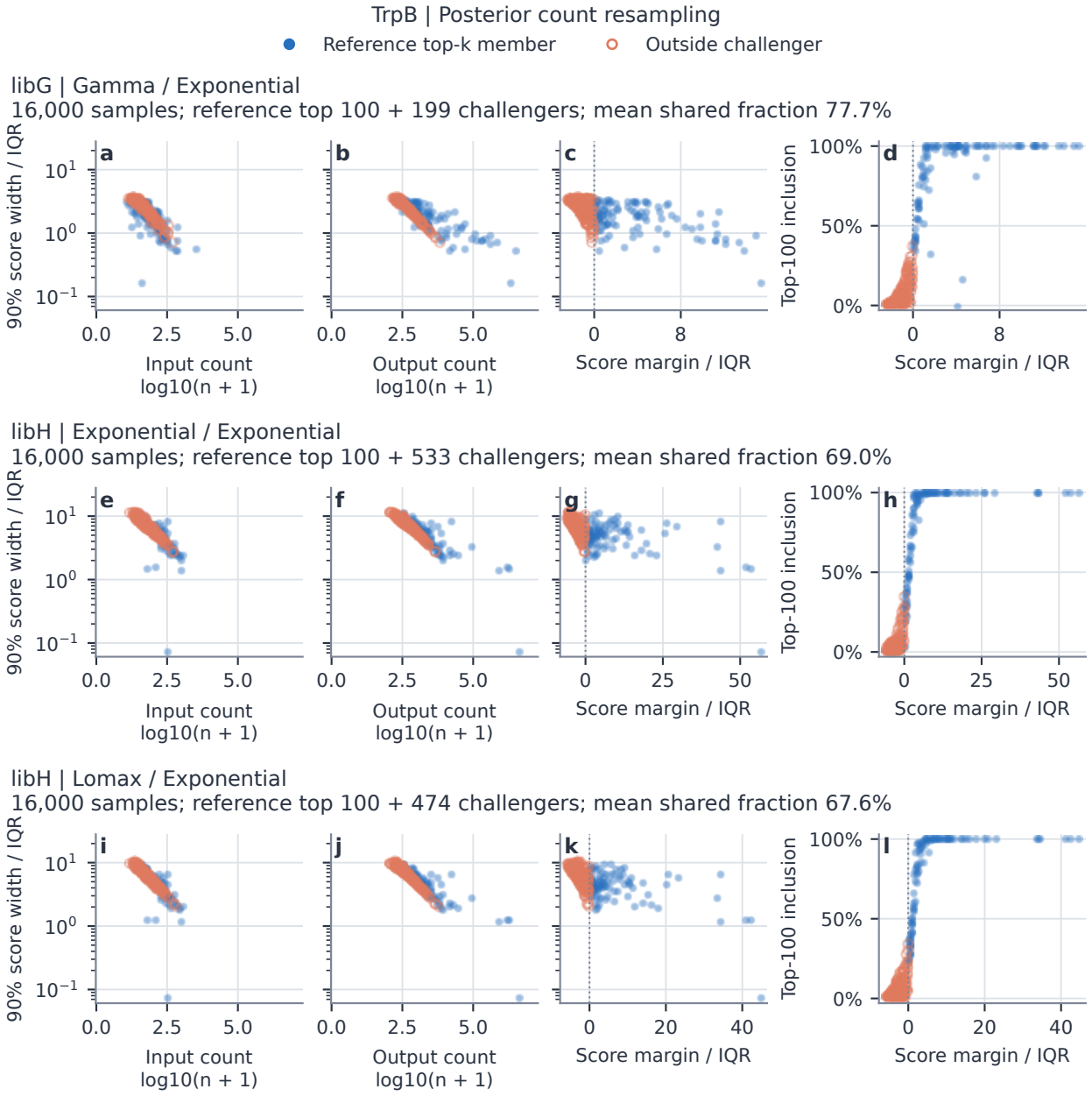


Figure S55: TrpB decision-cohort summaries: posterior count resampling. Rows, top to bottom: trpb-libG, Gamma–Exponential; trpb-libH, Exponential–Exponential; trpb-libH, Lomax–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

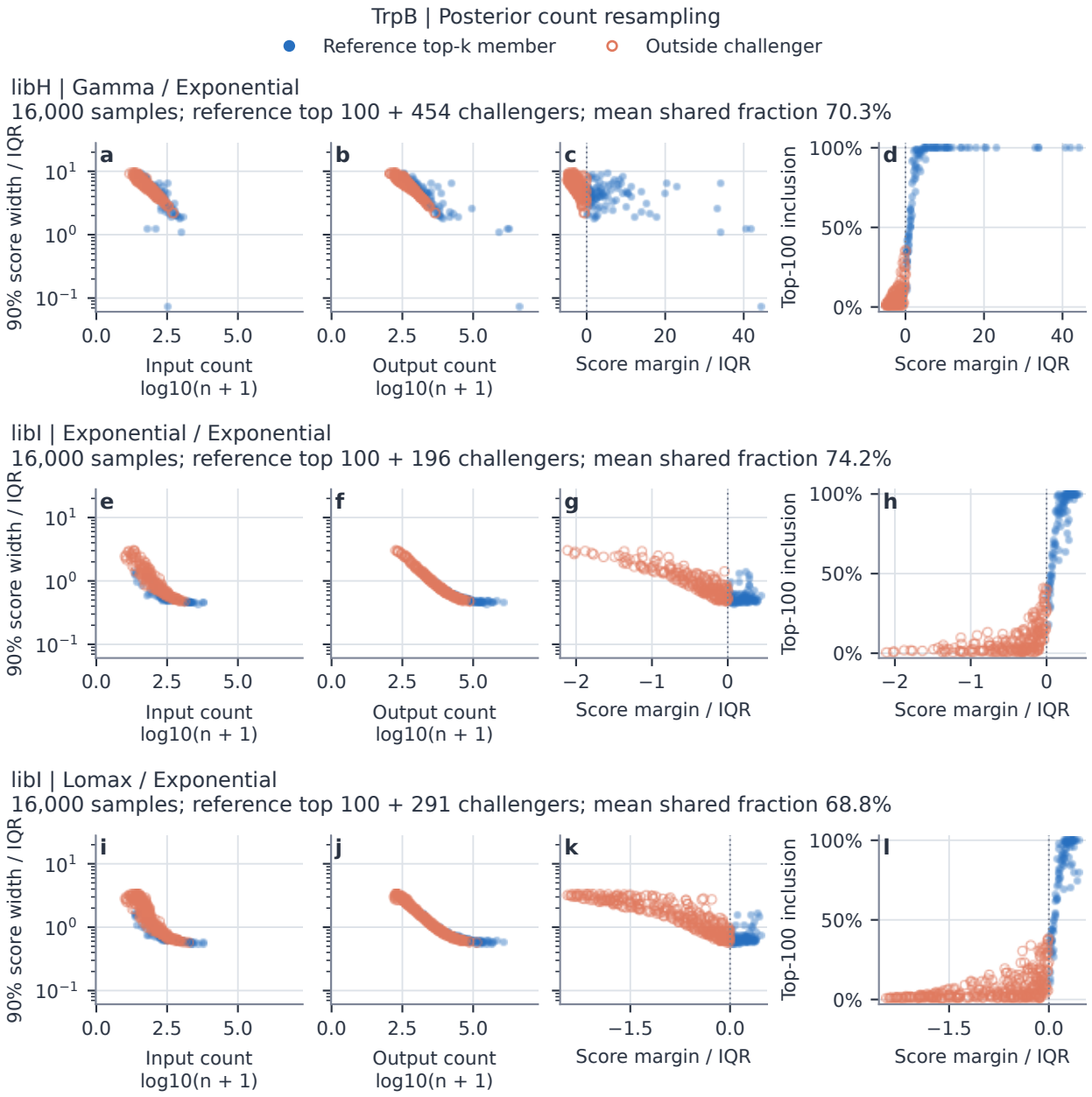


Figure S56: TrpB decision-cohort summaries: posterior count resampling. Rows, top to bottom: trpb-libH, Gamma–Exponential; trpb-libI, Exponential–Exponential; trpb-libI, Lomax–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

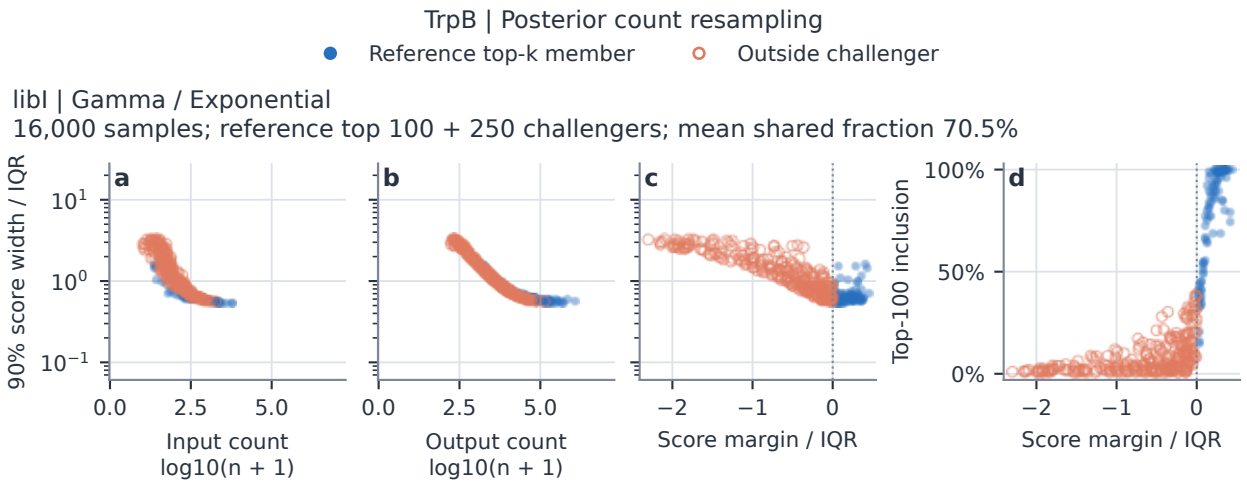


Figure S57: TrpB decision-cohort summaries: posterior count resampling. Rows, top to bottom: trpb-libI, Gamma–Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

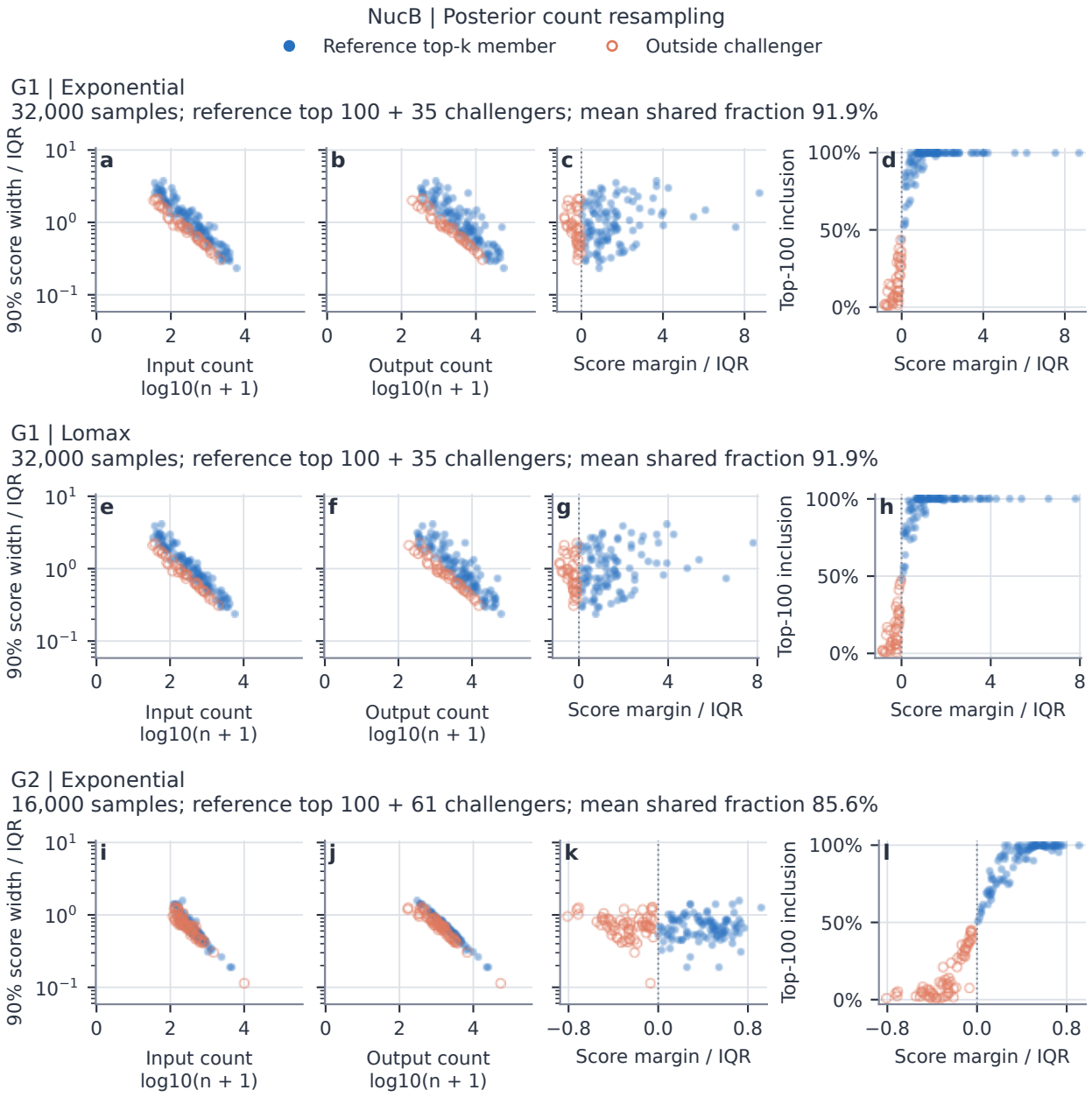


Figure S58: NucB decision-cohort summaries: posterior count resampling. Rows, top to bottom: nuc-G1, Exponential; nuc-G1, Lomax; nuc-G2, Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

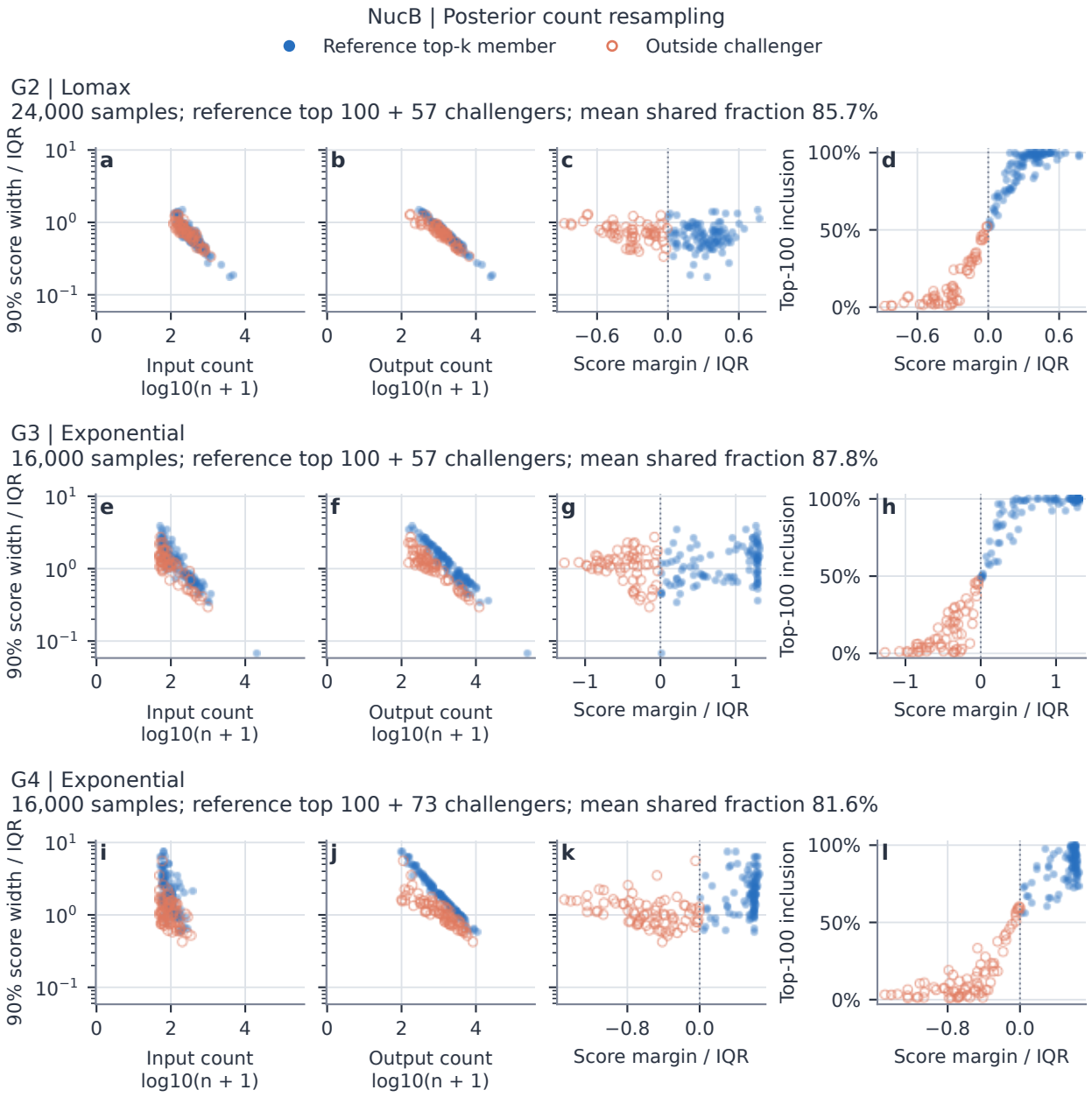


Figure S59: NucB decision-cohort summaries: posterior count resampling. Rows, top to bottom: nuc-G2, Lomax; nuc-G3, Exponential; nuc-G4, Exponential. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.

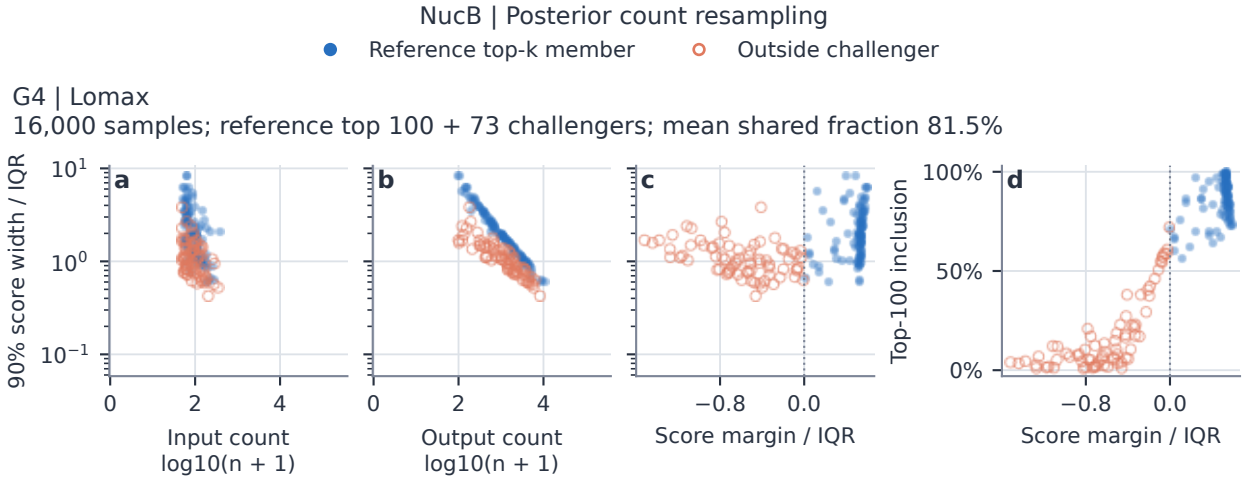


Figure S60: NucB decision-cohort summaries: posterior count resampling. Rows, top to bottom: nuc-G4, Lomax. Columns show normalized central 90% score width versus input count, output count, and median-score margin from the top-100 cutoff; then top-100 inclusion versus that margin. Widths and margins use the run's top-1,000 median-score interquartile range (IQR). Blue filled points are reference top-100 members; orange open points are outside candidates with inclusion probability at least 0.01. Count and score-width limits are shared within family and scheme; margin limits are shared within a row. Dotted lines mark the cutoff. No regression lines are fitted.